

The Bias of Nonlinear Two-Time-scale Stochastic Approximation under Constant Step-Sizes

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Abstract

Two-timescale stochastic approximation (TTSA) is a fundamental tool for analyzing coupled iterative algorithms in reinforcement learning, optimization, and stochastic control. However, finite-time guarantees for nonlinear two-timescale schemes remain difficult to obtain, especially under constant step-sizes. In this paper, we study nonlinear TTSA with step-sizes $\alpha \gg \beta$. Under standard stability, regularity, and Markovian noise assumptions, we upper bound the mean-squared error and the bias of both iterates around their limiting equilibria. Our bounds scale as $O(\alpha + \beta^2/\alpha^2)$, which we prove to be tight when $\beta \leq \alpha^{3/2}$. The analysis separates the contributions of initial conditions, fast-timescale tracking error, Markovian dependence, and timescale coupling, thereby clarifying the origin of the β^2/α^2 term. Our results reveal qualitative differences from the linear TTSA setting previously studied, showing that nonlinear dynamics introduce additional finite-time effects that are absent in the linear case.

1 Introduction

A broad family of machine learning and optimization algorithms can be formulated within the framework of Stochastic Approximation (SA) [4, 25, 3]. In the two-timescale setting [24, 29], two sequences of parameter vectors, $\{x_k\}_{k \geq 0}$ and $\{y_k\}_{k \geq 0}$, are generated from the following recursive update rule:

$$\begin{aligned} x_{k+1} &= x_k - \alpha_k f(x_k, y_k, \xi_{k+1}) \\ y_{k+1} &= y_k - \beta_k g(x_k, y_k, \xi_{k+1}), \end{aligned} \tag{1}$$

where, for a measurable space $(\Xi, \mathcal{S})$, the $\mathcal{S}$ -measurable functions $f : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \times \Xi \rightarrow \mathbb{R}^{d_x}$ and $g : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \times \Xi \rightarrow \mathbb{R}^{d_y}$ are the noisy dynamics of the system, $\{\xi_k\}_{k \geq 1}$ is the source of noise modeled as a stochastic process defined on Ξ and $\{\alpha_k\}_{k \geq 0}, \{\beta_k\}_{k \geq 0}$ are positive sequences representing the step-sizes. Some notable examples of algorithms that naturally exhibit this scheme of recursive updates include AdaGrad [13], Generative adversarial networks [16], Bi-level Optimization [20] and Reinforcement Learning [34, 23].

To study the dynamics of SA, a classical tool is to use the ODE method [5]. It consists in viewing (1) as a noisy Euler discretization of the singularly perturbed ODE

$$\begin{aligned} \dot{x}(t) &= -\frac{1}{\epsilon} \bar{f}(x(t), y(t)) \\ \dot{y}(t) &= -\bar{g}(x(t), y(t)), \end{aligned} \tag{2}$$

with $\epsilon = \beta_k/\alpha_k \approx 0$ [4, Chapter 6]. The small parameter ϵ induces a separation between the two timescales: The variable x evolves on the fast timescale, whereas y evolves slowly and is

therefore seen by the x -dynamics as quasi-static. This motivates the standard decoupling of the two ODEs: for each fixed y , the fast dynamics are described by $\dot{x}(t) = -\bar{f}(x(t), y)$. Under suitable stability assumptions, the trajectory solutions of this fast ODE converge to the asymptotically stable equilibrium $h(y)$ satisfying $\bar{f}(h(y), y) = 0$. Once this tracking property is established, the slow ODE is $\dot{y}(t) = -\bar{g}(h(y(t)), y(t))$. Under suitable assumptions, the solution of the ODE converges to a point y^* . The ODE method consists in assuming that $x_k \approx h(y_k)$ and $y_k \approx y^*$.

To guarantee the accuracy of the ODE method, it is therefore important to characterize the error that is made when approximating (1) by the solutions of the ODE (2). The classical convergence results for SA and TTSA typically rely on diminishing step-sizes. For two time-scale schemes, one usually assumes the step-sizes to have infinite ℓ^1 -norm, finite ℓ^2 -norm and that $\lim_{k \rightarrow \infty} \frac{\beta_k}{\alpha_k} = 0$ which ensures the time-scale separation of the associated limiting ODEs [4]. In many applications, however, constant step-sizes $\alpha_k \equiv \alpha$ and $\beta_k \equiv \beta$ are preferred, notably because they allow an exponential rate of forgetting initial conditions [30]. In this regime, one should not expect exact convergence to the equilibrium; instead, the iterates typically approach a neighborhood whose size depends primarily on the step-sizes and the noise.

This motivates a finite-time analysis of the constant-step-size TTSA under Markovian noise. Previous works have quantified the bias induced by constant step-sizes for nonlinear single time-scale SA [1] and for linear TTSA [19]. The nonlinear TTSA setting remains more delicate, because the fast tracking error and the slow dynamics interact through the nonlinear manifold $x = h(y)$. Our goal is to characterize this interaction quantitatively. We will measure the performance of the algorithm through two quantities: the *mean-squared error* (MSE), which is defined as

$$\text{MSE}_k^x \triangleq \mathbb{E} \left[\|x_k - h(y_k)\|^2 \right] \quad \text{and} \quad \text{MSE}_k^y \triangleq \mathbb{E} \left[\|y_k - y^*\|^2 \right],$$

and the *bias* of the iterates around their limiting equilibrium, which we formally define as

$$\text{Bias}_k^x \triangleq \mathbb{E} [x_k] - h(y^*) \quad \text{and} \quad \text{Bias}_k^y \triangleq \mathbb{E} [y_k] - y^*.$$

Our goal in this paper is to provide a finite-time analysis of nonlinear two time-scale stochastic approximation with constant step-sizes under a quite general Markovian noise. One original feature of our model is that we allow the kernel of the noise process ξ_k to depend on the iterates (x_k, y_k) . This is particularly relevant for applications to reinforcement learning where the exploration policy might depend on the current values of the parameters.

Contributions. Our first main result, Theorem 1, proves that the mean-squared errors of both time-scale components are bounded, up to exponentially decaying initialization terms, by $O(\alpha + \beta + \beta^2/\alpha^2)$, which reduces to $O(\alpha + \beta^2/\alpha^2)$ when $\beta \leq \alpha$. The proof of Theorem 1 combines a contraction argument for both iterates x_k and y_k with a decomposition of the Markovian noise based on the problem-specific Poisson equation which allows us to handle the dependencies of the transition kernel on the model's parameters. The bound we prove separates the exponentially vanishing contribution of the initialization from the persistent error induced by constant step-sizes. The resulting recursion makes explicit the different sources of error: the intrinsic fluctuations of the fast iterate, the slow motion of the quasi-stationary target $h(y_k)$, and the bias induced by the noise. This extends the approaches developed for nonlinear single time-scale SA [1] and nonlinear TTSA with diminishing step-sizes [10] to the constant-step-size non-linear TTSA regime. When $\beta \leq \alpha^{3/2}$, our MSE bounds are of order $O(\alpha)$ for both the fast and the slow iterate. In Theorem 2, we show that this bound cannot be improved by exhibiting an example whose MSE is of the order $\Omega(\alpha)$. This shows that the non-linear TTSA is significantly different from the linear TTSA for which the MSE of the slow iterate was shown to be $O(\beta + \alpha^2)$ in [26].

Our second main result, Theorem 3, establishes matching-order bounds on the bias of the fast and slow iterates. The bias analysis is substantially more delicate than the MSE analysis because the analysis of the first-order dynamics of the two bias components remains coupled. We handle this difficulty by linearizing the nonlinear drift around the limiting equilibrium and then applying a sequence of coordinate transformations that separates the dominant fast and slow modes. In these transformed coordinates, the bias satisfies a recursion formula, which leads to upper bounds involving the Markovian-noise residuals and the MSE bounds from Theorem 1. Transforming back to the original variables yields bias bounds of order $O(\alpha + \beta^2/\alpha^2)$. When $\beta \leq \alpha^{3/2}$, this again reduces to $O(\alpha)$ which we prove to be tight in Theorem 4.

Outline. The remainder of the paper is organized as follows. In Section 2, we detail related work on single and two time-scale stochastic approximation, focusing on finite-time guarantees obtained with constant step-sizes under Markovian noise. Section 3 introduces the nonlinear TTSA model and states the assumptions used throughout the paper. Section 4 presents Theorem 1, a finite-time MSE bound, together with a proof overview and a discussion of tightness. Section 5 presents Theorem 3, a finite-time bias bound, and outlines the main ideas behind its proof. Finally, Section 6 illustrates the theoretical bounds by creating an example of nonlinear TTSA showing how the term β^2/α^2 persists in the bounds of our MSE.

2 Related Work

The study of stochastic approximation (SA) algorithms traces back to the seminal work of [33], which introduced a framework for finding a root x that satisfies $M(x) = b$ for a given $b \in \mathbb{R}$. Here, when referring to SA, we mean *one* time-scale stochastic approximation, where only the variable x is present (as opposed to TTSA where two variables are present and evolve at different time-scales $\beta \ll \alpha$). In this classical setting, the monotonic function $M : \mathbb{R} \rightarrow \mathbb{R}$ is assumed to be unknown, requiring the algorithm to rely strictly on noisy observations. A pivotal development in the theoretical understanding of SA is through the ordinary differential equation (ODE) method [5]. This technique characterizes the asymptotic convergence properties of discrete SA updates by analyzing their continuous-time limit, represented by the analogous ODE $\dot{x} = -f(x)$. The ODE method is the method of choice for analyzing SA dynamics, with its underlying theory and applications extensively detailed across numerous foundational texts [4, 25, 3].

Constant step-size SA. A line of work has been developed around the constant step-size assumption in SA due to practical advantages in algorithm design. Starting with Linear SA (LSA), [27] have shown that the MSE of the Polyak-Ruppert (PR) averages [31] decreases with order $O(\frac{1}{k})$ under i.i.d noise. Previously, in [2] averaging was used on non-strongly convex SA to achieve the same rate. Under the same settings [15] proved that with probability at least $1 - \delta$, $|x_k - x^*|$ is bounded by $O(\sqrt{\alpha \log(1/\delta)})$, where x^* denotes the equilibrium of x . Later, the work of [14] generalized the high probability bounds of the latter result to PR averages under Markovian noise. In [19], it is shown that LSA under constant step size and Markovian noise is characterized by a bias of order $O(\alpha)$ that can be improved to higher orders by applying Richardson-Romberg extrapolation. Switching to non-linear SA, [7] showed an MSE of order $O(\alpha \log(1/\alpha))$. This bound is improved in [1] to $O(\alpha)$, with a characterization of the bias vector by $\alpha V + O(\alpha^2)$, where V is a solution vector of a problem-specific Lyapunov equation.

Linear TTSA. In the linear regime, the foundational work [24] established the asymptotic variance and normality of the iterates under i.i.d. noise and decreasing step-sizes. Subsequent research shifted toward finite-time analyses. In [21], the MSE of the fast iterate was shown to be of order $O(\alpha_k)$ and that of the slow iterate $O(\beta_k)$ for step-sizes satisfying $\alpha_k = O(k^{-v})$ for some $v < 1$ and $\beta_k = O(k^{-1})$. A convergence rate of $O(1/k^{2/3})$ under Markovian noise was derived in [9], which was later tightened to $O(1/k)$ in [17]. The constant step-sizes setting was studied in [26], which established an upper bound of order $O(\alpha)$ and $O(\beta + \alpha^2)$ for the fast and slow iterates, respectively. They further proved upper bounds of order $O(\alpha + \beta)$ for the bias of both iterates. Our results indicate that these bounds are specific to this setting, since they degrade after removing the linearity assumption.

Nonlinear TTSA. Parallel advancements have been made in non-linear TTSA. Extending asymptotic normality to this setting, [29] proved a Central Limit Theorem (CLT) for the iterates, a property recently expanded upon by [18] under decreasing step-sizes. Regarding finite-time guarantees, [10, 9] achieved an MSE rate of $O(1/k^{2/3})$ under both i.i.d. and Markovian noise. More recently, [6] analyzed contractive non-linear TTSA under Markovian noise, demonstrating that the MSE for both iterates is bounded by $O(\alpha_k + \beta_k^2/\alpha_k^2)$. All of those works assume decreasing step-sizes.

The remaining gap, which we address in this work, is to obtain finite-time bounds on both the MSE and the bias of nonlinear TTSA with constant step-sizes, under Markovian noise.

3 Assumptions

First, to guarantee algorithmic stability, we assume that the algorithm's iterates remain within a compact set. This is often enforced in practice via explicit projections [25, Chapter 5], [28].

A. 1. *There exist two compact sets $\mathcal{X} \subset \mathbb{R}^{d_x}$ and $\mathcal{Y} \subset \mathbb{R}^{d_y}$ such that for all $k \geq 0$, $x_k \in \mathcal{X}$ and $y_k \in \mathcal{Y}$ a.s.*

Our second assumption concerns the stochastic process $(\xi_k)_{k \geq 1}$, which we assume to be a *positive recurrent Markov chain*. This allows us to account for temporally correlated data, in contrast with the classical i.i.d. or martingale-difference noise models, and is particularly relevant for applications such as reinforcement learning. The price of this added flexibility is that the analysis must handle the bias and dependence induced by the Markovian dynamics.

A. 2. *$(\xi_k)_{k \geq 1}$ evolves as a Markov chain on a finite state space Ξ with a transition kernel $K_{(x_k, y_k)}$ that is allowed to depend on x_k, y_k , that is: for every $\xi', \xi \in \Xi$ and $x \in \mathcal{X}, y \in \mathcal{Y}$ we have:*

$$\mathbb{P}(\xi_{k+1} = \xi' | \xi_k = \xi, \mathcal{F}_k) = \mathbb{P}(\xi_{k+1} = \xi' | \xi_k = \xi, x_k = x, y_k = y) \triangleq K_{(x, y)}(\xi, \xi'),$$

where $\{\mathcal{F}_k\}_{k \geq 1}$ is the natural filtration of the process $(x_k, y_k, \xi_k)_{k \geq 1}$. We further assume that the kernel $K_{(x, y)}$ corresponds to a unichain Markov chain for all $(x, y) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$. Also, we assume that for all $\xi, \xi' \in \Xi$ the quantity $K_{(\cdot, \cdot)}(\xi, \xi')$ is L_K -Lipschitz continuous:

$$|K_{(x_1, y_1)}(\xi, \xi') - K_{(x_2, y_2)}(\xi, \xi')| \leq L_K(\|x_1 - x_2\| + \|y_1 - y_2\|).$$

The assumption of having a Markovian noise is standard in stochastic approximation and covers a broad class of algorithms [26, 18, 10, 7]. The fact that we allow the kernel at time k to depend on (x_k, y_k) is more original and is relevant in reinforcement learning algorithms where the exploration policy might depend on the current parameters [1]. Assumption A.2 guarantees the existence of a unique stationary distribution $\pi_{(x, y)}$ for each (x, y) , thereby allowing us to define the averaged functions $\bar{f}$ and $\bar{g}$ by integrating f and g with respect to π :

$$\bar{f}(x, y) \triangleq \mathbb{E}[f(x, y, \xi)] = \sum_{\xi \in \Xi} \pi_{(x, y)}(\xi) f(x, y, \xi) \quad \text{and} \quad \bar{g}(x, y) \triangleq \mathbb{E}[g(x, y, \xi)]. \quad (3)$$

Furthermore, the assumption on Lipschitzness of K implies that the stationary distribution $\pi_{(x, y)}$ is also Lipschitz-continuous in (x, y) , see e.g. [1]. Third, to ensure the existence and uniqueness of the limiting trajectories of the equivalent ODE, we need to assume Lipschitzness of f, g , which, combined with Lipschitzness of K and the definitions of $\bar{f}$ and $\bar{g}$ of (3) implies that the functions $\bar{f}$ and $\bar{g}$ are also Lipschitz continuous which is a fundamental prerequisite of the ODE method, ensuring the existence and uniqueness of solutions of the ODE. We start with our assumptions on f :

A. 3. *For all $\xi \in \Xi$, the function $f(\cdot, \cdot, \xi)$ is L_f -Lipschitz continuous:*

$$\|f(x_1, y_1, \xi) - f(x_2, y_2, \xi)\| \leq L_f(\|x_1 - x_2\| + \|y_1 - y_2\|)$$

In addition, there exists $\mu_f > 0$ such that $\bar{f}$ is μ_f -strongly monotonic in its first argument, i.e., for every $x_1, x_2 \in \mathcal{X}$ and $y \in \mathcal{Y}$ we have

$$\langle \bar{f}(x_1, y) - \bar{f}(x_2, y), x_1 - x_2 \rangle \geq \mu_f \|x_1 - x_2\|^2.$$

This strong monotonicity assumption guarantees that for a fixed y , the ODE $\dot{x}(t) = -\bar{f}(x(t), y)$ possesses a unique, exponentially stable equilibrium, which we denote by $h(y)$. The equilibrium map h is $\frac{L_f}{\mu_f}$ -Lipschitz from Lipschitzness and strong monotonicity of f :

$$\begin{aligned} \mu_f \|h(y_1) - h(y_2)\|^2 &\leq \langle \bar{f}(h(y_1), y_1) - \bar{f}(h(y_2), y_1), h(y_1) - h(y_2) \rangle \\ &= \langle \bar{f}(h(y_2), y_2) - \bar{f}(h(y_2), y_1), h(y_1) - h(y_2) \rangle \\ &\leq \|\bar{f}(h(y_2), y_2) - \bar{f}(h(y_2), y_1)\| \|h(y_1) - h(y_2)\| \\ &\leq L_f \|y_2 - y_1\| \|h(y_1) - h(y_2)\| \end{aligned}$$

The result follows when $h(y_1) \neq h(y_2)$, we denote $L_h \triangleq \frac{L_f}{\mu_f}$. Moving to the slow iterate, we need Lipschitzness of g :

A. 4. For all $\xi \in \Xi$, the function $g(\cdot, \cdot, \xi)$ is L_g -Lipschitz continuous:

$$\|g(x_1, y_1, \xi) - g(x_2, y_2, \xi)\| \leq L_g(\|x_1 - x_2\| + \|y_1 - y_2\|)$$

In addition, for $\mu_g > 0$, $\bar{g}$ is 1-point μ_g -strongly monotonic with respect to a point $y^* \in \mathcal{Y}$ that satisfies $\bar{g}(h(y^*), y^*) = 0$: for all $y \in \mathcal{Y}$:

$$\langle \bar{g}(h(y), y), y - y^* \rangle \geq \mu_g \|y - y^*\|^2.$$

This assumption implies that the function $\bar{g}(h(y), y)$ is also Lipschitz continuous and in particular implies that the ODE $\dot{y} = -\bar{g}(h(y), y)$ has a unique solution. To guarantee that this ODE has a unique stationary point y^* , we added the 1-point strong monotonicity at y^* , for instance, as in [11, 8]. It can be viewed as a generalization of the non-linear case of having Hurwitz matrices in the linear setting [26, 19, 15], which are the conditions required for the ODE trajectories to exhibit global convergence.

Notation Let $x \in \mathbb{R}^d$, $\|x\|$ is the Euclidean norm $\|x\| = (\sum_{i=1}^d x_i^2)^{1/2}$. For $y \in \mathbb{R}^d$ we denote the inner product between x and y by $\langle x, y \rangle$. For a square matrix $M \in \mathbb{R}^{n \times n}$ we define the operator norm $\|M\|_{\text{op}} = \sup_{\|x\|=1} \|Mx\|$. For a positive definite matrix L we define the condition number of L as $\kappa(L) = \frac{\lambda_{\max}(L)}{\lambda_{\min}(L)}$ where $\lambda_{\max}(L)$ and $\lambda_{\min}(L)$ are the largest and smallest eigenvalues of L . For a multivariate function $\bar{f} : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}^{d_x}$, $\nabla_x \bar{f}(x, y)$ denotes the partial Jacobian with respect to the first argument $(\nabla_x \bar{f}(x, y))_{i,j} = \frac{\partial \bar{f}_i(x, y)}{\partial x_j}$ and $\nabla_y \bar{f}(x, y)$ denotes the partial Jacobian with respect to the second argument.

4 The MSE Analysis of TTSA

Our first main result is a bound on the Mean Squared Error (MSE) for both the fast and the slow iterates.

4.1 Main Result

The following theorem provides non-asymptotic bounds on MSE_k^x and MSE_k^y , explicitly separating the transient initial condition from the steady-state residual error caused by the Markovian noise and the constant step-sizes.

Theorem 1. Assuming that A.1-3 hold, and that $\alpha\mu_f < 1$ and $\beta \leq \alpha$, then for any $k \geq 1$ we obtain the following upper bound on the MSE of the fast iterate,

$$\mathbb{E}[\|x_k - h(y_k)\|^2] \leq \underbrace{(1 - \alpha\mu_f)^k \|x_0 - h(y_0)\|^2}_{\text{Decaying initial condition}} + \underbrace{\alpha \left(\frac{B_1}{1 - \alpha\mu_f} + B_2 \right) + \beta B_3 + \frac{\beta^2}{\alpha^2} B_4}_{\text{MSE}_{\infty}^x(\alpha, \beta)}, \quad (4)$$

where the constants $(B_k)_{k \in [4]}$ are fully defined in appendix (Eq. (18)-(21)), and are independent of α, β and k : they only depend on the properties of the Markovian noise, the range, monotonicity and Lipschitzness parameters of the functions f, g, h . Further, assuming A.4, and $\beta\mu_g < 1$, for any $k \geq 1$ the MSE of the slow iterate is upper bounded by

$$\begin{aligned} \mathbb{E}[\|y_k - y^*\|^2] &\leq \underbrace{(1 - \beta\mu_g)^k \|y_0 - y^*\|^2 + D_1 \beta k (1 - (\alpha\mu_f \wedge \beta\mu_g))^k \|x_0 - h(y_0)\|^2}_{\text{Decaying initial conditions}} \\ &\quad + \underbrace{D_2 \text{MSE}_{\infty}^x(\alpha, \beta) + \beta \left(\frac{D_3}{1 - \beta\mu_g} + D_4 \right) + \alpha D_5}_{\text{MSE}_{\infty}^y(\alpha, \beta)}. \end{aligned}$$

where the constants $(D_k)_{k \in [5]}$ are also formally defined in appendix (Eq. (28)-(32)) and depend again on problem parameters but not on α, β, k .

Proof Overview. To prove the MSE bounds for the fast iterate x_k , we first use the strong monotonicity assumption A.3 to extract the following recursion on the tracking errors between steps k and $k + 1$,

$$\|x_{k+1} - h(y_{k+1})\|^2 \leq (1 - \alpha\mu_f) \|x_k - h(y_k)\|^2 + r_k^x, \quad (5)$$

where r_k^x is a residual term containing the deviation of the noisy dynamics from their stationary means in addition to higher order terms of α and β that we detail in the next paragraph. The goal of this recursion is to dissociate the impact of the initial conditions from the cumulative impact of the noise. Repeatedly applying the above inequality, we get

$$\|x_{k+1} - h(y_{k+1})\|^2 \leq (1 - \alpha\mu_f)^{k+1} \|x_0 - h(y_0)\|^2 + \sum_{j=0}^k (1 - \mu_f\alpha)^{k-j} r_j^x.$$

This first step is similar to the approach used for the decreasing step sizes in [10], except in the latter case the sum of the residuals under expectation vanishes as $k \rightarrow \infty$ which is not the case here due to the assumption of fixed step-sizes.

To bound this sum of residuals, we then use A.3 to show that each r_j^x is bounded by $2\alpha \langle x_j - h(y_j), f(x_j, y_j, \xi_j) - \bar{f}(x_j, y_j) \rangle$, plus a term $M_{\alpha, \beta}$ that is of order $O(\alpha^2 + \beta^2 + \alpha\beta + \beta^2/\alpha)$. The sum $\sum_{j=0}^k (1 - \mu_f\alpha)^{k-j} M_{\alpha, \beta}$ leads to a $O(\alpha + \beta + \beta^2/\alpha^2)$ error term. The main technical difficulty is therefore to control the sum induced by the first term. To do so, we follow the Poisson-equation technique used in [1, Lemma 8] to rewrite the noise as a Martingale difference term, a bounded term that scales linearly with the step-sizes, and a weighted telescoping term. Because of the geometric weights $(1 - \alpha\mu_f)^{k-j}$, the telescoping term is not exact and leaves additional terms. Additionally, since the kernel is depending on the current iterates, the Martingale term creates a drift in the kernel that we control due to Lipschitzness of K from A.2. Lemma 2 shows that all these contributions are of the required order.

To prove the MSE bound for the slow iterate, we apply a recipe similar to the one followed for the fast iterate. From assumption A.4 we obtain an equation for y similar to (5) with a contraction factor $1 - \beta\mu_g$, along with an additional term capturing the influence of the fast iterate:

$$\|y_{k+1} - y^*\|^2 \leq (1 - \beta\mu_g) \|y_k - y^*\|^2 + \beta \|x_k - h(y_k)\|^2 + r_k^y.$$

We then use the results of the fast iterate to bound $\|x_k - h(y_k)\|^2$. The treatment of r_k^y is then similar to the analysis done for the fast iterate. The full proof can be found in Appendix B. $\square$

4.2 Discussion and tightness

Under the constant step-size regime and the strong monotonicity assumptions A.3 and A.4, the initial conditions are forgotten at an exponential rate, respectively $(1 - \alpha\mu_f)^k$ for the fast iterate and $(1 - \beta\mu_g)^k$ for the slow iterate. This illustrates one of the main practical advantages of constant step-sizes: the iterates rapidly enter a neighborhood of the equilibrium, although the size of this neighborhood is controlled by the persistent stochastic error. The two time-scale structure also creates a transient coupling between the initial fast error and the slow iterate, which appears through the term

$$k(1 - (\alpha\mu_f \wedge \beta\mu_g))^k \|\hat{x}_0\|^2,$$

which may initially increase because of the prefactor k , but still vanishes exponentially as $k \rightarrow \infty$.

Once the transient terms have vanished, the iterates remain, in expectation, within a neighborhood of the equilibrium whose squared radius scales as $O(\alpha + \beta^2/\alpha^2)$. The presence of the term β^2/α^2 in both bounds is one of the main features of Theorem 1. It is consistent with finite-time results for nonlinear TTSA with decreasing step-sizes, where bounds of order $O(\alpha_k + \beta_k^2/\alpha_k^2)$ are obtained for both iterates [6]. This term quantifies the cost of coupling the fast and slow recursions: without sufficient time-scale separation, it may dominate the steady-state error.

However, when the slow step-size is sufficiently small relative to the fast one, the α -term becomes dominant. In particular, if $\beta \leq \alpha^{3/2}$, then $\beta^2/\alpha^2 \leq \alpha$, and Theorem 1 immediately yields that, for any TTSA satisfying its assumptions, there exist constants $C > 0$ and $\alpha_0 > 0$ such that, for all $\alpha < \alpha_0$ and all $\beta \leq \alpha^{3/2}$,

$$\limsup_{k \rightarrow \infty} \mathbb{E}[\|x_k - h(y_k)\|^2] \leq C\alpha, \quad \text{and} \quad \limsup_{k \rightarrow \infty} \mathbb{E}[\|y_k - y^*\|^2] \leq C\alpha. \quad (6)$$

We emphasize that the steady-state MSE bound for the slow iterate is substantially larger than what is obtained in the linear TTSA setting, where the corresponding rate is $O(\beta + \alpha^2)$ [26]. This raises a

natural question: is the $O(\alpha)$ term in the bound for the non-linear case unavoidable, or is it merely a proof artifact? The next result shows that it is not. It constructs a nonlinear TTSA instance satisfying the assumptions of Theorem 1 for which both the fast tracking error and the slow error have asymptotic MSE at least of order α . Thus, in the nonlinear setting, the rate in (6) cannot be improved in general: there is a qualitative separation from the linear TTSA setting, where sharper rates are possible for the slow iterate.

Theorem 2. *There exist constants $C > 0$, $\alpha_0 > 0$, and a TTSA instance satisfying all the assumptions of Theorem 1 such that, for all step-sizes $\beta \leq \alpha < \alpha_0$,*

$$\liminf_{k \rightarrow \infty} \mathbb{E} [\|x_k - h(y_k)\|^2] \geq C\alpha, \quad \text{and} \quad \liminf_{k \rightarrow \infty} \mathbb{E} [\|y_k - y^*\|^2] \geq C\alpha.$$

The proof is given in Appendix C and relies on a simple one-dimensional construction for both variables x_k and y_k .

5 The Bias Analysis of TTSA

Analyzing the bias requires a finer characterization of the dynamics. We therefore assume additional differentiability, which allows us to linearize the averaged update around the equilibrium. The first-order terms drive the bias recursion, while the second-order Taylor remainders are controlled by the MSE bounds from Theorem 1. We use the following notation: $J_{xx} \triangleq \nabla_x \bar{f}(h(y^*), y^*)$, $J_{xy} \triangleq \nabla_y \bar{f}(h(y^*), y^*)$, $J_{yx} \triangleq \nabla_x \bar{g}(h(y^*), y^*)$ and $J_{yy} \triangleq \nabla_y \bar{g}(h(y^*), y^*)$.

B.1. *$\bar{f}$ and $\bar{g}$ are at least two times differentiable and their second derivatives are bounded operators. In addition, the matrices $-J_{xx}$ and $-\Delta \triangleq -J_{yy} + J_{yx}J_{xx}^{-1}J_{xy}$ are Hurwitz.*

The differentiability part of B.1 is standard in analyses that rely on a local expansion of the stochastic-approximation dynamics. It is used, for instance, to characterize the constant-step-size bias in nonlinear SA [1], and it is also a natural assumption in CLT analyses for nonlinear TTSA [18]. In our proof, this assumption allows us to linearize the averaged dynamics around the equilibrium: the first-order terms determine the leading bias recursion, while the second-order remainders are controlled through the MSE bounds.

The Hurwitz condition on $-J_{xx}$ and $-\Delta$ is essential when studying TTSA settings [26, 17, 18, 21, 24]. In particular, for the linear settings, it is equivalent (up to a linear transformation) to monotonicity in A.3 and A.4. It is needed to ensure that the trajectories under the approximate linear model converge exponentially fast to the equilibrium.

5.1 Main result

We now introduce the second main result of this paper, which provides an upper bound on the bias of TTSA. The result shows that, once the initial conditions vanish, the bias is of the same order as the bound on the MSE obtained in Theorem 1.

Theorem 3. *Assuming A.1-4 and B.1, there exists α^* defined in (8), β^* defined in (9), and ω^* defined in (10) such that for $\alpha < \alpha^*$, $\beta < \beta^*$ and $\frac{\beta}{\alpha} < \omega^*$ we have the following bound on the bias of both iterates:*

$$\max\{\|\mathbb{E}[x_k - h(y^*)]\|, \|\mathbb{E}[y_k - y^*]\|\} \leq \|T^{-1}S^{-1}\|_{op} (\|s_k^x\| + \|s_k^y\|)$$

for some upper triangular matrix T defined in (35) and a lower triangular matrix S defined in (39), and where vectors s_k^x, s_k^y are some transformed bias vectors defined in (43), (44) respectively. Furthermore,

$$\begin{aligned} \|s_k^x\| + \|s_k^y\| &\leq \underbrace{e^{-kc\alpha} \sqrt{\kappa(L)} \|s_0^x\| + e^{-kd\beta} \sqrt{\kappa(G)} \|s_0^y\| + ae^{-\alpha\hat{c}(k-1)} \|\hat{x}_0\|^2 + be^{-\beta\hat{d}(k-1)} \|\hat{y}_0\|^2}_{\text{Decaying initial conditions}} \\ &\quad + \underbrace{F_1(\alpha + \beta) + F_2(\text{MSE}_x^\infty(\alpha, \beta) + \text{MSE}_y^\infty(\alpha, \beta)) + R_{\alpha, \beta}}_{\text{Asymptotic bias}}, \end{aligned} \quad (7)$$

where F_1, F_2 are positive constants independent of α, β and k , whose values may depend on the problem parameters and on ω^* , $R_{\alpha, \beta}$ contains terms of higher order $(\alpha + \beta)^2$. Additionally, L, G are

symmetric positive definite matrices solutions of the Lyapunov equations (45), (49), $c = (8 \|L\|_{op})^{-1}$ and $d = (8 \|G\|_{op})^{-1}$, constants $a, b, \hat{c}, \hat{d}$ are related to decaying initial errors from the MSE and can be found in (53), and the MSE terms are defined in Theorem 1.

We detail the proof of the theorem in Appendix D, and present its main arguments in the following proof overview.

Proof Overview. Recall that $\text{Bias}_k^x \triangleq \mathbb{E}[x_k - h(y^*)]$ and $\text{bias}_k^y \triangleq \mathbb{E}[y_k - y^*]$ are the bias vectors. As used in [29, 10], we define the concatenation of the two bias vectors as $b_k \triangleq (\text{Bias}_k^x, \text{bias}_k^y)^T$. This manipulation allows us to analyze the coupled dynamics and to control cross impacts of both iterates on each other. Using assumption B.1 we use a Taylor expansion of the dynamics $\bar{f}$ and $\bar{g}$ around $(h(y^*), y^*)$ to obtain

$$b_{k+1} = Hb_k + R_k,$$

where H represents the linear dynamics around the equilibrium and the remainder term R_k contains the higher order terms of the Taylor expansion plus the expected impact of the Markovian noise.

As we will see later, the remainder term R_k can be controlled somehow similarly to what we did for the MSE. The intricate part is to find a way to analyze the dynamics induced by H where the off diagonal term represents the cross influence between the biases of both iterates. To treat this term, we use a linear transformations of coordinates to decouple the fast and slow part, following techniques from control theory, see *e.g.* [22, Chapter 2]. First, we construct an upper triangular matrix T with identity on the diagonal and an unknown upper right block denoted by U , such that $\Pi \triangleq THT^{-1}$ is lower triangular with a zero in the upper right block. This leads to solving a Riccati equation for U [22, Eq. 2.13]. Second, we search for a transform of the matrix Π that would make it block-diagonal, by removing the lower left block. To do that, we build a lower triangular matrix S that has an unknown block X , and consider $\Lambda = S\Pi S^{-1}$ with the goal to make it diagonal matrix. We do that by solving a Sylvester equation in X [32, Lemma 8.3]. Using both transformations, we get

$$s_{k+1} = \Lambda s_k + STR_k,$$

where $s_k \triangleq STb_k$. By having the matrix Λ being block diagonal, with two blocks Λ_1, Λ_2 , we can unroll the recursion to give us access to individual updates of each component:

$$s_{k+1}^x = \Lambda_1^k s_0^x + \sum_{j=0}^k \Lambda_1^{k-j} R_j^x, \quad s_{k+1}^y = \Lambda_2^k s_0^y + \sum_{j=0}^k \Lambda_2^{k-j} R_j^y.$$

The last challenging part is to find the right norm so to ensure that the leading matrices behave as contractions for small enough step sizes. This is done by solving a set of Lyapunov equations to get symmetric positive definite matrices that contain information about the linear dynamics of the system and such that the leading matrices are contractions under the Lyapunov induced norms. We can then propagate back the bounds to the original bias vectors by applying the inverse operation $b_k = (ST)^{-1} s_k$. $\square$

5.2 Discussion

The asymptotic bias provided in Theorem 3 is of the order of $O(\alpha + \beta^2/\alpha^2)$. This is not a direct consequence of the MSE bound of Theorem 1 which, by Jensen's inequality, would only provide a $O(\sqrt{\alpha + \beta^2/\alpha^2})$ on the bias. To prove Theorem 3, we follow a different path from the one we used to prove the MSE: the approach we used was to treat the fast and slow iterates separately. We first obtained a bound on the MSE of x and then plugged this term in the analysis of y . For the bias, we believe that this approach does not work and that the two iterates must be treated at the same time. Hence, what we do in the proof of Theorem 3 is to couple the analysis of the fast and slow iterates by using a change of variables linked to linearization of $\bar{f}$ and $\bar{g}$ around $(h(y^*), y^*)$.

The first term in the bias bound (7) is an exponentially fast vanishing term. This term is similar to the one that we obtained for the MSE but is there because of a different reason. In the MSE analysis, this term came from the strong monotonicity assumption. Here, this term is linked to the Hurwitz stability of first order dynamics of the TTSA (Assumption B.1). The proof overview above explains how a

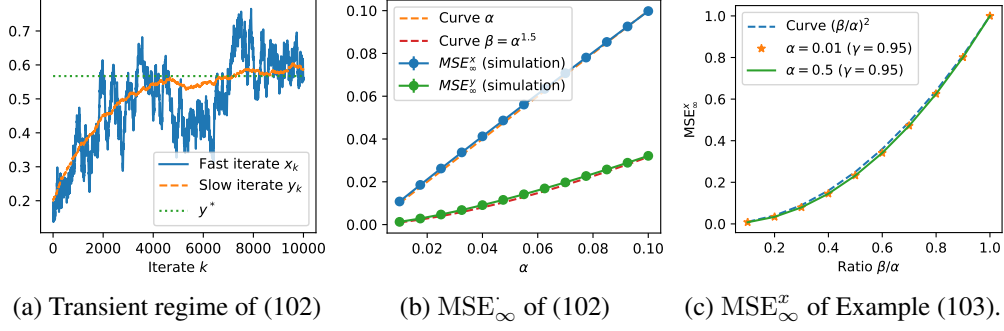


Figure 1: Illustration of the convergence results.

transformation of our bias vectors into new coordinates allowed us to get a contraction. Obtaining this contraction allows us to derive the two remaining terms that form the asymptotic bound on the bias. These terms emerge from two different sources: the first source is the Markovian noise and induces the term $F_1(\alpha + \beta)$. The other terms $\text{MSE}_\infty^x(\alpha, \beta) + \text{MSE}_\infty^y(\alpha, \beta)$ are inherited errors from the MSE due to the high order terms from the Taylor approximation.

Taken together, the bias of both iterates has a leading dependence of order $O(\alpha + \beta)$, which matches the results of linear TTSA under Markovian noise [26]. In addition to that, we get an extra term $\frac{\beta^2}{\alpha^2}$ due to non linearity of the dynamics. This term is injected from the MSE results and can not be improved unless we improve the bound on the MSE.

When assuming a clear separation of timescales such that the slow step-size β satisfies $\beta \leq \alpha^{3/2}$ the bound on the bias reduces to $O(\alpha)$ for both iterates. Similarly to the MSE bound, we can show that this bound is tight: We prove in Theorem 4 that there exists a TTSA that has a lower bound on the bias of order $\Omega(\alpha)$. The exact statement of the theorem along with its proof can be found in Appendix C. Note that, contrary to the MSE case where linear and non-linear TTSA behave very differently, the bias in non-linear TTSA is essentially of the same order as the bias in the linear setting of [26] where the order $O(\alpha + \beta)$ is proved to be tight.

6 Illustrations

In this section, we illustrate some implications of the main theorems of this paper. More numerical results are provided in Appendix F, which also contains the detailed definitions of the examples.

In a first example, we consider the TTSA defined in Eq. (102) of Appendix F. In Figure 1(a), we plot a trajectory of the iterates (x_k, y_k) as a function of k for $\alpha = 0.005$ and $\beta = \alpha^{1.5}$. In this example, $h(y) = y$ and $y^* \approx 0.57$. As observed on this figure, the fast iterate x_k tracks $h(y_k) = y_k$ but with a relatively large error. When k goes to infinity, both iterates get close to $y^* \approx 0.57$. This illustrates the vanishing term present in Theorems 1 and 3. In Figure 1(b), we show the MSE of x and y as a function of α , for the case $\beta = \alpha^{1.5}$. We observe that the MSE of x is close to α and that the MSE of y is close to $\beta = \alpha^{1.5}$. Hence, the MSE of y is smaller than what is suggested by Theorem 1. We believe that this is due to the fact that the dynamics of $\bar{f}$ and $\bar{g}$ are differentiable in y^* (and not just Lipschitz) and that for this particular case stronger guarantees for the MSE of y could be obtained.

In our second example, we consider the TTSA where the fast variable x_k tries to track y_k by applying $x_{k+1} = x_k + \alpha(y_k - x_k)$, while y_k tries to escape from x_k by applying $y_{k+1} = y_k + \beta \frac{(y_k - x_k)}{|y_k - x_k|^\gamma}$ for some $\gamma < 1$. As shown in Figure 1(c), for $\gamma = 0.95$, the MSE of x in this example is essentially equal to β^2/α^2 , independently of α (remark that the figure contains two curves for each of the values $\alpha = 0.01$ and $\alpha = 0.5$, but they are superposed). This shows that the dependence on β^2/α^2 cannot be avoided. To be more precise, this example does not satisfy A.4 (because g is not Lipschitz nor strong monotonic), but our method to obtain the bound on MSE^x (in (4)) does not use A.4. This means that the term β^2/α^2 that is present in (4) cannot be improved without using Assumption A.4. The bounds of all of our Theorems contain terms in β^2/α^2 that all come from MSE^x . As we were unable to find an example that has this β^2/α^2 term and that satisfies A.4, it is possible that the dependence

on β^2/α^2 could be removed under A.4 but this would require a radically different analysis. We refer to Appendix F for a detailed discussion.

7 Conclusion

In this paper, we carry out a finite time analysis on the bias and MSE of nonlinear two time-scale stochastic approximation (TTSA) under constant step-sizes. We consider a general model with Markovian noise where the transition kernel can depend on the iterates (x_k, y_k) , this allows for broad applications, for instance in reinforcement learning. We show that the bias and MSE are upper bounded by vanishing initialization terms plus a residual term that scales as $O(\alpha + \beta^2/\alpha^2)$ and we show that our analysis is tight. While our current framework establishes robust theoretical limits for TTSA, there are several directions for future work. The first is to improve the bounds by (slightly) strengthening the assumptions: for instance, could the MSE bounds be improved if we add B.1 to Theorem 1? Also, one may want to relax the finite state-space restriction on the Markovian noise. This will require careful technical adjustments, particularly in validating the Poisson equation solutions required to bound the noise impact in continuous or unbounded spaces. Furthermore, replacing the strict compactness assumption on the iterate trajectory with non-uniform bounds would capture a larger class of practical, unconstrained algorithms.

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A Main notations

In this section, we recall some notation and main variables that are used in all appendix.

Spaces and norms Let $x \in \mathbb{R}^d$. The quantity $\|x\|$ is the Euclidean norm $\|x\| = (\sum_{i=1}^d x_i^2)^{1/2}$. For $y \in \mathbb{R}^d$ we denote the inner product between x and y by $\langle x, y \rangle$. For a square matrix $M \in \mathbb{R}^{n \times n}$ we define the operator norm $\|M\|_{\text{op}} = \sup_{\|x\|=1} \|Mx\|$. For a positive definite matrix L we define the norm induced by L as $\|x\|_L = \sqrt{x^T L x}$. We define the operator norm induced by L by $\|X\|_L = \sup_{\|u\|_L=1} \|Xu\|_L$. The quantity $\sigma_{\max}(L)$ denotes the maximal singular value of L and $\lambda_{\max}(L)$ is its maximal eigenvalue, and the condition number of L is $\kappa(L) = \frac{\lambda_{\max}(L)}{\lambda_{\min}(L)}$.

Probabilities For a discrete random variable ξ in a countable space Ξ , we have for a probability measure π on Ξ the notation $\mathbb{E}_{\xi \sim \pi}[f(x, \xi)] = \sum_{\xi \in \Xi} f(x, \xi) \pi(\xi)$ and for a kernel $P : \Xi \times \Xi \rightarrow [0, 1]$ we have for $\epsilon \in \Xi$, $\mathbb{E}_{\xi \sim P(\cdot|\epsilon)}[f(x, \xi)] = \sum_{\xi \in \Xi} f(x, \xi) P(\xi|\epsilon)$. For an event E the quantity $\mathbb{1}_{\{E\}}$ equals 1 if E is true and 0 otherwise.

Derivatives For a multivariate function $\bar{f} : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}^{d_x}$, $\nabla_x \bar{f}(x, y)$ denotes the partial Jacobian with respect to the first argument $(\nabla_x \bar{f}(x, y))_{i,j} = \frac{\partial \bar{f}_i(x, y)}{\partial x_j}$ and $\nabla_y \bar{f}(x, y)$ denotes the partial Jacobian with respect to the second argument.

Important variables. In what follows, we will use the notation:

$$\begin{aligned} \hat{x}_k &\triangleq x_k - h(y_k) && \text{Tracking error of } x_k \text{ with respect to } h(y_k). \text{ Note that } \text{MSE}_k^x = \mathbb{E}[\|\hat{x}_k\|^2] \\ \tilde{x}_k &\triangleq x_k - h(y^*) && \text{Tracking error of } x_k \text{ with respect to } h(y^*). \text{ Note that } \text{Bias}_k^x = \mathbb{E}[\tilde{x}_k] \\ \hat{y}_k &\triangleq y_k - y^* && \text{Tracking error of } y_k. \text{ Note that } \text{MSE}_k^y = \mathbb{E}[\|\hat{y}_k\|^2] \text{ and } \text{Bias}_k^y = \mathbb{E}[\hat{y}_k] \end{aligned}$$

Step sizes To obtain our result on the bias (Theorem 3), we require a few conditions on the step sizes from the contraction under the Lyapunov norms passing by the existence and boundedness of the solutions of the Riccati and Sylvester equations to the invertibility conditions on the bias dynamics. We compile all the conditions here so we can refer to them easily in the statement of our Theorem 3. In what follows, we define $R_{\text{ric}} = \frac{1}{4\|G\|_{\text{op}}\sqrt{\kappa(G)M_J}}$ where M_J is a positive constant satisfying $\|J_{yx}\|_{\text{op}} \leq M_J$.

We start with the conditions on α^* :

$$\alpha^* \triangleq \min \left\{ \frac{1}{\mu_f}, \frac{1}{4\|L\|_{\text{op}}(\|J_{xx}\|_L^2 + \varepsilon_4^{*2}\kappa(L)(R_{\text{ric}} + \|U_0\|_{\text{op}})^2\|J_{yx}\|_L^2)}, \frac{1}{\|J_{xx}\|_{\text{op}} + (R_{\text{ric}} + \|U_0\|_{\text{op}})\|J_{yx}\|_{\text{op}}} \right\}, \quad (8)$$

where the first upper bound comes from the MSE Theorem, the second bound gives us the contraction of the fast bias dynamics and can be found in (81) where $\varepsilon_4^* = \min\{(74), (75)\}$ and the last condition is for the invertibility of the fast bias dynamics and can be found in (90).

Second, we enumerate the conditions on β^* :

$$\beta^* \triangleq \min \left\{ \frac{1}{\mu_g}, \frac{1}{4\|G\|_{\text{op}}(2\kappa(G)\|\Delta\|_{\text{op}}M_JR_{\text{ric}} + \|\Delta\|_G^2 + \kappa(G)M_J^2R_{\text{ric}}^2)}, \frac{1}{\|\Delta\|_{\text{op}} + R_{\text{ric}}\|J_{yx}\|_{\text{op}}} \right\}, \quad (9)$$

where the first condition is required for the results of the MSE theorem to hold, the second condition is required by the contraction of the slow bias dynamics and can be found in (84), the last condition is about the invertibility of the slow dynamics and can be found in (97).

Finally, we define ω^* to be:

$$\omega^* \triangleq \min \left\{ 1, \frac{R_{ric}}{\|J_{xx}^{-1}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}}) (\|J_{yy}\|_{\text{op}} + (R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}})}, \frac{1}{\|J_{xx}^{-1}\|_{\text{op}} \left(\|J_{yy}\|_{\text{op}} + 2 \|J_{yx}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}}) \right)}, \frac{1}{\|J_{yy}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2(R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}}, \frac{1}{2 \|L\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}}) (2\tilde{\alpha}^* \kappa(L) \|J_{yx}\|_{\text{op}} \|J_{xx}\|_{\text{op}} + 2\kappa(L)^{(1/2)} M_J)} \right\} \quad (10)$$

where the first condition is to make the results of the MSE hold, the second and third conditions come from the existence and boundedness of the Riccati equation and can be found in (74), (75), the fourth condition, alongside the third, is for the existence and boundedness of the solution of the Sylvester equation and can be found in (78), the fifth condition is needed for the contraction of the fast bias dynamics and can be found in (82) with noting that $\tilde{\alpha}^*$ in the denominator is independent of α and β and can be found in (81).

B Proof of Theorem 1: MSE of TTSA

In this section we prove the upper bound of Theorem 1 on the MSE of TTSA. We restate the theorem, before detailing the proof.

Theorem 1. *Assuming that A.1-3 hold, and that $\alpha\mu_f < 1$ and $\beta \leq \alpha$, then for any $k \geq 1$ we obtain the following upper bound on the MSE of the fast iterate,*

$$\mathbb{E}[\|x_k - h(y_k)\|^2] \leq \underbrace{(1 - \alpha\mu_f)^k \|x_0 - h(y_0)\|^2}_{\text{Decaying initial condition}} + \underbrace{\alpha \left(\frac{B_1}{1 - \alpha\mu_f} + B_2 \right) + \beta B_3 + \frac{\beta^2}{\alpha^2} B_4}_{\text{MSE}_{\infty}^x(\alpha, \beta)}, \quad (4)$$

where the constants $(B_k)_{k \in [4]}$ are fully defined in appendix (Eq. (18)-(21)), and are independent of α, β and k : they only depend on the properties of the Markovian noise, the range, monotonicity and Lipschitzness parameters of the functions f, g, h . Further, assuming A.4, and $\beta\mu_g < 1$, for any $k \geq 1$ the MSE of the slow iterate is upper bounded by

$$\mathbb{E}[\|y_k - y^*\|^2] \leq \underbrace{(1 - \beta\mu_g)^k \|y_0 - y^*\|^2 + D_1 \beta k (1 - (\alpha\mu_f \wedge \beta\mu_g))^k \|x_0 - h(y_0)\|^2}_{\text{Decaying initial conditions}} + \underbrace{D_2 \text{MSE}_{\infty}^x(\alpha, \beta) + \beta \left(\frac{D_3}{1 - \beta\mu_g} + D_4 \right) + \alpha D_5}_{\text{MSE}_{\infty}^y(\alpha, \beta)}.$$

where the constants $(D_k)_{k \in [5]}$ are also formally defined in appendix (Eq. (28)-(32)) and depend again on problem parameters but not on α, β, k .

Proof. We divide the proof into two main parts, the first part is dedicated to the fast iterate MSE and the second part is for the slow iterate.

Upper bounding the MSE of the fast iterate We recall that $\hat{x}_k \triangleq x_k - h(y_k)$. We also introduce the shorthand

$$\psi_k \triangleq f(x_k, y_k, \xi_{k+1}) - \bar{f}(x_k, y_k), \quad \Delta_k^h \triangleq h(y_{k+1}) - h(y_k).$$

Using the update $x_{k+1} = x_k - \alpha f(x_k, y_k, \xi_{k+1})$, we obtain

$$\hat{x}_{k+1} = x_{k+1} - h(y_{k+1}) = \hat{x}_k - \alpha f(x_k, y_k, \xi_{k+1}) - \Delta_k^h,$$

so expanding the squared norm gives

$$\begin{aligned} \|\hat{x}_{k+1}\|^2 &= \|x_{k+1} - h(y_{k+1})\|^2 \\ &= \|\hat{x}_k - \alpha f(x_k, y_k, \xi_{k+1}) - \Delta_k^h\|^2 \\ &= \|\hat{x}_k\|^2 + \alpha^2 \|f(x_k, y_k, \xi_{k+1})\|^2 - 2\alpha \langle \hat{x}_k, \bar{f}(x_k, y_k) \rangle \\ &\quad + \|\Delta_k^h\|^2 - 2\langle \hat{x}_k, \Delta_k^h \rangle + 2\alpha \langle f(x_k, y_k, \xi_{k+1}), \Delta_k^h \rangle - 2\alpha \langle \hat{x}_k, \psi_k \rangle \end{aligned}$$

Rearranging the terms to facilitate the organization of the bound:

$$\|\hat{x}_{k+1}\|^2 = \|\hat{x}_k\|^2 - 2\alpha \langle \hat{x}_k, \bar{f}(x_k, y_k) \rangle - 2\langle \hat{x}_k, \Delta_k^h \rangle \quad (11)$$

$$+ \alpha^2 \|f(x_k, y_k, \xi_{k+1})\|^2 \quad (12)$$

$$+ \|\Delta_k^h\|^2 + 2\alpha \langle f(x_k, y_k, \xi_{k+1}), \Delta_k^h \rangle \quad (13)$$

$$- 2\alpha \langle \hat{x}_k, \psi_k \rangle. \quad (14)$$

We first control the bounded second-order terms. By Lemma 1, there exist constants $M_f, M_h > 0$ such that

$$\|f(x_k, y_k, \xi_{k+1})\| \leq M_f, \quad \|\Delta_k^h\| \leq \beta M_h.$$

Therefore, by the Cauchy–Schwarz inequality,

$$\begin{aligned} (12) + (13) &\leq \alpha^2 \|f(x_k, y_k, \xi_{k+1})\|^2 + \|\Delta_k^h\|^2 + 2\alpha \|f(x_k, y_k, \xi_{k+1})\| \|\Delta_k^h\| \\ &\leq \alpha^2 M_f^2 + \beta^2 M_h^2 + 2\alpha \beta M_f M_h. \end{aligned}$$

We now extract the contraction term. By property of h , for every $y \in \mathcal{Y}$,

$$\bar{f}(h(y), y) = 0.$$

Hence, recalling that $\hat{x}_k = x_k - h(y_k)$, we can write

$$\begin{aligned} -2\alpha \langle \hat{x}_k, \bar{f}(x_k, y_k) \rangle &= -2\alpha \langle x_k - h(y_k), \bar{f}(x_k, y_k) - \bar{f}(h(y_k), y_k) \rangle \\ &\leq -2\alpha \mu_f \|x_k - h(y_k)\|^2 \\ &= -2\alpha \mu_f \|\hat{x}_k\|^2, \end{aligned}$$

where the inequality follows from the strong monotonicity assumption A.3 applied to the first argument of $\bar{f}$, with $y = y_k$.

The third term in line (11) is controlled by Cauchy–Schwarz and Young’s inequality. Precisely:

$$-2\langle \hat{x}_k, \Delta_k^h \rangle \leq 2\|\hat{x}_k\| \|\Delta_k^h\| \leq \alpha \mu_f \|\hat{x}_k\|^2 + \frac{1}{\alpha \mu_f} \|\Delta_k^h\|^2 \leq \alpha \mu_f \|\hat{x}_k\|^2 + \frac{\beta^2 M_h^2}{\alpha \mu_f},$$

where the last inequality follows from Lemma 1. Combining the previous bounds on (11)–(13), we obtain

$$\|\hat{x}_{k+1}\|^2 \leq (1 - \alpha \mu_f) \|\hat{x}_k\|^2 - 2\alpha \langle \hat{x}_k, \psi_k \rangle + \alpha^2 M_f^2 + \beta^2 M_h^2 + 2\alpha \beta M_f M_h + \frac{\beta^2 M_h^2}{\alpha \mu_f}.$$

For brevity, set

$$M_{\alpha, \beta} \triangleq \alpha^2 M_f^2 + \beta^2 M_h^2 + 2\alpha \beta M_f M_h + \frac{\beta^2 M_h^2}{\alpha \mu_f}.$$

Iterating the previous inequality gives

$$\|\hat{x}_{k+1}\|^2 \leq (1 - \alpha \mu_f)^{k+1} \|\hat{x}_0\|^2 - 2\alpha \sum_{j=0}^k (1 - \alpha \mu_f)^{k-j} \langle \hat{x}_j, \psi_j \rangle + \sum_{j=0}^k (1 - \alpha \mu_f)^{k-j} M_{\alpha, \beta}.$$

Taking expectations yields

$$\mathbb{E} \left[\|\hat{x}_{k+1}\|^2 \right] \leq (1 - \alpha\mu_f)^{k+1} \|\hat{x}_0\|^2 \quad (15)$$

$$+ M_{\alpha,\beta} \sum_{j=0}^k (1 - \alpha\mu_f)^j \quad (16)$$

$$- 2\alpha \mathbb{E} \left[\sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, \psi_j \rangle \right]. \quad (17)$$

The first term (15) is the contribution of the initial condition and decays geometrically. The two remaining terms correspond to the steady-state error induced by the constant step-size. We first bound the deterministic geometric term (16):

$$M_{\alpha,\beta} \sum_{j=0}^k (1 - \alpha\mu_f)^j = M_{\alpha,\beta} \frac{1 - (1 - \alpha\mu_f)^{k+1}}{\alpha\mu_f} \leq \frac{M_{\alpha,\beta}}{\alpha\mu_f}.$$

It remains to control the Markovian-noise term (17). To do so, we invoke Lemma 2, that we detail and prove in Appendix E.

$$\begin{aligned} \left| \alpha \mathbb{E} \left[\sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, \psi_j \rangle \right] \right| &\leq \alpha \left(M_\xi \left(\frac{3 - \alpha\mu_f}{1 - \alpha\mu_f} + \frac{M_f(1 + |\Xi| M_S L_K)}{\mu_f} \right) + \frac{L_{v_f} M_f (M_S + M_h)}{\mu_f} \right) \\ &\quad + \beta \left(\frac{M_\xi M_h + L_{v_f} M_g (M_S + M_h) + |\Xi| M_S L_K M_\xi M_g}{\mu_f} \right) \end{aligned}$$

Combining the above results provide

$$\begin{aligned} \mathbb{E} \left[\|\hat{x}_{k+1}\|^2 \right] &\leq (1 - \alpha\mu_f)^{k+1} \|\hat{x}_0\|^2 \\ &\quad + \alpha \left(\frac{4M_\xi}{1 - \alpha\mu_f} + 2M_\xi + \frac{M_f^2 + 2(M_\xi(1 + |\Xi| M_S L_K) + L_{v_f} (M_S + M_h)) M_f}{\mu_f} \right) \\ &\quad + \beta \left(\frac{2(M_\xi + M_f) M_h + 2L_{v_f} M_g (M_S + M_h) + |\Xi| M_S L_K M_\xi M_g}{\mu_f} \right) + \frac{\beta^2}{\alpha} \frac{M_h^2}{\mu_f} + \frac{\beta^2 M_h^2}{\alpha^2 \mu_f^2}. \end{aligned}$$

Using that $\frac{\beta^2}{\alpha} \leq \beta$ because $\beta \leq \alpha$, we are ready to formally define the constants

$$B_1 := 4M_\xi, \quad (18)$$

$$B_2 := 2M_\xi + \frac{M_f^2 + 2(M_\xi(1 + |\Xi| M_S L_K) + L_{v_f} (M_S + M_h)) M_f}{\mu_f}, \quad (19)$$

$$B_3 := \frac{2(M_\xi + M_f) M_h + 2L_{v_f} M_g (M_S + M_h) + 2|\Xi| M_S L_K M_\xi M_g + M_h^2}{\mu_f}, \text{ and} \quad (20)$$

$$B_4 := \frac{M_h^2}{\mu_f^2}. \quad (21)$$

This proves the desired MSE bound for the fast iterate, and allows us to define the non-vanishing term of the bound,

$$\text{MSE}_\infty^x(\alpha, \beta) := \alpha \left(\frac{B_1}{1 - \alpha\mu_f} + B_2 \right) + \beta B_3 + \frac{\beta^2}{\alpha^2} B_4.$$

MSE of the slow iterate We now turn to the MSE of the slow iterate. The argument follows the same structure as for the fast iterate, with one additional difficulty: the update of y_k is evaluated at (x_k, y_k) , whereas the contraction assumption is available for the reduced dynamics evaluated at $(h(y_k), y_k)$. We therefore need to control the tracking error of the fast iterate, which enters the slow recursion as an additional residual term.

For convenience, introduce

$$\Delta_k^g \triangleq \bar{g}(x_k, y_k) - \bar{g}(h(y_k), y_k), \quad \phi_k \triangleq g(x_k, y_k, \xi_{k+1}) - \bar{g}(x_k, y_k).$$

Thus, Δ_k^g measures the error due to replacing x_k by $h(y_k)$ in the noisy averaged dynamics, while ϕ_k is the Markovian-noise fluctuations around the averaged slow dynamics.

Using the update $y_{k+1} = y_k - \beta g(x_k, y_k, \xi_{k+1})$, we have

$$\hat{y}_{k+1} = \hat{y}_k - \beta g(x_k, y_k, \xi_{k+1}).$$

Expanding the squared norm and adding/subtracting the reduced dynamics gives

$$\begin{aligned} \|\hat{y}_{k+1}\|^2 &= \|\hat{y}_k - \beta g(x_k, y_k, \xi_{k+1})\|^2 \\ &= \|\hat{y}_k\|^2 - 2\beta \langle \hat{y}_k, g(x_k, y_k, \xi_{k+1}) \rangle + \beta^2 \|g(x_k, y_k, \xi_{k+1})\|^2 \\ &= \|\hat{y}_k\|^2 - 2\beta \langle \hat{y}_k, \bar{g}(h(y_k), y_k) \rangle - 2\beta \langle \hat{y}_k, \bar{g}(x_k, y_k) - \bar{g}(h(y_k), y_k) \rangle \\ &\quad + \beta^2 \|g(x_k, y_k, \xi_{k+1})\|^2 - 2\beta \langle \hat{y}_k, g(x_k, y_k, \xi_{k+1}) - \bar{g}(x_k, y_k) \rangle \\ &= \|\hat{y}_k\|^2 - 2\beta \langle \hat{y}_k, \bar{g}(h(y_k), y_k) \rangle - 2\beta \langle \hat{y}_k, \Delta_k^g \rangle \\ &\quad + \beta^2 \|g(x_k, y_k, \xi_{k+1})\|^2 \end{aligned} \tag{22}$$

$$- 2\beta \langle \hat{y}_k, \phi_k \rangle. \tag{23}$$

$$- 2\beta \langle \hat{y}_k, \phi_k \rangle. \tag{24}$$

By the strong monotonicity assumption A.4, the first two terms in (22) yield a contraction. The additional term involving Δ_k^g measures the error induced by evaluating the slow update at x_k instead of $h(y_k)$. More precisely,

$$\begin{aligned} (22) &= \|\hat{y}_k\|^2 - 2\beta \langle \hat{y}_k, \bar{g}(h(y_k), y_k) \rangle - 2\beta \langle \hat{y}_k, \Delta_k^g \rangle \\ &\leq \|\hat{y}_k\|^2 - 2\beta \mu_g \|\hat{y}_k\|^2 + 2\beta \|\hat{y}_k\| \|\Delta_k^g\| \\ &\leq \|\hat{y}_k\|^2 - 2\beta \mu_g \|\hat{y}_k\|^2 + \beta \mu_g \|\hat{y}_k\|^2 + \frac{\beta}{\mu_g} \|\Delta_k^g\|^2 \\ &\leq (1 - \beta \mu_g) \|\hat{y}_k\|^2 + \frac{\beta}{\mu_g} \|\Delta_k^g\|^2 \\ &\leq (1 - \beta \mu_g) \|\hat{y}_k\|^2 + \beta \frac{L^2}{\mu_g} \|\hat{x}_k\|^2. \end{aligned}$$

In the first inequality, we used the strong monotonicity of the reduced slow dynamics and Cauchy-Schwarz. In the second inequality, we used Young's inequality $2ab \leq \epsilon a^2 + b^2/\epsilon$ with $\epsilon = \mu_g$. The last inequality follows from the Lipschitz property of $\bar{g}(\cdot, \cdot)$:

$$\|\Delta_k^g\| = \|\bar{g}(x_k, y_k) - \bar{g}(h(y_k), y_k)\| \leq L_g \|x_k - h(y_k)\| = L_g \|\hat{x}_k\|$$

We take $\beta < \mu_g^{-1}$ so that the factor $1 - \beta \mu_g$ is strictly contractive and bounded away from zero. The remaining deterministic term in (23) is bounded using Lemma 1: there exists $M_g > 0$ such that

$$\beta^2 \|g(x_k, y_k, \xi_{k+1})\|^2 \leq \beta^2 M_g^2.$$

Combining the previous bounds on (22)–(24), we obtain

$$\|\hat{y}_{k+1}\|^2 \leq (1 - \beta \mu_g) \|\hat{y}_k\|^2 + \frac{L_g^2}{\mu_g} \beta \|\hat{x}_k\|^2 + \beta^2 M_g^2 - 2\beta \langle \hat{y}_k, \phi_k \rangle.$$

Iterating this recursion from time 0 to time k , and then taking expectations, gives

$$\mathbb{E} [\|\hat{y}_{k+1}\|^2] \leq (1 - \beta \mu_g)^{k+1} \|\hat{y}_0\|^2 + M_g^2 \beta^2 \sum_{j=0}^k (1 - \beta \mu_g)^j \tag{25}$$

$$+ \frac{L_g^2}{\mu_g} \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \mathbb{E} [\|\hat{x}_j\|^2] \tag{26}$$

$$- 2\beta \mathbb{E} \left[\sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle \hat{y}_j, \phi_j \rangle \right]. \tag{27}$$

The first term in (25) is the contribution of the initial condition and decays geometrically. The second term in (25) is a geometric sum, and therefore

$$\begin{aligned} M_g^2 \beta^2 \sum_{j=0}^k (1 - \beta \mu_g)^j &= M_g^2 \beta^2 \frac{1 - (1 - \beta \mu_g)^{k+1}}{\beta \mu_g} \\ &\leq \frac{M_g^2}{\mu_g} \beta. \end{aligned}$$

We now control the term (26), which is the geometrically weighted accumulation of the fast-iterate MSEs. Using the upper bound we obtained for the fast iterate, we get

$$\begin{aligned} \frac{L_g^2}{\mu_g} \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \mathbb{E} \left[\|\hat{x}_j\|^2 \right] &\leq \frac{L_g^2}{\mu_g} \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \left((1 - \alpha \mu_f)^{j+1} \|\hat{x}_0\|^2 + \text{MSE}_\infty^x(\alpha, \beta) \right) \\ &\leq \frac{L_g^2}{\mu_g^2} \text{MSE}_\infty^x(\alpha, \beta) \\ &\quad + \frac{L_g^2}{\mu_g} \beta (1 - \alpha \mu_f) \|\hat{x}_0\|^2 \underbrace{\sum_{j=0}^k (1 - \beta \mu_g)^{k-j} (1 - \alpha \mu_f)^j}_{S_k}. \end{aligned}$$

For the term S_k , we simply remark that, for $j \in [k]$, it holds that

$$(1 - \beta \mu_g)^{k-j} (1 - \alpha \mu_f)^j \leq (1 - (\alpha \mu_f \wedge \beta \mu_g))^k.$$

Therefore, we can upper bound each term of the sum uniformly and obtain that

$$S_k \leq (k+1)(1 - (\alpha \mu_f \wedge \beta \mu_g))^k.$$

Note that the bound is tight for $\alpha \mu_f = \beta \mu_g$. This leads to

$$\frac{L_g^2}{\mu_g} \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \mathbb{E} \left[\|\hat{x}_j\|^2 \right] \leq \frac{L_g^2}{\mu_g^2} \text{MSE}_\infty^x(\alpha, \beta) + \beta \frac{L_g^2}{\mu_g} \|\hat{x}_0\|^2 (k+1)(1 - (\alpha \mu_f \wedge \beta \mu_g))^{k+1}.$$

It remains to control the Markovian-noise term (27). As in the fast-iterate analysis, we use the Poisson-equation decomposition from Lemma 2. In particular,

$$\begin{aligned} -2\beta \mathbb{E} \left[\sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle \hat{y}_j, \phi_j \rangle \right] &\leq 2\beta \left| \mathbb{E} \left[\sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle \hat{y}_j, \phi_j \rangle \right] \right| \\ &\leq 2\beta \left(\frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_g}{\mu_g} + M_\xi \left(\frac{3 - \beta \mu_g}{1 - \beta \mu_g} + \frac{M_g}{\mu_g} \right) \right) \\ &\quad + 2\alpha \frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_f}{\mu_g} \end{aligned}$$

Combining this bound with the bounds on (25) and (26), we obtain

$$\begin{aligned} \mathbb{E} \left[\|\hat{y}_{k+1}\|^2 \right] &\leq (1 - \beta \mu_g)^{k+1} \|\hat{y}_0\|^2 + \frac{M_g^2}{\mu_g} \beta \\ &\quad + \beta \frac{L_g^2}{\mu_g} \|\hat{x}_0\|^2 (k+1)(1 - (\alpha \mu_f \wedge \beta \mu_g))^{k+1} \\ &\quad + \frac{L_g^2}{\mu_g^2} \text{MSE}_\infty^x(\alpha, \beta) + 2\beta \left(\frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_g}{\mu_g} + \frac{2M_\xi}{1 - \beta \mu_g} + M_\xi \right) \\ &\quad + 2\alpha \frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_f}{\mu_g}. \end{aligned}$$

It remains to identify the constants,

$$D_1 := \frac{L_g^2}{\mu_g}, \quad (28)$$

$$D_2 := \frac{L_g^2}{\mu_g^2}, \quad (29)$$

$$D_3 := 4M_\xi, \text{ and} \quad (30)$$

$$D_4 := \frac{M_g^2}{\mu_g} + 2 \left(\frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_g + M_\xi M_g}{\mu_g} + M_\xi \right). \quad (31)$$

$$D_5 := 2 \frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_f}{\mu_g}. \quad (32)$$

This proves the desired MSE bound for the slow iterate and concludes the proof of the theorem. $\square$

C Proof of Theorem 2: Lower bounds

Theorem 2 shows that there exists a TTSA whose MSE is larger than $\Omega(\alpha)$ for both the fast and the slow iterates. This result is a consequence of Theorem 4 below where we exhibit an example that attains this lower bound, and where we also provide an example has an MSE of order $\Omega(\alpha)$ for both x and y .

Theorem 4. *There exist a constant C and a constant $\alpha_0 > 0$ and:*

(i) *a TTSA satisfying all Assumptions of Theorem 1 such that for all $\beta \leq \alpha \leq \alpha_0$:*

$$\liminf_{k \rightarrow \infty} \mathbb{E} [\|x_k - h(y_k)\|^2] \geq C\alpha \quad \text{and} \quad \liminf_{k \rightarrow \infty} \mathbb{E} [\|y_k - y^*\|^2] \geq C\alpha;$$

(ii) *a TTSA satisfying all Assumptions of Theorem 3 such that for all $\beta \leq \alpha \leq \alpha_0$:*

$$\liminf_{k \rightarrow \infty} \|\mathbb{E}[x_k] - h(y^*)\| \geq C(\alpha + \beta) \quad \text{and} \quad \liminf_{k \rightarrow \infty} \|\mathbb{E}[y_k] - y^*\| \geq C(\alpha + \beta).$$

Proof. The proof of (i) and (ii) uses the same fast iterate (x_k) that is defined as:

$$x_{k+1} = x_k - \alpha(f(x_k) + \xi_{k+1}),$$

where $f(x) = \frac{x}{1+x}$. The sequence of noises (ξ_k) is a sequence of zero-mean *i.i.d.* scaled Rademacher random variables (*i.e.*, such that $\mathbb{P}[\xi_k = 1/2] = \mathbb{P}[\xi_k = -1/2] = 0.5$). Note that the function f is Lipschitz continuous with a Lipschitz-continuous derivative. We assume that $\alpha \in (0, 1)$ and we set $x_0 = 0$.

We first show by induction on k that the sequence x_k is bounded almost surely, and more precisely that $x_k \in [-1/2, 1]$. Assume that this holds for some k and recall that

$$x_{k+1} = x_k - \alpha \left(\frac{x_k}{1+x_k} + \xi_{k+1} \right).$$

We distinguish two cases:

1. If $\frac{x_k}{1+x_k} + \xi_k \geq 0$, then

$$x_{k+1} \leq x_k \leq 1$$

$$x_{k+1} \geq x_k - \left(\frac{x_k}{1+x_k} + \xi_k \right) = \frac{x_k^2}{1+x_k} - \frac{1}{2} \geq -\frac{1}{2},$$

where we used that $\alpha \leq 1$ in the second line and the fact that $x^2/(1+x) \geq 0$.

2. If $\frac{x_k}{1+x_k} + \xi_k \leq 0$, then

$$x_{k+1} \geq x_k \geq -\frac{1}{2}$$

$$x_{k+1} \leq x_k - \left(\frac{x_k}{1+x_k} + \xi_k \right) = \frac{x_k^2}{1+x_k} + \frac{1}{2} \leq 1,$$

As the behavior of x_k does not depend on the slow iterates $(y_k)_k$, the sequence (x_k) can be analyzed with standard tools from one-time-scale stochastic approximation. Here, we have $x^* = h(y_k) = 0$. In particular, since near 0 this function has a second order approximation of $x - x^2$, from [1], we know that the bias and the MSE of x_k are of order α and asymptotically equal to:

$$\lim_{k \rightarrow \infty} \mathbb{E}[x_k] = \lim_{k \rightarrow \infty} \mathbb{E}[(x_k)^2] = \frac{\alpha}{2c} + O(\alpha^2).$$

for some positive constant c . Moreover, from [25, Chapter 10], when α is small and k is large, the variable x_k is close to a Gaussian distribution of variance $\alpha/2c$, which implies

$$\lim_{k \rightarrow \infty} \mathbb{E}[|x_k|] = \sqrt{\alpha/d\pi} + O(\alpha) \quad (33)$$

for some positive constant d . For the variable y_k , we consider two different cases to prove either (i) or (ii):

(i) To prove (i), we set

$$y_{k+1} = y_k - \beta(y_k - |x_k|).$$

Unrolling the recurrence equation $y_{k+1} = y_k(1 - \beta) + \beta|x_k|$ shows that

$$y_{k+1} = \beta \sum_{i=0}^k (1 - \beta)^{k-i} |x_i|.$$

Combined with (33), this shows that

$$\begin{aligned} \lim_{k \rightarrow \infty} \mathbb{E}[y_k] &= \beta \sum_{i=0}^{\infty} (1 - \beta)^i \lim_{k \rightarrow \infty} \mathbb{E}[|x_k|] \\ &= \lim_{k \rightarrow \infty} \mathbb{E}[|x_k|] \\ &= \sqrt{\frac{\alpha}{d\pi}} + O(\alpha). \end{aligned}$$

By using Jensen's inequality, this shows that:

$$\begin{aligned} \lim_{k \rightarrow \infty} \mathbb{E}[(y_k)^2] &\geq \lim_{k \rightarrow \infty} (\mathbb{E}[y_k])^2 \\ &= \frac{\alpha}{d\pi} + O(\alpha^{3/2}). \end{aligned}$$

(ii) To prove (ii), we define y_k as:

$$y_{k+1} = y_k - \beta(y_k - x_k)$$

This shows that $\lim_{k \rightarrow \infty} \mathbb{E}[y_k] = \lim_{k \rightarrow \infty} \mathbb{E}[x_k] = \alpha/2c + O(\alpha^2)$.

□

The proof of the above theorem exhibits an example where the MSE of y can be of order $\Omega(\alpha)$. To construct this example, we used a function g such that $\bar{g}(x, y) = y - |x|$ that satisfies all assumptions of Theorem 1 but not the assumptions of Theorem 3 because it is not differentiable. In fact, we believe that having a non-differentiable function is needed to obtain the $\Omega(\alpha)$ lower bound for MSE_y^∞ , and that under a stronger assumption of differentiability, one may obtain a tighter bound for the MSE. This is an interesting question for future work.

D Proof of Theorem 3 : Bias of TTSA

We start by restating the theorem, before detailing its proof.

Theorem 3. Assuming A.1-4 and B.1, there exists α^* defined in (8), β^* defined in (9), and ω^* defined in (10) such that for $\alpha < \alpha^*$, $\beta < \beta^*$ and $\frac{\beta}{\alpha} < \omega^*$ we have the following bound on the bias of both iterates:

$$\max\{\|\mathbb{E}[x_k - h(y^*)]\|, \|\mathbb{E}[y_k - y^*]\|\} \leq \|T^{-1}S^{-1}\|_{op} (\|s_k^x\| + \|s_k^y\|)$$

for some upper triangular matrix T defined in (35) and a lower triangular matrix S defined in (39), and where vectors s_k^x, s_k^y are some transformed bias vectors defined in (43), (44) respectively. Furthermore,

$$\begin{aligned} \|s_k^x\| + \|s_k^y\| &\leq \underbrace{e^{-kc\alpha} \sqrt{\kappa(L)} \|s_0^x\| + e^{-kd\beta} \sqrt{\kappa(G)} \|s_0^y\| + ae^{-\alpha\hat{c}(k-1)} \|\hat{x}_0\|^2 + be^{-\beta\hat{d}(k-1)} \|\hat{y}_0\|^2}_{\text{Decaying initial conditions}} \\ &\quad + \underbrace{F_1(\alpha + \beta) + F_2(\text{MSE}_x^\infty(\alpha, \beta) + \text{MSE}_y^\infty(\alpha, \beta)) + R_{\alpha, \beta}}_{\text{Asymptotic bias}}, \end{aligned} \quad (7)$$

where F_1, F_2 are positive constants independent of α, β and k , whose values may depend on the problem parameters and on ω^* , $R_{\alpha, \beta}$ contains terms of higher order $(\alpha + \beta)^2$. Additionally, L, G are symmetric positive definite matrices solutions of the Lyapunov equations (45), (49), $c = (8\|L\|_{op})^{-1}$ and $d = (8\|G\|_{op})^{-1}$, constants $a, b, \hat{c}, \hat{d}$ are related to decaying initial errors from the MSE and can be found in (53), and the MSE terms are defined in Theorem 1.

Proof. We start by detailing the general outline of the proof. We begin by concatenating the TTSA updates to jointly analyze the bias of the fast and slow variables. Exploiting the differentiability of the underlying dynamics (Assumption B. 1), we apply a first-order Taylor expansion to linearize the updates, which yields a highly coupled linear dynamical system. To mathematically decouple these dynamics, we introduce a two-stage coordinate transformation. First, we block-triangularize the system by solving an algebraic Riccati equation. Then, we fully diagonalize the system by solving a Sylvester equation.

This transformation projects the bias into a new coordinate space governed by a diagonalized recursion. While this recursion features strong contracting terms, it is corrupted by higher-order perturbation residuals arising from the diagonalization process and the multiplication by the transformation matrices. However, the order of these perturbations will not affect the nature of the contractions due to our assumptions on the step sizes. The diagonal structure then allows us to unroll the recursion and bound each transformed component independently. Finally, because our transformation matrices possess finite norms, and by a simple triangular inequality we get the final bounds on the bias.

Concatenating the iterates. We start by deriving a recursion for the joint bias vector. In this proof, we work with deviations from the equilibrium and write

$$\tilde{x}_k \triangleq x_k - h(y^*), \quad \text{and we recall that } \hat{y}_k = y_k - y^*.$$

We also introduce the centered Markovian-noise terms

$$\psi_k \triangleq f(x_k, y_k, \xi_{k+1}) - \bar{f}(x_k, y_k), \quad \varphi_k \triangleq g(x_k, y_k, \xi_{k+1}) - \bar{g}(x_k, y_k).$$

For the fast iterate, we obtain

$$\begin{aligned} \mathbb{E}[\tilde{x}_{k+1}] &= \mathbb{E}[\tilde{x}_k] - \alpha \mathbb{E}[\bar{f}(x_k, y_k)] - \alpha \mathbb{E}[\psi_k] \\ &= \mathbb{E}[\tilde{x}_k] - \alpha \mathbb{E}[\bar{f}(h(y^*) + \tilde{x}_k, y^* + \hat{y}_k)] - \alpha \mathbb{E}[\psi_k] \\ &= \mathbb{E}[\tilde{x}_k] - \alpha \mathbb{E}[\bar{f}(h(y^*), y^*) + J_{xx}\tilde{x}_k + J_{xy}\hat{y}_k + R_f(\tilde{x}_k, \hat{y}_k)] - \alpha \mathbb{E}[\psi_k] \\ &= \mathbb{E}[\tilde{x}_k] - \alpha J_{xx} \mathbb{E}[\tilde{x}_k] - \alpha J_{xy} \mathbb{E}[\hat{y}_k] - \alpha \mathbb{E}[R_f(\tilde{x}_k, \hat{y}_k)] - \alpha \mathbb{E}[\psi_k], \end{aligned}$$

where we used $\bar{f}(h(y^*), y^*) = 0$ in the last line.

Similarly, for the slow iterate,

$$\begin{aligned} \mathbb{E}[\hat{y}_{k+1}] &= \mathbb{E}[\hat{y}_k] - \beta \mathbb{E}[\bar{g}(x_k, y_k)] - \beta \mathbb{E}[\varphi_k] \\ &= \mathbb{E}[\hat{y}_k] - \beta \mathbb{E}[\bar{g}(h(y^*) + \tilde{x}_k, y^* + \hat{y}_k)] - \beta \mathbb{E}[\varphi_k] \\ &= \mathbb{E}[\hat{y}_k] - \beta \mathbb{E}[\bar{g}(h(y^*), y^*) + J_{yx}\tilde{x}_k + J_{yy}\hat{y}_k + R_g(\tilde{x}_k, \hat{y}_k)] - \beta \mathbb{E}[\varphi_k] \\ &= \mathbb{E}[\hat{y}_k] - \beta J_{yx} \mathbb{E}[\tilde{x}_k] - \beta J_{yy} \mathbb{E}[\hat{y}_k] - \beta \mathbb{E}[R_g(\tilde{x}_k, \hat{y}_k)] - \beta \mathbb{E}[\varphi_k], \end{aligned}$$

where we used $\bar{g}(h(y^*), y^*) = 0$.

In both expansions, we applied a first-order Taylor expansion of the averaged dynamics around the equilibrium $(h(y^*), y^*)$. Since the second derivatives of $\bar{f}$ and $\bar{g}$ are bounded by Assumption B.1, there exists constant $L_R > 0$ such that, for all $k \geq 0$,

$$\|R_f(\tilde{x}_k, \hat{y}_k)\| \leq L_R(\|\tilde{x}_k\| + \|\hat{y}_k\|)^2, \quad \|R_g(\tilde{x}_k, \hat{y}_k)\| \leq L_R(\|\tilde{x}_k\| + \|\hat{y}_k\|)^2.$$

Concatenating the two bias vectors yields the joint recursion

$$\begin{bmatrix} \mathbb{E}[\tilde{x}_{k+1}] \\ \mathbb{E}[\hat{y}_{k+1}] \end{bmatrix} = \underbrace{\begin{bmatrix} I - \alpha J_{xx} & -\alpha J_{xy} \\ -\beta J_{yx} & I - \beta J_{yy} \end{bmatrix}}_{H_{\alpha, \beta}} \begin{bmatrix} \mathbb{E}[\tilde{x}_k] \\ \mathbb{E}[\hat{y}_k] \end{bmatrix} + \underbrace{\begin{bmatrix} -\alpha \mathbb{E}[R_f(\tilde{x}_k, \hat{y}_k)] - \alpha \mathbb{E}[\psi_k] \\ -\beta \mathbb{E}[R_g(\tilde{x}_k, \hat{y}_k)] - \beta \mathbb{E}[\varphi_k] \end{bmatrix}}_{\mathcal{R}_k}. \quad (34)$$

The matrix $H_{\alpha, \beta}$ is the linearized discrete-time dynamics of the joint bias. Its block structure shows that the two components are still coupled: the fast bias depends on the slow bias through the block J_{xy} , while the slow bias depends on the fast bias through the block J_{yx} . This coupling is the main reason why one cannot simply analyze the two bias coordinates separately.

The next step is therefore to find a change of coordinates adapted to this two-time-scale linear system. More precisely, following ideas from singular perturbation theory [22, Chapter 2], we construct invertible transformations that turn $H_{\alpha, \beta}$ into a block-diagonal matrix. In the transformed coordinates, the fast and slow bias components can then be controlled separately, with their respective contraction rates. In the following, we drop the subscript α, β from the matrix H for conciseness of notation.

Triangularization through a Riccati equation. The first step is to remove the upper-right coupling in the linearized dynamics. We look for an invertible change of coordinates T such that

$$\Pi \triangleq THT^{-1}$$

is block lower triangular, or equivalently $H = T^{-1}\Pi T$.

We choose T to be upper triangular with identity blocks on the diagonal:

$$T = \begin{bmatrix} I & U \\ 0 & I \end{bmatrix}, \quad T^{-1} = \begin{bmatrix} I & -U \\ 0 & I \end{bmatrix}, \quad (35)$$

where $U \in \mathbb{R}^{d_x \times d_y}$ is to be determined. To simplify the notation, write the block decomposition of H as

$$H = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

Then a direct computation gives

$$\begin{aligned} THT^{-1} &= \begin{bmatrix} I & U \\ 0 & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} I & -U \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} A + UC & R(U) \\ C & D - CU \end{bmatrix}, \end{aligned}$$

where

$$R(U) \triangleq -(A + UC)U + B + UD.$$

Thus, the upper-right block is zero if and only if U solves

$$R(U) = -(A + UC)U + B + UD = 0. \quad (36)$$

This is an algebraic Riccati equation for the unknown block U [22, Eq. 2.8]. Any solution of (36) yields the block lower-triangular form

$$\Pi = THT^{-1} = \begin{bmatrix} A + UC & 0 \\ C & D - CU \end{bmatrix}.$$

Writing the Riccati equation in terms of the blocks of the TTSA dynamics gives

$$\frac{\beta}{\alpha} U J_{yx} U + J_{xx} U - \frac{\beta}{\alpha} U J_{yy} = J_{xy}. \quad (37)$$

This equation is the singular-perturbation Riccati equation associated with the linearized two-time-scale dynamics. Under Assumption B.1, it is shown in [22, Chapter 2, Eq. 2.13] that when β/α small enough, this Riccati equation admits a solution of the form

$$U = \sum_{\ell=0}^{\infty} \left(\frac{\beta}{\alpha} \right)^{\ell} \tilde{U}_{\ell},$$

where the matrices $(\tilde{U}_{\ell})_{\ell \geq 0}$ are independent of α and β . The matrix $\tilde{U}_0$ can be found by setting $\beta/\alpha = 0$ in (37) which gives

$$J_{xx} \tilde{U}_0 = J_{xy}.$$

Since $-J_{xx}$ is Hurwitz by Assumption B.1, the matrix J_{xx} is invertible and $\tilde{U}_0 = J_{xx}^{-1} J_{xy}$. This shows that

$$\begin{aligned} U &= \tilde{U}_0 + \sum_{\ell=1}^{\infty} \left(\frac{\beta}{\alpha} \right)^{\ell} \tilde{U}_{\ell} \\ &= J_{xx}^{-1} J_{xy} + \frac{\beta}{\alpha} \hat{U}_{\alpha, \beta}, \end{aligned}$$

where $\hat{U}_{\alpha, \beta} \triangleq \sum_{\ell=1}^{\infty} \left(\frac{\beta}{\alpha} \right)^{\ell-1} \tilde{U}_{\ell}$.

From Assumption B.1 and since we are assuming that $\frac{\beta}{\alpha} < \min\{\varepsilon_0^*, \varepsilon_1^*\}$ where $\varepsilon_0^*, \varepsilon_1^*$ are defined in (74), (75) and are dependent on the predefined radius $R_{ric} = \frac{1}{4\|G\|_{\text{op}}\sqrt{\kappa(G)M_J}}$ we can apply Lemma 3 to show that the Riccati equation accepts a unique solution that lives inside the closed ball $B_{R_{ric}}^{U_0} = \{U : \|U - \tilde{U}_0\| \leq R_{ric}\}$. This implies that for the selected range of the ratio $\frac{\beta}{\alpha}$ we have :

$$\|U\|_{\text{op}} \leq R_{ric} + \|U_0\|_{\text{op}}$$

We now apply the transformation to the linear recursion. Recall that

$$\begin{bmatrix} b_k^x \\ b_k^y \end{bmatrix} = \begin{pmatrix} \mathbb{E}[\tilde{x}_k] \\ \mathbb{E}[\tilde{y}_k] \end{pmatrix}.$$

Using $H = T^{-1}\Pi T$, the recursion (34) becomes

$$\begin{bmatrix} b_{k+1}^x \\ b_{k+1}^y \end{bmatrix} = \begin{bmatrix} I & -U \\ 0 & I \end{bmatrix} \begin{bmatrix} I - \alpha J_{xx} - \beta U J_{yx} & 0 \\ -\beta J_{yx} & I - \beta(J_{yy} - J_{yx}U) \end{bmatrix} \begin{bmatrix} I & U \\ 0 & I \end{bmatrix} \begin{bmatrix} b_k^x \\ b_k^y \end{bmatrix} + \mathcal{R}_k$$

Define the transformed bias

$$\begin{bmatrix} r_k^x \\ r_k^y \end{bmatrix} \triangleq T \begin{bmatrix} b_k^x \\ b_k^y \end{bmatrix} = \begin{bmatrix} I & U \\ 0 & I \end{bmatrix} \begin{bmatrix} b_k^x \\ b_k^y \end{bmatrix}.$$

Multiplying the bias recursion by T gives

$$\begin{bmatrix} r_{k+1}^x \\ r_{k+1}^y \end{bmatrix} = \begin{bmatrix} I - \alpha J_{xx}^{\alpha, \beta} & 0 \\ -\beta J_{yx} & I - \beta J_{yy}^{\alpha, \beta} \end{bmatrix} \begin{bmatrix} r_k^x \\ r_k^y \end{bmatrix} + T \mathcal{R}_k, \quad (38)$$

where

$$\begin{aligned} J_{xx}^{\alpha, \beta} &\triangleq J_{xx} + \frac{\beta}{\alpha} U J_{yx} \\ J_{yy}^{\alpha, \beta} &\triangleq J_{yy} - J_{yx} J_{xx}^{-1} J_{xy} - \frac{\beta}{\alpha} J_{yx} \hat{U}_{\alpha, \beta}. \end{aligned}$$

In particular, the leading-order matrix for the slow variable's bias is

$$\Delta \triangleq J_{yy} - J_{yx} J_{xx}^{-1} J_{xy}.$$

This matrix is the effective first-order drift of the slow dynamics. The assumption that $-\Delta$ is Hurwitz is standard in two-time-scale stochastic approximation and appears, for instance, in the linear and CLT analyses of TTSA [26, 17, 18, 21].

For lighter notation, we write (38) as

$$r_{k+1} = \Pi r_k + T \mathcal{R}_k,$$

where $r_k \triangleq \begin{bmatrix} r_k^x \\ r_k^y \end{bmatrix}$, and $\Pi \triangleq \begin{bmatrix} I - \alpha J_{xx}^{\alpha, \beta} & 0 \\ -\beta J_{yx} & I - \beta J_{yy}^{\alpha, \beta} \end{bmatrix}$.

Block diagonalization through a Sylvester equation. After the Riccati transformation, the linear part of the recursion is block lower triangular. The remaining coupling is the bottom left block $-\beta J_{yx}$, which still lets the fast transformed bias influence the slow transformed bias. We now remove this coupling by a second change of coordinates. Let

$$S \triangleq \begin{bmatrix} I & 0 \\ X & I \end{bmatrix}, \quad S^{-1} = \begin{bmatrix} I & 0 \\ -X & I \end{bmatrix}, \quad (39)$$

where $X \in \mathbb{R}^{d_y \times d_x}$ is to be determined. A direct computation gives

$$S\Pi S^{-1} = \begin{bmatrix} I - \alpha J_{xx}^{\alpha,\beta} & 0 \\ X(I - \alpha J_{xx}^{\alpha,\beta}) - (I - \beta J_{yy}^{\alpha,\beta})X - \beta J_{yx} & I - \beta J_{yy}^{\alpha,\beta} \end{bmatrix}.$$

Hence, the bottom left block vanishes if and only if

$$X(I - \alpha J_{xx}^{\alpha,\beta}) - (I - \beta J_{yy}^{\alpha,\beta})X - \beta J_{yx} = 0.$$

Equivalently, after simplifying and dividing by α ,

$$X J_{xx}^{\alpha,\beta} - \frac{\beta}{\alpha} J_{yy}^{\alpha,\beta} X = -\frac{\beta}{\alpha} J_{yx}. \quad (40)$$

This is a Sylvester equation for the unknown block X [22, Chapter 2, Eq 4.3], [32, Lemma 8.3]. From [32, Lemma 8.3]. Since we are assuming B.1 and $\frac{\beta}{\alpha} < \min\{\varepsilon_0^*, \varepsilon_1^*, \varepsilon_2^*, \varepsilon_3^*\}$ that are dependent on R_{ric} and are defined in (74), (75), (78), (79) then the above Sylvester equation accepts a unique solution X that lives inside a closed ball $B_{\varepsilon R_{syl}} = \{X : \|X\|_{\text{op}} \leq \varepsilon R_{syl}\}$ where $R_{syl} = 2 \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}$, this directly implies that:

$$\|X\|_{\text{op}} \leq \frac{\beta}{\alpha} R_{syl} \quad (41)$$

Furthermore, the solution admits the expansion

$$X = \sum_{\ell=1}^{\infty} \left(\frac{\beta}{\alpha}\right)^{\ell} \tilde{X}_{\ell},$$

We can solve for $\tilde{X}_1$ by replacing the sum into (40) then dividing by $\frac{\beta}{\alpha}$ then setting it to zero to get:

$$\tilde{X}_1 = -J_{yx} J_{xx}^{-1}$$

With this choice of X , the transformed matrix $\Lambda \triangleq S\Pi S^{-1}$ is block diagonal:

$$\Lambda = \begin{bmatrix} I - \alpha J_{xx}^{\alpha,\beta} & 0 \\ 0 & I - \beta J_{yy}^{\alpha,\beta} \end{bmatrix}.$$

We now apply the second change of coordinates to the transformed bias vector r_k . Define

$$\begin{bmatrix} s_k^x \\ s_k^y \\ s_k \end{bmatrix} \triangleq S \begin{bmatrix} r_k^x \\ r_k^y \\ r_k \end{bmatrix} = \begin{bmatrix} I & 0 \\ X & I \end{bmatrix} \begin{bmatrix} r_k^x \\ r_k^y \\ r_k \end{bmatrix}.$$

By the choice of X , the linear part of the recursion is block diagonal. Hence, using

$$ST = \begin{bmatrix} I & U \\ X & XU + I \end{bmatrix},$$

we obtain

$$\begin{bmatrix} s_{k+1}^x \\ s_{k+1}^y \\ s_{k+1} \end{bmatrix} = \Lambda \begin{bmatrix} s_k^x \\ s_k^y \\ s_k \end{bmatrix} - \begin{bmatrix} I & U \\ X & XU + I \end{bmatrix} \begin{bmatrix} \alpha(\mathbb{E}[R_f(\tilde{x}_k, \hat{y}_k)] + \mathbb{E}[\psi_k]) \\ \beta(\mathbb{E}[R_g(\tilde{x}_k, \hat{y}_k)] + \mathbb{E}[\varphi_k]) \end{bmatrix}.$$

Hence, the concatenated transformed bias s_k satisfies the compact recursion

$$s_{k+1} = \Lambda s_k + ST\mathcal{R}_k. \quad (42)$$

Recursion on the transformed system. We now roll out the transformed recursion (42). Since the matrix Λ is block diagonal, we obtain

$$s_{k+1} = \Lambda^{k+1} s_0 + \sum_{j=0}^k \Lambda^{k-j} S T \mathcal{R}_j.$$

This representation is the main payoff of the two transformations: the linear part of the recursion is now decoupled into a fast block and a slow block. Writing the two components separately gives

$$\begin{aligned} s_{k+1}^x &= (I - \alpha J_{xx}^{\alpha, \beta})^{k+1} s_0^x - \alpha \sum_{j=0}^k (I - \alpha J_{xx}^{\alpha, \beta})^{k-j} (\mathbb{E}[R_f(\tilde{x}_j, \hat{y}_j)] + \mathbb{E}[\psi_j]) \\ &\quad - \beta \sum_{j=0}^k (I - \alpha J_{xx}^{\alpha, \beta})^{k-j} U (\mathbb{E}[R_g(\tilde{x}_j, \hat{y}_j)] + \mathbb{E}[\varphi_j]), \end{aligned} \quad (43)$$

$$\begin{aligned} s_{k+1}^y &= (I - \beta J_{yy}^{\alpha, \beta})^{k+1} s_0^y - \alpha \sum_{j=0}^k (I - \beta J_{yy}^{\alpha, \beta})^{k-j} X (\mathbb{E}[R_f(\tilde{x}_j, \hat{y}_j)] + \mathbb{E}[\psi_j]) \\ &\quad - \beta \sum_{j=0}^k (I - \beta J_{yy}^{\alpha, \beta})^{k-j} (XU + I) (\mathbb{E}[R_g(\tilde{x}_j, \hat{y}_j)] + \mathbb{E}[\varphi_j]). \end{aligned} \quad (44)$$

The original bias recursion was fully coupled: the fast and slow biases influenced each other already at the level of the linear dynamics. After the changes of coordinates defined by T and S , this coupling has been pushed into the residual terms, while the leading linear dynamics are block diagonal. This is precisely what allows us to study the two components separately.

The next step is to show that the two diagonal blocks are contractions in suitable Lyapunov norms. More precisely, for our choice on the step sizes, the matrices describing the perturbed dynamics:

$$I - \alpha J_{xx}^{\alpha, \beta} \quad \text{and} \quad I - \beta J_{yy}^{\alpha, \beta}$$

inherit the stability of $I - \alpha J_{xx}$ and $I - \beta \Delta$, respectively. Their powers therefore produce the exponentially decaying initial-condition terms in the final bias bound, with decay rates of order α for the fast component and β for the slow component.

Bound of the fast transformed component. We start from the recursion (43). For readability, write

$$A_x \triangleq I - \alpha J_{xx}^{\alpha, \beta}.$$

Applying the triangle inequality, submultiplicativity of the operator norm, and Jensen's inequality gives

$$\begin{aligned} \|s_{k+1}^x\| &\leq \|A_x^{k+1} s_0^x\| + \alpha \sum_{j=0}^k \|A_x^{k-j}\|_{\text{op}} \mathbb{E}[\|R_f(\tilde{x}_j, \hat{y}_j)\|] \\ &\quad + \beta \sum_{j=0}^k \|A_x^{k-j}\|_{\text{op}} \|U\|_{\text{op}} \mathbb{E}[\|R_g(\tilde{x}_j, \hat{y}_j)\|] \\ &\quad + \alpha \left\| \sum_{j=0}^k A_x^{k-j} \mathbb{E}[\psi_j] \right\| + \beta \left\| \sum_{j=0}^k A_x^{k-j} U \mathbb{E}[\varphi_j] \right\|. \end{aligned}$$

In the two terms involving R_f and R_g , we used $\|\mathbb{E}[Z]\| \leq \mathbb{E}[\|Z\|]$. By contrast, we keep the Markovian-noise terms as weighted sums, since these will be controlled later through the Poisson equation argument.

The Euclidean operator norm does not directly reveal the contraction of $A_x = I - \alpha J_{xx}^{\alpha, \beta}$. We therefore use a Lyapunov norm adapted to the unperturbed fast linear dynamics. Since $-J_{xx}$ is Hurwitz by Assumption B.1, there exists a unique symmetric positive definite matrix L solving

$$L J_{xx} + J_{xx}^\top L = I. \quad (45)$$

We denote the associated norm by

$$\|z\|_{\mathbf{L}} \triangleq \sqrt{z^\top \mathbf{L} z}.$$

Since we are assuming B.1, $\alpha < \alpha^*(R_{ric})$ as defined in (81) and the ratio satisfies $\frac{\beta}{\alpha} < \min\{\varepsilon_0^*, \varepsilon_1^*, \varepsilon_5^*\}$ where $\varepsilon_0^*, \varepsilon_1^*, \varepsilon_5^*$ depend on R_{ric} and are fully defined in (74), (75), (82) then after applying submultiplicativity of the norm we can apply Lemma 5 to get: we have:

$$\|(I - \alpha J_{xx} - \beta U J_{yx})^k\|_{\mathbf{L}} \leq \|I - \alpha J_{xx} - \beta U J_{yx}\|_{\mathbf{L}}^k \leq e^{-kc\alpha}$$

where $c = (8\|\mathbf{L}\|_{\text{op}})^{-1}$, see Lemma 5 and the proof for full derivation. Equivalently,

$$\|A_x^k\|_{\mathbf{L}} \leq e^{-kc\alpha}$$

which yields for every $\ell \geq 0$,

$$\|A_x^\ell\|_{\text{op}} \leq \sqrt{\kappa(\mathbf{L})} \|A_x^\ell\|_{\mathbf{L}} \leq \sqrt{\kappa(\mathbf{L})} e^{-c\alpha\ell}.$$

Applying this bound to (43) gives

$$\|s_{k+1}^x\| \leq \sqrt{\kappa(\mathbf{L})} e^{-(k+1)c\alpha} \|s_0^x\| + \alpha \sqrt{\kappa(\mathbf{L})} \sum_{j=0}^k e^{-c\alpha(k-j)} \mathbb{E}[\|R_f(\tilde{x}_j, \hat{y}_j)\|] \quad (46)$$

$$+ \beta \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} \sum_{j=0}^k e^{-c\alpha(k-j)} \mathbb{E}[\|R_g(\tilde{x}_j, \hat{y}_j)\|] \quad (47)$$

$$+ \alpha \left\| \sum_{j=0}^k A_x^{k-j} \mathbb{E}[\psi_j] \right\| + \beta \left\| \sum_{j=0}^k A_x^{k-j} U \mathbb{E}[\varphi_j] \right\|. \quad (48)$$

We now bound the terms on the right hand side separately. The first term in (46) is the exponentially decaying contribution of the initial condition. The second term involves the Taylor remainder. Since R_f is a second order remainder, there exists a constant $L_R > 0$ such that:

$$\begin{aligned} \mathbb{E}[\|R_f(\tilde{x}_j, \hat{y}_j)\|] &\leq L_R \mathbb{E}[(\|\tilde{x}_j\| + \|\hat{y}_j\|)^2] \\ &\leq 2L_R \mathbb{E}[\|\tilde{x}_j\|^2 + \|\hat{y}_j\|^2] \\ &= 2L_R \mathbb{E}[\|x_j - h(y_j) + h(y_j) - h(y^*)\|^2 + \|\hat{y}_j\|^2] \\ &\leq 2L_R \mathbb{E}[2\|x_j - h(y_j)\|^2 + 2\|h(y_j) - h(y^*)\|^2 + \|\hat{y}_j\|^2] \\ &\leq 2L_R \mathbb{E}[2\|x_j - h(y_j)\|^2 + (2L_h^2 + 1)\|\hat{y}_j\|^2]. \end{aligned}$$

Setting:

$$d_1 \triangleq \max\{2, 2L_h^2 + 1\},$$

we obtain:

$$\mathbb{E}[\|R_f(\tilde{x}_j, \hat{y}_j)\|] \leq 2L_R d_1 \mathbb{E}[\|x_j - h(y_j)\|^2 + \|\hat{y}_j\|^2].$$

Plugging this into the second term of (46) yields:

$$\alpha \sqrt{\kappa(\mathbf{L})} \sum_{j=0}^k e^{-c\alpha(k-j)} \mathbb{E}[\|R_f(\tilde{x}_j, \hat{y}_j)\|] \leq 2\alpha \sqrt{\kappa(\mathbf{L})} L_R d_1 \sum_{j=0}^k e^{-c\alpha(k-j)} \mathbb{E}[\|x_j - h(y_j)\|^2 + \|\hat{y}_j\|^2].$$

By Lemma 8, and using the notation

$$\mathcal{M}_{\alpha, \beta} \triangleq \text{MSE}_x^\infty(\alpha, \beta) + \text{MSE}_y^\infty(\alpha, \beta),$$

we obtain

$$\begin{aligned} \alpha \sqrt{\kappa(\mathbf{L})} \sum_{j=0}^k e^{-c\alpha(k-j)} \mathbb{E}[\|R_f(\tilde{x}_j, \hat{y}_j)\|] &\leq \underbrace{\alpha 2\sqrt{\kappa(\mathbf{L})} L_R d_1}_{\triangleq C_{x,1}} (\mathcal{M}_{\alpha, \beta} + u_k^{c, \alpha} + w_k^{c, \alpha}) \\ &\quad + \underbrace{2\sqrt{\kappa(\mathbf{L})} L_R d_1 c^{-1}}_{\triangleq C_{x,2}} \mathcal{M}_{\alpha, \beta}. \end{aligned}$$

The term involving R_g is controlled in the same way. We denote $M_U \triangleq R_{ric} + \|U_0\|_{\text{op}}$. Thus,

$$(47) \leq \underbrace{\beta 2\sqrt{\kappa(\mathbf{L})}M_U L_R d_1}_{\triangleq C_{x,3}} (\mathcal{M}_{\alpha,\beta} + u_k^{c,\alpha} + w_k^{c,\alpha}) + \frac{\beta}{\alpha} \underbrace{2\sqrt{\kappa(\mathbf{L})}M_U L_R d_1 c^{-1}}_{C_{x,4}} \mathcal{M}_{\alpha,\beta}.$$

It remains to control the two weighted Markovian-noise terms in (48). By Lemma 7,

$$(48) \leq \alpha A_1 + \beta(A_2 + B_1) + \alpha^2 A_3 + \beta^2 B_2 + \alpha\beta(A_4 + B_3) + \frac{\beta^2}{\alpha} B_4.$$

where the constants are defined in detail in the proof of the lemma. Merging all the bounds from above we obtain

$$\begin{aligned} \|s_{k+1}^x\| &\leq e^{-(k+1)c\alpha} \sqrt{\kappa(\mathbf{L})} \|s_0^x\| + (\alpha + \beta) F_x^{(1)} + F_x^{(2)} \mathcal{M}_{\alpha,\beta} + (\alpha + \beta) F_x^{(3)} (u_k^{c,\alpha} + w_k^{c,\alpha}) \\ &\quad + R_{\alpha,\beta}^x \end{aligned}$$

where the quantities $F_x^{(1)}, F_x^{(2)}, F_x^{(3)}, R_{\alpha,\beta}^x$ are defined by:

$$\begin{aligned} F_x^{(1)} &:= A_1 \vee (A_2 + B_1) \vee \omega^* B_4, \\ F_x^{(2)} &:= C_{x,2} + \omega^* C_{x,4} + F_x^{(3)} (\alpha^* + \beta^*), \\ F_x^{(3)} &:= C_{x,3} \vee C_{x,1}, \\ R_{\alpha,\beta}^x &:= \alpha^2 A_3 + \beta^2 B_2 + \alpha\beta(A_4 + B_3). \end{aligned}$$

Bound of the slow transformed component. We now turn to the slow component. Let

$$A_y \triangleq I - \beta J_{yy}^{\alpha,\beta}, \quad I' \triangleq I + XU.$$

Applying the triangle inequality, submultiplicativity, and Jensen's inequality to (44) gives

$$\begin{aligned} \|s_{k+1}^y\| &\leq \|A_y^{k+1} s_0^y\| + \alpha \sum_{j=0}^k \|A_y^{k-j}\|_{\text{op}} \|X\|_{\text{op}} \mathbb{E} [\|R_f(\tilde{x}_j, \hat{y}_j)\|] \\ &\quad + \beta \sum_{j=0}^k \|A_y^{k-j}\|_{\text{op}} \|I'\|_{\text{op}} \mathbb{E} [\|R_g(\tilde{x}_j, \hat{y}_j)\|] \\ &\quad + \alpha \left\| \sum_{j=0}^k A_y^{k-j} X \mathbb{E} [\psi_j] \right\| + \beta \left\| \sum_{j=0}^k A_y^{k-j} I' \mathbb{E} [\varphi_j] \right\|. \end{aligned}$$

We use a Lyapunov norm adapted to the effective slow matrix

$$\Delta \triangleq J_{yy} - J_{yx} J_{xx}^{-1} J_{xy}.$$

Since $-\Delta$ is Hurwitz by Assumption B.1, there exists a unique symmetric positive definite matrix G solving the following Lyapunov equation:

$$G\Delta + \Delta^\top G = I. \quad (49)$$

Since we are assuming B.1, and we have $\beta \leq \beta^*$ as defined in (84) that also satisfies $\frac{\beta}{\alpha} < \min\{\varepsilon_0^*, \varepsilon_1^*\}$ then we can apply Lemma 6 to have:

$$\|I - \beta(\Delta - J_{yx}(U - U_0))\|_G \leq e^{-d\beta}$$

where $d = (8 \|G\|_{\text{op}})^{-1}$, see Lemma 6 and its proof for full derivation. Equivalently,

$$\|A_y^\ell\|_G \leq e^{-d\beta\ell}, \quad \forall \ell \geq 0.$$

we obtain, for every $\ell \geq 0$,

$$\|A_y^\ell\|_{\text{op}} \leq \sqrt{\kappa(G)} \|A_y^\ell\|_G \leq \sqrt{\kappa(G)} e^{-d\beta\ell}.$$

Therefore,

$$\begin{aligned} \|s_{k+1}^y\| &\leq e^{-d\beta(k+1)} \sqrt{\kappa(\mathbf{G})} \|s_0^y\| \\ &\quad + \alpha \|X\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} \sum_{j=0}^k e^{-d\beta(k-j)} \mathbb{E} [\|R_f(\tilde{x}_j, \hat{y}_j)\|] \end{aligned} \quad (50)$$

$$+ \beta \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} \sum_{j=0}^k e^{-d\beta(k-j)} \mathbb{E} [\|R_g(\tilde{x}_j, \hat{y}_j)\|] \quad (51)$$

$$+ \alpha \left\| \sum_{j=0}^k A_y^{k-j} X \mathbb{E} [\psi_j] \right\| + \beta \left\| \sum_{j=0}^k A_y^{k-j} I' \mathbb{E} [\varphi_j] \right\|. \quad (52)$$

We now bound the Taylor remainder terms. As in the fast component analysis, there exists a constant $L_R > 0$ such that

$$\mathbb{E} [\|R_f(\tilde{x}_j, \hat{y}_j)\|] \leq 2L_R d_1 \mathbb{E} [\|x_j - h(y_j)\|^2 + \|\hat{y}_j\|^2], \quad d_1 \triangleq \max\{2, 2L_h^2 + 1\}.$$

The same bound holds for R_g , up to increasing L_R if necessary.

By Lemma 8,

$$\sum_{j=0}^k e^{-d\beta(k-j)} \mathbb{E} [\|R_f(\tilde{x}_j, \hat{y}_j)\|] \leq 2L_R d_1 \left[\left(1 + \frac{1}{d\beta}\right) \mathcal{M}_{\alpha,\beta} + u_k^{d,\beta} + w_k^{d,\beta} \right].$$

At first sight, the factor $1/\beta$ is problematic, however, the term (50) still has a multiplication by $\alpha \|X\|_{\text{op}}$. Since we are assuming B.1, we showed previously in (41) that

$$\alpha \|X\|_{\text{op}} \leq \beta R_{syl}$$

with $R_{syl} = 2 \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}$. Consequently,

$$\begin{aligned} (50) &\leq \underbrace{\beta 2R_{syl} \sqrt{\kappa(\mathbf{G})} L_R d_1}_{\triangleq C_{y,1}} \left(\mathcal{M}_{\alpha,\beta} + u_k^{d,\beta} + w_k^{d,\beta} \right) \\ &\quad + \underbrace{2R_{syl} \sqrt{\kappa(\mathbf{G})} L_R d_1 d^{-1}}_{\triangleq C_{y,2}} \mathcal{M}_{\alpha,\beta}. \end{aligned}$$

The term (51) is controlled similarly. For $\frac{\beta}{\alpha} < \min\{\varepsilon_0^*, \varepsilon_1^*, \varepsilon_2^*, \varepsilon_3^*\}$ we have

$$\|I'\|_{\text{op}} \leq \|I\|_{\text{op}} + \|X\|_{\text{op}} \|U\|_{\text{op}} \leq 1 + R_{syl}(R_{ric} + \|U_0\|_{\text{op}}) \triangleq M_{I'}$$

Lemma 8 gives

$$\begin{aligned} (51) &\leq \underbrace{\beta 2M_{I'} \sqrt{\kappa(\mathbf{G})} L_R d_1}_{\triangleq C_{y,3}} \left(\mathcal{M}_{\alpha,\beta} + u_k^{d,\beta} + w_k^{d,\beta} \right) \\ &\quad + \underbrace{2M_{I'} \sqrt{\kappa(\mathbf{G})} L_R d_1 d^{-1}}_{\triangleq C_{y,4}} \mathcal{M}_{\alpha,\beta}. \end{aligned}$$

It remains to bound the weighted Markovian-noise terms in (52). By Lemma 7,

$$(52) \leq \alpha(C_2 + D_4) + \beta(C_1 + D_1) + \alpha\beta(C_4 + D_3) + \beta^2(C_3 + D_2).$$

We define :

$$\begin{aligned} F_y^{(1)} &:= (C_2 + D_4) \vee (C_1 + D_1), \\ F_y^{(2)} &:= C_{y,2} + C_{y,4} + \beta^*(C_{y,1} + C_{y,3}), \\ F_y^{(3)} &:= C_{y,3} + C_{y,1}, \\ R_{\alpha,\beta}^y &:= \alpha\beta(C_4 + D_3) + \beta^2(C_3 + D_2). \end{aligned}$$

We wrap all the bounds in the final one

$$\begin{aligned} \|s_{k+1}^y\| &\leq e^{-d\beta(k+1)} \sqrt{\kappa(\mathbf{G})} \|s_0^y\| + (\alpha + \beta) F_y^{(1)} + F_y^{(2)} \mathcal{M}_{\alpha,\beta} + \beta F_y^{(3)} (u_k^{d,\beta} + w_k^{d,\beta}) \\ &\quad + R_{\alpha,\beta}^y \end{aligned}$$

we merge the bounds of $\|s_k^x\|$ and $\|s_k^y\|$ to get

$$\begin{aligned} \|s_{k+1}^x\| + \|s_{k+1}^y\| &\leq e^{-(k+1)c\alpha} \sqrt{\kappa(\mathbf{L})} \|s_0^x\| + e^{-d\beta(k+1)} \sqrt{\kappa(\mathbf{G})} \|s_0^y\| \\ &\quad + \underbrace{F_x^{(1)} + F_y^{(1)}(\alpha + \beta)}_{\triangleq F_1} + \underbrace{(F_x^{(2)} + F_y^{(2)}) \mathcal{M}_{\alpha,\beta}}_{\triangleq F_2} + (\alpha + \beta) F_x^{(3)} (u_k^{c,\alpha} + w_k^{c,\alpha}) \\ &\quad + \beta F_y^{(3)} (u_k^{d,\beta} + w_k^{d,\beta}) + \underbrace{R_{\alpha,\beta}^x + R_{\alpha,\beta}^y}_{\triangleq R_{\alpha,\beta}} \end{aligned}$$

We define the new sequence $\hat{S}_k^{\text{MSE}}$ to be:

$$\hat{S}_k^{\text{MSE}} = \beta F_y^{(3)} (u_k^{d,\beta} + w_k^{d,\beta}) + (\alpha + \beta) F_x^{(3)} (u_k^{c,\alpha} + w_k^{c,\alpha})$$

where we have showed that by Lemma 8 that the limit exist and it is zero

$$\lim_{k \rightarrow \infty} \hat{S}_k^{\text{MSE}} = 0$$

which yields our bound:

$$\begin{aligned} \|s_{k+1}^x\| + \|s_{k+1}^y\| &\leq e^{-(k+1)c\alpha} \sqrt{\kappa(\mathbf{L})} \|s_0^x\| + e^{-d\beta(k+1)} \sqrt{\kappa(\mathbf{G})} \|s_0^y\| \\ &\quad + F_1(\alpha + \beta) + F_2 \mathcal{M}_{\alpha,\beta} + \hat{S}_k^{\text{MSE}} + R_{\alpha,\beta} \end{aligned}$$

where $R_{\alpha,\beta}$ contains terms of higher order $O((\alpha + \beta)^2)$. Moreover, from the definition of $u_k^{c,\alpha}, w_k^{c,\alpha}$ in Lemma 8 there exists $a, b, \hat{c}, \hat{d}$ such that:

$$\hat{S}_k^{\text{MSE}} \leq a e^{-\alpha \hat{c} k} \|\hat{x}_0\|^2 + b e^{-\beta \hat{d} k} \|\hat{y}_0\|^2 \quad (53)$$

So our final bound will be

$$\begin{aligned} \|s_{k+1}^x\| + \|s_{k+1}^y\| &\leq e^{-(k+1)c\alpha} \sqrt{\kappa(\mathbf{L})} \|s_0^x\| + e^{-d\beta(k+1)} \sqrt{\kappa(\mathbf{G})} \|s_0^y\| + a e^{-\alpha \hat{c} k} \|\hat{x}_0\|^2 + b e^{-\beta \hat{d} k} \|\hat{y}_0\|^2 \\ &\quad + F_1(\alpha + \beta) + F_2 \mathcal{M}_{\alpha,\beta} + R_{\alpha,\beta} \end{aligned}$$

Unwrapping the results. It remains to return to the original bias coordinates. By construction,

$$s_k = \begin{bmatrix} s_k^x \\ s_k^y \end{bmatrix} = S T \begin{bmatrix} b_k^x \\ b_k^y \end{bmatrix}.$$

Since the matrices S and T are invertible, we can write equivalently,

$$\begin{bmatrix} b_k^x \\ b_k^y \end{bmatrix} = T^{-1} S^{-1} s_k.$$

Therefore, by applying the norm on both sides,

$$\begin{aligned} \left\| \begin{bmatrix} b_k^x \\ b_k^y \end{bmatrix} \right\| &\leq \|T^{-1} S^{-1} s_k\| \\ &\leq \|T^{-1} S^{-1}\|_{\text{op}} \|s_k\| \\ &\leq \|T^{-1} S^{-1}\|_{\text{op}} (\|s_k^x\| + \|s_k^y\|). \end{aligned}$$

where in the last inequality we have used the triangular inequality since

$$\left\| \begin{bmatrix} s_k^x \\ b_k^y \end{bmatrix} \right\| = \left\| \begin{bmatrix} s_k^x \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b_k^y \end{bmatrix} \right\| \leq \|s_k^x\| + \|s_k^y\|$$

Plugging the bounds obtained above for the fast and slow transformed components gives the desired bias bound:

$$\max \{\|b_k^x\|, \|b_k^y\|\} \leq \|T^{-1} S^{-1}\|_{\text{op}} (\|s_k^x\| + \|s_k^y\|).$$

which concludes the proof of our theorem. $\square$

E Supporting Lemmas

The first lemma is merely a collection of various Lipschitzness/boundedness properties induced by the assumptions we make in the paper, introducing several constants that we use in the proofs of the main theorems.

Lemma 1. *Under assumptions A.1-4, there exist finite non-negative constants $M_f, M_g, M_h, M_\xi, M_S$ dependent on the problem settings satisfying :*

1. $\|\bar{f}(x_k, y_k)\| \leq 2L_f M_S + \|\bar{f}(0, 0)\| \leq M_f$
2. $\|\bar{g}(x_k, y_k)\| \leq 2L_g M_S + \|\bar{g}(0, 0)\| \leq M_g$
3. $\|f(x_k, y_k, \xi_{k+1})\| \leq 2L_f M_S + M_S \leq M_f$
4. $\|g(x_k, y_k, \xi_{k+1})\| \leq 2L_g M_S + M_S \leq M_g$
5. $\|\Delta_k^h\| \leq \beta L_h M_g \leq \beta M_h$
6. $\|\psi_k\| \leq M_\xi$
7. $\|h(y_k)\| \leq L_h M_S + \|h(y_0)\| \leq M_h$
8. *Given assumption A.2-4, for every $x_1, x_2 \in \mathbb{R}^{d_x}, y_1, y_2 \in \mathbb{R}^{d_y}, v_f$ and v_g (as defined in (54) and (65)) are L_{v_f} and L_{v_g} -Lipschitz continuous:*

$$\begin{aligned} \|v_f(x_1, y_1, \xi) - v_f(x_2, y_2, \xi)\| &\leq L_{v_f}(\|x_1 - x_2\| + \|y_1 - y_2\|) \\ \|v_g(x_1, y_1, \xi) - v_g(x_2, y_2, \xi)\| &\leq L_{v_g}(\|x_1 - x_2\| + \|y_1 - y_2\|) \end{aligned}$$
9. $\|v_f(x_1, y_1, \xi)\| \leq 2L_f M_S + f_0 \leq M_\xi$
10. $\langle \hat{x}_k, v_f(x_k, y_k, \xi_{k+1}) \rangle \leq M_\xi$

Proof. We provide one line proof for each point:

1. $\|\bar{f}(x_k, y_k)\| \leq \|\bar{f}(x_k, y_k) - \bar{f}(0, 0)\| + \|\bar{f}(0, 0)\| \leq L_f(\|x_k\| + \|y_k\|) + \|\bar{f}(0, 0)\| \leq 2L_f M_S + \|\bar{f}(0, 0)\| \leq M_f$
2. $\|\bar{g}(x_k, y_k)\| \leq 2L_g M_S + \|\bar{g}(0, 0)\| \leq M_g$
3. $\|f(x_k, y_k, \xi_{k+1})\| \leq 2L_f M_S + M_S \leq M_f$
4. $\|g(x_k, y_k, \xi_{k+1})\| \leq 2L_g M_S + M_S \leq M_g$
5. $\|\Delta_k^h\| = \|h(y_{k+1}) - h(y_k)\| \leq L_h \|y_{k+1} - y_k\| \leq \beta M_h$
6. $\|\psi_k\| = \|f(x_k, y_k, \xi_{k+1}) - \bar{f}(x_k, y_k)\| \leq \|f(x_k, y_k, \xi_{k+1})\| + \|\bar{f}(x_k, y_k)\| \leq 2M_f \leq M_\xi$
7. $\|h(y_k)\| \leq L_h M_S + \|h(y_0)\| \leq M_h$
8. See [1, Lemma 8] and [17, Lemma D.4].
9. $\|v_f(x_1, y_1, \xi)\| \leq 2L_f M_S + f_0 \leq M_\xi$
10. $\langle \hat{x}_k, v_f(x_k, y_k, \xi_{k+1}) \rangle \leq \|\hat{x}_k\| \|v_f(x_k, y_k, \xi_{k+1})\| \leq M_\xi$

□

The next lemma proves bounds on the terms containing the weighted sum of deviations of the dynamics estimators from their stationary averages $f(x_k, y_k, \xi_{k+1}) - \bar{f}(x_k, y_k)$ and $g(x_k, y_k, \xi_{k+1}) - \bar{g}(x_k, y_k)$ in Theorem 1.

Lemma 2. Taking the sequence $(x_k, y_k)_{k \geq 0}$ defined by (1) and under assumptions A.1-3, for every $\alpha\mu_f < 1$ and $\beta\mu_g < 1$:

$$\begin{aligned} \left| \alpha \mathbb{E} \left[\sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, \psi_j \rangle \right] \right| &\leq \alpha \left(M_\xi \left(\frac{3 - \alpha\mu_f}{1 - \alpha\mu_f} + \frac{M_f(1 + |\Xi| M_S L_K)}{\mu_f} \right) + \frac{L_{v_f} M_f (M_S + M_h)}{\mu_f} \right) \\ &\quad + \beta \left(\frac{M_\xi M_h + L_{v_f} M_g (M_S + M_h) + |\Xi| M_S L_K M_\xi M_g}{\mu_f} \right) \\ \left| \beta \mathbb{E} \left[\sum_{j=0}^k (1 - \beta\mu_g)^{k-j} \langle \hat{y}_j, \phi_j \rangle \right] \right| &\leq \beta \left(\frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_g}{\mu_g} + M_\xi \left(\frac{3 - \beta\mu_g}{1 - \beta\mu_g} + \frac{M_g}{\mu_g} \right) \right) \\ &\quad + \alpha \frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_f}{\mu_g} \end{aligned}$$

where $\psi_k = f(x_k, y_k, \xi_{k+1}) - \bar{f}(x_k, y_k)$ and $\phi_k = g(x_k, y_k, \xi_{k+1}) - \bar{g}(x_k, y_k)$.

Proof. To prove this lemma, we use the same technique in [1, Lemma 8] and [21] by invoking the solution of a Poisson equation that allow us to decompose the deviation terms $f(x_j, y_j, \xi_{j+1}) - \bar{f}(x_j, y_j)$ inside the sum into a Martingale Difference Sequence (MDS), a bounded term and a telescoping term. Since our terms in the sum are weighted by $(1 - \alpha\mu_f)^{k-j}$ we don't have exactly the telescoping terms that we want, instead, we have additional residuals that we needed to deal with. Moreover, since our transition kernel K is depending on the iterates, we will have a drift between the Poisson solution time step and the required term to build the Martingale Difference. In this proof we first start by introducing the core idea of the Poisson equation, then we derive each component individually to get their respective bounds. Moreover, through this lemma we bound images of the functions g and $\bar{g}$ by continuity and compactness and not using A. 4 and we use the same upper bound constant M_g . Under assumption A.2, by [12, Definition 21.2.1] and [1, Lemma 8] v_f is a solution of the following Poisson equation:

$$v_f(x_k, y_k, \xi_{k+1}) - \sum_{\xi' \in \Xi} K_{(x_k, y_k)}(\xi_{k+1}, \xi') v_f(x_k, y_k, \xi') = f(x_k, y_k, \xi_{k+1}) - \bar{f}(x_k, y_k) \quad (54)$$

In the following, we use the notation, with $K_k \triangleq K_{(x_k, y_k)}$

$$\mathbb{E}_{\xi' \sim K_k(\xi_{k+1}, \cdot)}[v_f(x_k, y_k, \xi')] \triangleq \sum_{\xi' \in \Xi} K_{(x_k, y_k)}(\xi_{k+1}, \xi') v_f(x_k, y_k, \xi').$$

It is shown in [1, Lemma 8] that v_f is Lipschitz-continuous. Our goal from introducing this solution is to be able to write this decomposition:

$$\sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, \psi_j \rangle = \sum_{j=0}^k \text{MDS} + \text{Telescoping Term} + \text{Bounded Term} + \text{Kernel Drift}$$

First, we start by substituting (54) in the sum:

$$\sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, \psi_j \rangle = \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, v_f(x_j, y_j, \xi_{j+1}) - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] \rangle$$

We notice that the term $\mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')]$ does not make a Martingale Difference with the term $v_f(x_j, y_j, \xi_{j+1})$ since ξ' is sampled at the step $j + 2$, we add and subtract $v_f(x_j, y_j, \xi_{j+2})$:

$$\begin{aligned} v_f(x_j, y_j, \xi_{j+1}) - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] &= v_f(x_j, y_j, \xi_{j+2}) - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] \\ &\quad + v_f(x_j, y_j, \xi_{j+1}) - v_f(x_j, y_j, \xi_{j+2}) \end{aligned}$$

However, this is not enough since $\xi_{j+2} \sim K_{(x_{j+1}, y_{j+1})}(\xi_{j+1}, \cdot)$ and not $K_{(x_j, y_j)}(\xi_{j+1}, \cdot)$. To obtain a Martingale Difference we add and subtract $\mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')]$. Finally, to use

Lipschitzness of v_f and get a telescoping term we add and subtract $v_f(x_{j+1}, y_{j+1}, \xi_{j+2})$. This will give us the following decomposition of the Markovian noise deviation

$$\langle \hat{x}_j, \psi_j \rangle = \left\langle \hat{x}_j, v_f(x_j, y_j, \xi_{j+2}) - \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] \right\rangle \quad (55)$$

$$+ v_f(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_f(x_j, y_j, \xi_{j+2}) \quad (56)$$

$$+ v_f(x_j, y_j, \xi_{j+1}) - v_f(x_{j+1}, y_{j+1}, \xi_{j+2}) \quad (57)$$

$$+ \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] \rangle \quad (58)$$

the first term (55) is an MDS, so it reduces to zero under expectation:

$$\mathbb{E} \left[\sum_{j=0}^k (1 - \alpha \mu_f)^{k-j} \langle \hat{x}_j, v_f(x_j, y_j, \xi_{j+2}) - \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] \rangle \right] = 0$$

The second term (56) scales linearly with the sum of step-sizes, since f and K are Lipschitz, so v_f will be L_{v_f} -Lipschitz too for some positive constant L_{v_f} and this is due to Lemma 1. So we can bound (56) by:

$$\begin{aligned} |(56)| &\leq \|\hat{x}_j\| \|v_f(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_f(x_j, y_j, \xi_{j+2})\| \\ &= \|x_j - h(y_j)\| \|v_f(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_f(x_j, y_j, \xi_{j+2})\| \\ &\leq L_{v_f} \|x_j - h(y_j)\| (\|x_{j+1} - x_j\| + \|y_{j+1} - y_j\|) \\ &\leq L_{v_f} (\|x_j\| + \|h(y_j)\|) (\|x_{j+1} - x_j\| + \|y_{j+1} - y_j\|) \\ &\leq L_{v_f} (M_S + M_h) (\|\alpha f(x_j, y_j, \xi_{j+1})\| + \|\beta g(x_j, y_j, \xi_{j+1})\|) \\ &\leq L_{v_f} (M_S + M_h) (\alpha M_f + \beta M_g) \end{aligned}$$

For brevity we set $z_j = \langle \hat{x}_j, v_f(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_f(x_j, y_j, \xi_{j+2}) \rangle$. Now we apply our bound on the sum to have

$$\begin{aligned} \left| \alpha \sum_{j=0}^k (1 - \alpha \mu_f)^{k-j} z_j \right| &\leq \alpha \sum_{j=0}^k (1 - \alpha \mu_f)^{k-j} |z_j| \\ &\leq \alpha \sum_{j=0}^k (1 - \alpha \mu_f)^{k-j} L_{v_f} (M_S + M_h) (\alpha M_f + \beta M_g) \\ &= \alpha L_{v_f} (M_S + M_h) (\alpha M_f + \beta M_g) \left(\frac{1 - (1 - \alpha \mu_f)^{k+1}}{\mu_f \alpha} \right) \\ &\leq \frac{L_{v_f} (M_S + M_h)}{\mu_f} (\alpha M_f + \beta M_g) \end{aligned}$$

To see why the term (58) is bounded we use our assumptions A 2 and A 3 about Lipschitzness of K and f

$$\begin{aligned} (58) &= \left\| \sum_{\xi' \in \Xi} (K_{j+1}(\xi_{j+1}, \xi') - K_j(\xi_{j+1}, \xi')) v_f(x_j, y_j, \xi') \right\| \\ &\leq \sum_{\xi' \in \Xi} \|K_{j+1}(\xi_{j+1}, \xi') - K_j(\xi_{j+1}, \xi')\| \|v_f(x_j, y_j, \xi')\| \\ &\leq \sum_{\xi' \in \Xi} L_K (\|x_{j+1} - x_j\| + \|y_{j+1} - y_j\|) M_\xi \\ &\leq L_K M_\xi (\alpha M_f + \beta M_g) |\Xi| \end{aligned}$$

which gives the bound

$$\begin{aligned} \alpha \left| \sum_{j=0}^k (1 - \mu_f \alpha)^{k-j} \langle \hat{x}_j, \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_f(x_j, y_j, \xi')] \rangle \right| \\ \leq \frac{|\Xi| M_S L_K M_\xi (\alpha M_f + \beta M_g)}{\mu_f} \end{aligned}$$

Finally, we handle the third term (57) which is supposed to be a telescoping term inside the sum:

$$\alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, v_f(x_j, y_j, \xi_{j+1}) - v_f(x_{j+1}, y_{j+1}, \xi_{j+2}) \rangle \quad (59)$$

This is not perfectly telescoping. This hurdle emerges from two quantities: first, in the inner product itself we have $\hat{x}_j$ which depends on j and second, the weights $(1 - \alpha\mu_f)^{k-j}$ also depend on j . To get over this, we will proceed in two steps. First, we fix the telescoping term inside the inner product by using the update rule in (1). Due to this step, we will have a remainder term dependent on the dynamics of x_j and controlled by α . The second step, is to handle the weights to have our telescoping terms. These two steps, will introduce remainders that we will need to control. Now we detail every step, so we start with the inner product and we set $v_j = v_f(x_j, y_j, \xi_{j+1})$ so we can easily write the computations:

$$\begin{aligned} \langle \hat{x}_j, v_{j+1} - v_j \rangle &= \langle x_j - h(y_j), v_{j+1} \rangle - \langle \hat{x}_j, v_j \rangle \\ &= \langle x_{j+1} + \alpha f(x_j, y_j, \xi_{j+1}) - h(y_j), v_{j+1} \rangle - \langle \hat{x}_j, v_j \rangle \\ &= \langle \hat{x}_{j+1} + \alpha f(x_j, y_j, \xi_{j+1}) + \Delta_j^h, v_{j+1} \rangle - \langle \hat{x}_j, v_j \rangle \\ &= \langle \hat{x}_{j+1}, v_{j+1} \rangle - \langle \hat{x}_j, v_j \rangle + \langle \alpha f(x_j, y_j, \xi_{j+1}) + \Delta_j^h, v_{j+1} \rangle \end{aligned}$$

where $\Delta_j^h = h(y_{j+1}) - h(y_j)$. Now, we have a telescoping term plus a remainder. Substituting the last line into (59):

$$\begin{aligned} (59) &= \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} (\langle \hat{x}_{j+1}, v_{j+1} \rangle - \langle \hat{x}_j, v_j \rangle) \\ &\quad + \alpha^2 \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle f(x_j, y_j, \xi_{j+1}), v_{j+1} \rangle \\ &\quad + \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \Delta_j^h, v_{j+1} \rangle \end{aligned} \quad (60)$$

Starting with the first term (60). Because of the weights in the sum $(1 - \alpha\mu_f)^{k-j}$ we can't directly have a telescoping sum. An extra step has to be done for this term to be able to make it telescope with the cost of introducing new remainders. Hopefully, these remainders can be controlled. We proceed by:

$$\begin{aligned} (60) &= \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_{j+1}, v_{j+1} \rangle - \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, v_j \rangle \\ &= \alpha(1 - \alpha\mu_f) \sum_{j=0}^k (1 - \alpha\mu_f)^{k-(j+1)} \langle \hat{x}_{j+1}, v_{j+1} \rangle - \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, v_j \rangle \\ &= \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-(j+1)} \langle \hat{x}_{j+1}, v_{j+1} \rangle - \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, v_j \rangle \\ &\quad - \alpha^2 \mu_f \sum_{j=0}^k (1 - \alpha\mu_f)^{k-(j+1)} \langle \hat{x}_{j+1}, v_{j+1} \rangle \\ &= \alpha(1 - \alpha\mu_f)^{-1} \langle \hat{x}_{k+1}, v_{k+1} \rangle - \alpha(1 - \alpha\mu_f)^k \langle \hat{x}_0, v_0 \rangle \\ &\quad - \alpha^2 \mu_f \sum_{j=0}^k (1 - \alpha\mu_f)^{k-(j+1)} \langle \hat{x}_{j+1}, v_{j+1} \rangle \end{aligned}$$

where in the second equality we factored out $1 - \alpha\mu_f$ to make the inside weight match the weight in the other sum. In the last line, we removed the telescoping terms to end up with the final three terms.

Now, into (59) we apply absolute value to get:

$$|(59)| \leq \left| \alpha \left((1 - \alpha\mu_f)^{-1} \langle \hat{x}_{k+1}, v_{k+1} \rangle - (1 - \alpha\mu_f)^k \langle \hat{x}_0, v_0 \rangle \right) \right| \quad (61)$$

$$+ \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} |\langle \Delta_j^h, v_{j+1} \rangle| \quad (62)$$

$$+ \alpha^2 \mu_f \sum_{j=0}^k (1 - \alpha\mu_f)^{k-(j+1)} |\langle \hat{x}_{j+1}, v_{j+1} \rangle| \quad (63)$$

$$+ \alpha^2 \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} |\langle f(x_j, y_j, \xi_{j+1}), v_{j+1} \rangle| \quad (64)$$

We bound each term independently. Starting with the first term (61):

$$(61) \leq \alpha \frac{M_\xi}{1 - \alpha\mu_f} + \alpha M_\xi = \alpha M_\xi \left(\frac{2 - \alpha\mu_f}{1 - \alpha\mu_f} \right)$$

the second term (62)

$$\begin{aligned} (62) &\leq \alpha \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \|\Delta_j^h\| \|v_{j+1}\| \leq \alpha \beta M_\xi M_h \sum_{j=0}^k (1 - \alpha\mu_f)^j \\ &= \alpha \beta M_\xi M_h \frac{1 - (1 - \alpha\mu_f)^{k+1}}{\alpha\mu_f} \leq \beta \frac{M_\xi M_h}{\mu_f} \end{aligned}$$

the third term (63)

$$\begin{aligned} (63) &\leq \alpha^2 \mu_f M_\xi \sum_{j=0}^k (1 - \alpha\mu_f)^{k-(j+1)} = \alpha^2 \mu_f M_\xi (1 - \alpha\mu_f)^{-1} \sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \\ &\leq \alpha \frac{M_\xi}{1 - \alpha\mu_f} \end{aligned}$$

and the last term (64)

$$(64) \leq \alpha^2 M_\xi M_f \sum_{j=0}^k (1 - \alpha\mu_f)^j \leq \alpha \frac{M_\xi M_f}{\mu_f}$$

After bounding each term individually, using bounds from Lemma 1, we apply these bounds on the main sum (59)

$$|(59)| \leq \alpha M_\xi \left(\frac{3 - \alpha\mu_f}{1 - \alpha\mu_f} + \frac{M_f}{\mu_f} \right) + \beta \frac{M_\xi M_h}{\mu_f}$$

Since expectation preserves order, we apply it on both sides to get the final result:

$$\begin{aligned} \left| \alpha \mathbb{E} \left[\sum_{j=0}^k (1 - \alpha\mu_f)^{k-j} \langle \hat{x}_j, \psi_j \rangle \right] \right| &\leq \alpha \left(M_\xi \left(\frac{3 - \alpha\mu_f}{1 - \alpha\mu_f} + \frac{M_f(1 + |\Xi| M_S L_K)}{\mu_f} \right) + \frac{L_{v_f} M_f (M_S + M_h)}{\mu_f} \right) \\ &\quad + \beta \left(\frac{M_\xi M_h + L_{v_f} M_g (M_S + M_h) + |\Xi| M_S L_K M_\xi M_g}{\mu_f} \right) \end{aligned}$$

which concludes the proof of the first part.

As above, let v_g be the solution of the Poisson equation:

$$g(x_j, y_j, \xi_{j+1}) - \bar{g}(x_j, y_j) = v_g(x_j, y_j, \xi_{j+1}) - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)} [v_g(x_j, y_j, \xi')] \quad (65)$$

we rewrite:

$$\langle \hat{y}_j, \phi_j \rangle = \langle \hat{y}_j, v_g(x_j, y_j, \xi_{j+2}) - \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)} [v_g(x_j, y_j, \xi')] \rangle \quad (66)$$

$$+ v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_g(x_j, y_j, \xi_{j+2}) \quad (67)$$

$$+ v_g(x_j, y_j, \xi_{j+1}) - v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) \quad (68)$$

$$+ \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)} [v_g(x_j, y_j, \xi')] - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)} [v_g(x_j, y_j, \xi')] \quad (69)$$

the first term (66) is 0 under expectation because it is a Martingale Difference. The last term (69) is treated in the same way as in (58) and it has the following bound

$$\begin{aligned} \beta \left| \sum_{j=0}^k (1 - \mu_g \beta)^{k-j} \langle \hat{y}_j, \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)} [v_g(x_j, y_j, \xi')] - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)} [v_g(x_j, y_j, \xi')] \rangle \right| \\ \leq \frac{L_K M_\xi M_S |\Xi| (\alpha M_f + \beta M_g)}{\mu_g} \end{aligned}$$

The second term (67) is of order $O(\alpha + \beta)$:

$$\begin{aligned} |(67)| &\leq \|\hat{y}_j\| \|v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_g(x_j, y_j, \xi_{j+2})\| \\ &\leq L_{v_g} M_S (\|y_{j+1} - y_j\| + \|x_{j+1} - x_j\|) \\ &\leq L_{v_g} M_S (\|\alpha f(x_j, y_j, \xi_{j+1})\| + \|\beta g(x_j, y_j, \xi_{j+1})\|) \\ &\leq L_{v_g} M_S (\alpha M_f + \beta M_g) \end{aligned}$$

where in the third line we have used the definition of the updates and in the last line we have used the bounds from Lemma 1. So we have:

$$\begin{aligned} \left| \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle \hat{y}_j, v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_g(x_j, y_j, \xi_{j+2}) \rangle \right| &\leq \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} |(67)| \\ &\leq \frac{L_{v_g} M_S (\alpha M_f + \beta M_g)}{\mu_g} \end{aligned}$$

where in the first inequality we have used the triangular inequality. Moving to the third term (68). Let $w_j = v_g(x_j, y_j, \xi_{j+1})$, the term in the sum becomes:

$$\begin{aligned} -(68) &= \langle \hat{y}_j, w_{j+1} - w_j \rangle = \langle \hat{y}_j, w_{j+1} \rangle - \langle \hat{y}_j, w_j \rangle \\ &= \langle \hat{y}_{j+1} + \beta g(x_j, y_j, \xi_{j+1}), w_{j+1} \rangle - \langle \hat{y}_j, w_j \rangle \\ &= \langle \hat{y}_{j+1}, w_{j+1} \rangle - \langle \hat{y}_j, w_j \rangle + \langle \beta g(x_j, y_j, \xi_{j+1}), w_{j+1} \rangle \end{aligned}$$

We substitute in the sum:

$$\begin{aligned} \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle \hat{y}_j, w_{j+1} - w_j \rangle &= \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} (\langle \hat{y}_{j+1}, w_{j+1} \rangle - \langle \hat{y}_j, w_j \rangle) \\ &\quad + \beta^2 \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle g(x_j, y_j, \xi_{j+1}), w_{j+1} \rangle \end{aligned} \quad (70)$$

treating the first term (70):

$$\begin{aligned} (70) &= \beta (1 - \beta \mu_g)^{-1} \langle \hat{y}_{k+1}, w_{k+1} \rangle - \beta (1 - \beta \mu_g)^k \langle \hat{y}_0, w_0 \rangle \\ &\quad - \beta^2 \mu_g \sum_{j=0}^k (1 - \beta \mu_g)^{k-(j+1)} \langle \hat{y}_{j+1}, w_{j+1} \rangle \end{aligned}$$

substituting:

$$\left| \beta \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle \hat{y}_j, w_{j+1} - w_j \rangle \right| = \beta \left| (1 - \beta \mu_g)^{-1} \langle \hat{y}_{k+1}, w_{k+1} \rangle - (1 - \beta \mu_g)^k \langle \hat{y}_0, w_0 \rangle \right| \quad (71)$$

$$+ \beta^2 \mu_g \left| \sum_{j=0}^k (1 - \beta \mu_g)^{k-(j+1)} \langle \hat{y}_{j+1}, w_{j+1} \rangle \right| \quad (72)$$

$$+ \beta^2 \left| \sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle g(x_j, y_j, \xi_{j+1}), w_{j+1} \rangle \right| \quad (73)$$

we bound the term (71):

$$(71) \leq \beta M_\xi \left(\frac{2 - \beta \mu_g}{1 - \beta \mu_g} \right)$$

the second term (72):

$$(72) \leq \beta \frac{M_\xi}{1 - \beta \mu_g}$$

and the last term (73):

$$(73) \leq \beta \frac{M_\xi M_g}{\mu_g}$$

where we have used the bounds from Lemma 1 as we have done in detail for the previous section regarding the fast iterate. The last bound can be written as:

$$\begin{aligned} \left| \beta \mathbb{E} \left[\sum_{j=0}^k (1 - \beta \mu_g)^{k-j} \langle \hat{y}_j, \phi_j \rangle \right] \right| &\leq \beta \left(\frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_g}{\mu_g} + M_\xi \left(\frac{3 - \beta \mu_g}{1 - \beta \mu_g} + \frac{M_g}{\mu_g} \right) \right) \\ &\quad + \alpha \frac{(L_{v_g} + M_\xi L_K |\Xi|) M_S M_f}{\mu_g} \end{aligned}$$

which concludes the proof. $\square$

The following two lemmas are describing a uniform upper bounds of the solutions of the Riccati equation (37) and Sylvester equation (40) for a range of values of $\frac{\beta}{\alpha}$.

Lemma 3. *Let $\varepsilon > 0$ and let $J_{yx} \in \mathbb{R}^{m \times n}$, $J_{xx} \in \mathbb{R}^{n \times n}$, $J_{yy} \in \mathbb{R}^{m \times m}$, $J_{xy} \in \mathbb{R}^{n \times m}$ with J_{xx} being invertible, for every $R_{ric} > 0$ we define $\varepsilon_0^*, \varepsilon_1^*$ as follows :*

$$\varepsilon_0^* = \frac{R_{ric}}{\|J_{xx}^{-1}\|_{op} (R_{ric} + \|U_0\|_{op}) (\|J_{yy}\|_{op} + (R_{ric} + \|U_0\|_{op}) \|J_{yx}\|_{op})}, \quad (74)$$

$$\varepsilon_1^* = \frac{1}{\|J_{xx}^{-1}\|_{op} (\|J_{yy}\|_{op} + 2 \|J_{yx}\|_{op} (R_{ric} + \|U_0\|_{op}))} \quad (75)$$

with $U_0 = J_{xx}^{-1} J_{xy}$. Then for every $\varepsilon < \min\{\varepsilon_0^*, \varepsilon_1^*\}$ there exists a matrix $U \in \mathbb{R}^{n \times m}$ that belongs to the closed ball $B_{R_{ric}}^{U_0} = \{U : \|U - U_0\|_{op} \leq R_{ric}\}$ that is the unique solution of the following Riccati equation:

$$\varepsilon U J_{yx} U + J_{xx} U - \varepsilon U J_{yy} = J_{xy} \quad (76)$$

Proof. We multiply the Riccati equation (76) by J_{xx}^{-1} , since we assumed it is invertible:

$$\varepsilon J_{xx}^{-1} U J_{yx} U + U - \varepsilon J_{xx}^{-1} U J_{yy} = \underbrace{J_{xx}^{-1} J_{xy}}_{\triangleq U_0}$$

which gives the fixed point equation:

$$U = U_0 + \varepsilon (J_{xx}^{-1} U J_{yy} - J_{xx}^{-1} U J_{yx} U) \quad (77)$$

We define T to be the operator of the fixed point equation. We define the closed ball:

$$B_{R_{ric}}^{U_0} = \{U : \|U - U_0\|_{op} \leq R_{ric}\}$$

First let's find ε_0^* such that $T(B_{R_{ric}}^{U_0}) \subseteq B_{R_{ric}}^{U_0}$. Let $M \in B_{R_{ric}}^{U_0}$:

$$T(M) = U_0 + \varepsilon (J_{xx}^{-1} M J_{yy} - J_{xx}^{-1} M J_{yx} M)$$

calculating the norm

$$\begin{aligned}
\|T(M) - U_0\|_{\text{op}} &= \varepsilon \|J_{xx}^{-1} M J_{yy} - J_{xx}^{-1} M J_{yx} M\|_{\text{op}} \\
&\leq \varepsilon (\|J_{xx}^{-1} M J_{yy}\|_{\text{op}} + \|J_{xx}^{-1} M J_{yx} M\|_{\text{op}}) \\
&= \varepsilon (\|J_{xx}^{-1} (M - U_0 + U_0) J_{yy}\|_{\text{op}} + \|J_{xx}^{-1} (M - U_0 + U_0) J_{yx} (M - U_0 + U_0)\|_{\text{op}}) \\
&\leq \varepsilon (\|J_{xx}^{-1}\|_{\text{op}} (\|M - U_0\|_{\text{op}} + \|U_0\|_{\text{op}}) \|J_{yy}\|_{\text{op}} \\
&\quad + \|J_{xx}^{-1}\|_{\text{op}} (\|M - U_0\|_{\text{op}} + \|U_0\|_{\text{op}})^2 \|J_{yx}\|_{\text{op}}) \\
&\leq \varepsilon (\|J_{xx}^{-1}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}}) \|J_{yy}\|_{\text{op}} + \|J_{xx}^{-1}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}})^2 \|J_{yx}\|_{\text{op}})
\end{aligned}$$

now we condition on ε to make $T(M) \in B_{R_{ric}}^{U_0}$, we need $\varepsilon \leq \varepsilon_0^*$ such that :

$$\varepsilon_0^* = \frac{R_{ric}}{\|J_{xx}^{-1}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}}) (\|J_{yy}\|_{\text{op}} + (R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}})}$$

Now we prove T is a contraction. Let $U_1, U_2 \in B_{R_{ric}}^{U_0}$, we calculate the norm of their difference:

$$\|T(U_1) - T(U_2)\|_{\text{op}} = \|\varepsilon (J_{xx}^{-1} U_1 J_{yy} - J_{xx}^{-1} U_1 J_{yx} U_1) - \varepsilon (J_{xx}^{-1} U_2 J_{yy} - J_{xx}^{-1} U_2 J_{yx} U_2)\|_{\text{op}}$$

We have:

$$U_1 A U_1 - U_2 A U_2 = U_1 A U_1 - U_1 A U_2 + U_1 A U_2 - U_2 A U_2 = U_1 A (U_1 - U_2) + (U_1 - U_2) A U_2$$

We replace:

$$\begin{aligned}
\|T(U_1) - T(U_2)\|_{\text{op}} &= \varepsilon \|J_{xx}^{-1} ((U_1 - U_2) J_{yy} - ((U_1 - U_2) J_{yx} U_2 + U_1 J_{yx} (U_1 - U_2)))\|_{\text{op}} \\
\|T(U_1) - T(U_2)\|_{\text{op}} &\leq \varepsilon \|J_{xx}^{-1}\|_{\text{op}} \|U_1 - U_2\|_{\text{op}} (\|J_{yy}\|_{\text{op}} + \|J_{yx}\|_{\text{op}} (\|U_1\|_{\text{op}} + \|U_2\|_{\text{op}}))
\end{aligned}$$

Bounding the norms:

$$\begin{aligned}
\|U_1\| &\leq \|U_1 - U_0\| + \|U_0\| \leq R_{ric} + \|U_0\|_{\text{op}} \\
\|U_2\| &\leq \|U_2 - U_0\| + \|U_0\| \leq R_{ric} + \|U_0\|_{\text{op}} \\
\|T(U_1) - T(U_2)\| &\leq \varepsilon \underbrace{\left(\|J_{xx}^{-1}\|_{\text{op}} (\|J_{yy}\|_{\text{op}} + 2 \|J_{yx}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}})) \right)}_{r(\varepsilon)} \|U_1 - U_2\|_{\text{op}}
\end{aligned}$$

To make $r(\varepsilon) < 1$ we need:

$$\varepsilon < \frac{1}{\|J_{xx}^{-1}\|_{\text{op}} (\|J_{yy}\|_{\text{op}} + 2 \|J_{yx}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}}))} = \varepsilon_1^*$$

For $\varepsilon < \min\{\varepsilon_0^*, \varepsilon_1^*\}$, the solution U of the Riccati equation is unique and is inside the ball $B_{R_{ric}}^{U_0}$. It can be obtained by successive application of T on U_0 . This shows that the uniform bound on $\|U\| \leq R_{ric} + \|U_0\|_{\text{op}}$ is proved. $\square$

Lemma 4. Let $\varepsilon > 0$ and let $J_{yx} \in \mathbb{R}^{m \times n}$, $J_{xx} \in \mathbb{R}^{n \times n}$, $J_{yy} \in \mathbb{R}^{m \times m}$, $J_{xy} \in \mathbb{R}^{n \times m}$ with J_{xx} being invertible, define $U_0 = J_{xx}^{-1} J_{xy}$, and for all $R_{ric} > 0$ there exists ε^* such that for all $\varepsilon < \varepsilon^*$ the matrix $U_\varepsilon \in \mathbb{R}^{n \times m}$ is bounded by $\|U_\varepsilon\|_{\text{op}} \leq R_{ric} + \|U_0\|_{\text{op}}$, we set $R_{syl} = 2 \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}$ and we define $\varepsilon_2^*, \varepsilon_3^*$:

$$\varepsilon_2^* = \frac{1}{\|J_{yy}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2(R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}}, \quad (78)$$

$$\varepsilon_3^* = \frac{1}{\|J_{xx}^{-1}\|_{\text{op}} (\|J_{yy}\|_{\text{op}} + 2 \|J_{yx}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}}))} \quad (79)$$

then for all $\varepsilon < \min\{\varepsilon^*, \varepsilon_2^*, \varepsilon_3^*\}$ there exists a matrix $X \in \mathbb{R}^{m \times n}$ that belongs to the closed ball $B_{\varepsilon R_{syl}} = \{X : \|X\|_{\text{op}} \leq \varepsilon R_{syl}\}$ which is the unique solution of the following Sylvester equation:

$$X(J_{xx} + \varepsilon U_\varepsilon J_{yx}) - \varepsilon (J_{yy} - J_{yx} U_\varepsilon) X = -\varepsilon J_{yx}$$

Proof. To make the notation lighter we drop ε from U_ε such that $U = U_\varepsilon$. We multiply the Sylvester equation by J_{xx}^{-1} on the right:

$$X + \varepsilon XU J_{yx} J_{xx}^{-1} - \varepsilon J_{yy} X J_{xx}^{-1} + \varepsilon J_{yx} U X J_{xx}^{-1} = -\varepsilon J_{yx} J_{xx}^{-1}$$

which gives the fixed point equation:

$$X = \varepsilon \underbrace{(J_{yy} X J_{xx}^{-1} - XU J_{yx} J_{xx}^{-1} - J_{yx} U X J_{xx}^{-1} - J_{yx} J_{xx}^{-1})}_{T(X)}$$

We define the ball $B_{\varepsilon R_{syl}} = \{X : \|X\|_{\text{op}} \leq \varepsilon R_{syl}\}$ where $R_{syl} = 2 \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}$ (This choice is explained in the next lines). Let $M \in B_{\varepsilon R_{syl}}$ and $U \in B_{R_{ric}}^{U_0}$:

$$\begin{aligned} \|T(M)\|_{\text{op}} &= \varepsilon \|J_{yy} M J_{xx}^{-1} - MU J_{yx} J_{xx}^{-1} - J_{yx} U M J_{xx}^{-1} - J_{yx} J_{xx}^{-1}\|_{\text{op}} \\ &\leq \varepsilon (\|J_{yy}\|_{\text{op}} \|M\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2 \|M\|_{\text{op}} \|U\|_{\text{op}} \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}) \\ &= \varepsilon (\|M\|_{\text{op}} (\|J_{yy}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2 \|U\|_{\text{op}} \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}) + \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}) \\ &\leq \varepsilon R_{syl} \frac{\|J_{yy}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2(R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}}{R_{syl}} \end{aligned}$$

a first condition to set is that

$$\frac{\varepsilon R_{syl} (\|J_{yy}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2(R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}) + \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}}{R_{syl}} \leq 1$$

to make the operator T maps into $B_{\varepsilon R_{syl}}$ which is equivalent to

$$\varepsilon < \frac{R_{syl} - \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}}{R_{syl} (\|J_{yy}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2(R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}})}$$

where we extract two conditions, the first is that $R_{syl} > \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}$ since ε is strictly positive, here we justify the choice of the radius $R_{syl} = 2 \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}$, once we have done that we can have a condition on $\varepsilon \leq \varepsilon_2^*$ to ensure that image of T stays in the ball $B_{\varepsilon R_{syl}}$ such that:

$$\varepsilon_2^* = \frac{1}{\|J_{yy}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}} + 2(R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}} \|J_{xx}^{-1}\|_{\text{op}}}$$

Now we move to the contraction part of T . Let $X_1, X_2 \in B_{\varepsilon R_{syl}}$:

$$\begin{aligned} \|T(X_1) - T(X_2)\|_{\text{op}} &\leq \varepsilon \|J_{xx}^{-1}\|_{\text{op}} (\|J_{yy}\|_{\text{op}} + \|U J_{yx}\|_{\text{op}} + \|J_{yx} U\|_{\text{op}}) \|X_1 - X_2\|_{\text{op}} \\ &\leq \varepsilon \|J_{xx}^{-1}\|_{\text{op}} (\|J_{yy}\|_{\text{op}} + 2 \|J_{yx}\|_{\text{op}} (R_{ric} + \|U_0\|_{\text{op}})) \|X_1 - X_2\|_{\text{op}} \end{aligned}$$

The condition for the contraction is the same as the Riccati equation

$$\varepsilon_3^* = \varepsilon_1^*$$

then by Banach fixed point theorem, the map T has a fixed point solution X that is unique inside the ball $B_{\varepsilon R_{syl}}$ for every $\varepsilon < \min\{\varepsilon_2^*, \varepsilon_3^*\}$. $\square$

In the next lemmas, we extract contractions of the bias dynamics using Lyapunov norms suitable for a given range of step sizes. The choice of the Lyapunov equations (80) and (83) appears in [22, Chapter 7] to study the stability of the equivalent singular perturbation system. The bound on the L-norm of $I - \gamma A$ is common and appears in [17, 15, 26, 21]. However, in our case we deal with nonlinear systems so we have extra perturbations emerging from the coupling between the two variables, so we extend the classical way to prove the contraction property under the suitable Lyapunov norms to account for the new added perturbations which are dependent on the solutions of the Riccati and the Sylvester equations that we showed previously to exist and to be bounded under certain assumptions on the step sizes.

Lemma 5. Let $-J_{xx}$ be a Hurwitz matrix in $\mathbb{R}^{d \times d}$, $J_{yx} \in \mathbb{R}^{n \times d}$ such that $\|J_{yx}\|_{op} \leq M_J$ for some positive constant M_J , $L \in \mathbb{R}^{d \times d}$ a symmetric positive definite matrix a solution to the following Lyapunov equation:

$$J_{xx}^T L + L J_{xx} = I. \quad (80)$$

Let $R_{ric} > 0$, there exists $\alpha^* = \alpha^*(R_{ric})$ where $\alpha^*(R_{ric})$ is defined by:

$$\alpha^*(R_{ric}) = \frac{1}{4 \|L\|_{op} (\|J_{xx}\|_L^2 + \varepsilon_4^{*2} \kappa(L)(R_{ric} + \|U_0\|_{op})^2 \|J_{yx}\|_L^2)} \quad (81)$$

such that for every $0 < \alpha \leq \alpha^*$ and for every β such that $\frac{\beta}{\alpha} < \min\{\varepsilon_0^*, \varepsilon_1^*, \varepsilon_5^*\}$ where $\varepsilon_0^*, \varepsilon_1^*, \varepsilon_5^*$ depend on R_{ric} and are fully defined in (74), (75), (82), we define the matrix $U_{\frac{\beta}{\alpha}} \in \mathbb{R}^{d \times n}$ to be bounded by $\|U_{\frac{\beta}{\alpha}}\| \leq R_{ric} + \|U_0\|_{op}$, then we have:

$$\left\| I - \alpha J_{xx} - \beta U_{\frac{\beta}{\alpha}} J_{yx} \right\|_L \leq e^{-c\alpha}$$

where $c = (8 \|L\|_{op})^{-1}$.

Proof. Under the Lyapunov norm:

$$\begin{aligned} \|I - \alpha J_{xx} - \beta U J_{yx}\|_L^2 &= \sup_{\|u\|_L=1} u^T (I - \alpha J_{xx} - \beta U J_{yx})^T L (I - \alpha J_{xx} - \beta U J_{yx}) u \\ &= \sup_{\|u\|_L=1} \{u^T (L - \alpha J_{xx}^T L - \beta (U J_{yx})^T L - \alpha L J_{xx} + \alpha^2 J_{xx}^T L J_{xx} \\ &\quad + \alpha \beta J_{yx}^T U^T L J_{xx} - \beta L U J_{yx} + \alpha \beta J_{xx}^T L U J_{yx} + \beta^2 J_{yx}^T U^T L U J_{yx}) u\} \\ &= \sup_{\|u\|_L=1} \{u^T L u - \alpha u^T (J_{xx}^T L + L J_{xx}) u + \alpha^2 u^T J_{xx}^T L J_{xx} u \\ &\quad - \beta u^T (\underbrace{J_{yx}^T U^T L + L U J_{yx}}_{A_1}) u + \alpha \beta u^T (\underbrace{J_{yx}^T U^T L J_{xx} + J_{xx}^T L U J_{yx}}_{A_2}) u \\ &\quad + \beta^2 u^T (J_{yx}^T U^T L U J_{yx}) u\} \\ &= 1 + \sup_{\|u\|_L=1} \{-\alpha u^T u + \alpha^2 u^T J_{xx}^T L J_{xx} u - \beta u^T A_1 u + \alpha \beta u^T A_2 u \\ &\quad + \beta^2 u^T (J_{yx}^T U^T L U J_{yx}) u\} \\ &\leq 1 + \sup_{\|u\|_L=1} \{-\alpha u^T u\} + \alpha^2 \|J_{xx}\|_L^2 + \beta^2 \|U J_{yx}\|_L^2 + \alpha \beta \left\| L^{-1/2} A_2 L^{-1/2} \right\|_{op} \\ &\quad + \beta \sup_{\|u\|_L=1} -u^T A_1 u \\ &\leq 1 - \alpha (2 \|L\|_{op})^{-1} + \alpha^2 \|J_{xx}\|_L^2 + \beta^2 \|U J_{yx}\|_L^2 \\ &\quad - \alpha (2 \|L\|_{op})^{-1} + \alpha \beta \left\| L^{-1/2} A_2 L^{-1/2} \right\|_{op} + \beta \sup_{\|u\|_L=1} -u^T A_1 u \end{aligned}$$

We have used the fact that $\inf_{\|u\|_L=1} \{u^T u\} = \|L\|_{op}^{-1} = (2 \|L\|_{op})^{-1} + (2 \|L\|_{op})^{-1}$. We will use the symmetry of A_1 to bound the last term such that

$$\beta \sup_{\|u\|_L=1} -u^T A_1 u \leq \beta \sup_{\|u\|_L=1} |u^T A_1 u| = \beta \sup_{\|z\|=1} |z^T L^{-1/2} A_1 L^{-1/2} z| = \beta \left\| L^{-1/2} A_1 L^{-1/2} \right\|_{op}$$

we have

$$\begin{aligned} \|I - \alpha J_{xx} - \beta U J_{yx}\|_L^2 &\leq 1 - \alpha (2 \|L\|_{op})^{-1} + \alpha^2 (\|J_{xx}\|_L^2 + \frac{\beta^2}{\alpha^2} \|U J_{yx}\|_L^2) \\ &\quad - \alpha (2 \|L\|_{op})^{-1} + \beta \alpha \left\| L^{-1/2} A_2 L^{-1/2} \right\|_{op} + \beta \left\| L^{-1/2} A_1 L^{-1/2} \right\|_{op} \end{aligned}$$

Since U is a solution of the Riccati equation (37) then by Lemma 3 for all $R_{ric} > 0$ there exists ε_0^* and ε_1^* defined in (74) and (75), respectively, such that for $\frac{\beta}{\alpha} < \varepsilon_4^* \triangleq \min\{\varepsilon_0^*, \varepsilon_1^*\}$ we have $\|U\| \leq R_{ric} + \|U_0\|_{\text{op}}$. We apply the result of lemma:

$$\begin{aligned} \|I - \alpha J_{xx} - \beta U J_{yx}\|_{\text{L}}^2 &\leq 1 - \alpha(2\|L\|_{\text{op}})^{-1} + \alpha^2(\|J_{xx}\|_{\text{L}}^2 + \varepsilon_4^{*2}\kappa(L)(R_{ric} + \|U_0\|_{\text{op}})^2\|J_{yx}\|_{\text{L}}^2) \\ &\quad - \alpha(2\|L\|_{\text{op}})^{-1} + \beta\alpha\left\|L^{-1/2}A_2L^{-1/2}\right\|_{\text{op}} + \beta\left\|L^{-1/2}A_1L^{-1/2}\right\|_{\text{op}} \\ &\leq 1 - \alpha(2\|L\|_{\text{op}})^{-1} + \alpha^2(\|J_{xx}\|_{\text{L}}^2 + \varepsilon_4^{*2}\kappa(L)(R_{ric} + \|U_0\|_{\text{op}})^2\|J_{yx}\|_{\text{L}}^2) \\ &\quad - \alpha(2\|L\|_{\text{op}})^{-1} + 2\beta\alpha\kappa(L)(\|J_{yx}\|_{\text{op}}\|J_{xx}\|_{\text{op}}(R_{ric} + \|U_0\|_{\text{op}})) \\ &\quad + 2\beta\kappa(L)^{1/2}(M_J(R_{ric} + \|U_0\|_{\text{op}})) \end{aligned}$$

First, we look for values of α such that the first line is less than one to get the contraction property. This is true for the choice of $\alpha \leq \alpha^*(R_{ric})$ defined in the body of the lemma so we have:

$$\begin{aligned} \|I - \alpha J_{xx} - \beta U J_{yx}\|_{\text{L}}^2 &\leq 1 - c'\alpha \\ &\quad - \alpha(2\|L\|_{\text{op}})^{-1} + 2\beta\alpha^*\kappa(L)(\|J_{yx}\|_{\text{op}}\|J_{xx}\|_{\text{op}}(R_{ric} + \|U_0\|_{\text{op}})) \\ &\quad + 2\beta\kappa(L)^{1/2}(M_J(R_{ric} + \|U_0\|_{\text{op}})) \end{aligned}$$

with $c' = (4\|L\|_{\text{op}})^{-1}$. Second, we need the term in the last two lines to be negative so this will imply a condition of the type:

$$\frac{\beta}{\alpha} \leq \varepsilon_5^* \triangleq \frac{1}{2\|L\|_{\text{op}}(R_{ric} + \|U_0\|_{\text{op}})(2\alpha^*\kappa(L)\|J_{yx}\|_{\text{op}}\|J_{xx}\|_{\text{op}} + 2\kappa(L)^{1/2}M_J)} \quad (82)$$

Once this is established we have:

$$\|I - \alpha J_{xx} - \beta U J_{yx}\|_{\text{L}}^2 \leq 1 - c'\alpha \leq e^{-c\alpha}$$

this implies that:

$$\|I - \alpha J_{xx} - \beta U J_{yx}\|_{\text{L}} \leq e^{-\frac{c'}{2}\alpha}$$

which establishes the proof. $\square$

Lemma 6. Let $-\Delta$ be a Hurwitz matrix in $\mathbb{R}^{n \times n}$, $J_{yx} \in \mathbb{R}^{n \times m}$ such that $\|J_{yx}\|_{\text{op}} \leq M_J$ for some positive constant M_J , $G \in \mathbb{R}^{n \times n}$ a symmetric positive definite matrix a solution to the following Lyapunov equation:

$$\Delta^T G + G \Delta = I. \quad (83)$$

Let $R_{ric} = \frac{1}{4\|G\|_{\text{op}}\sqrt{\kappa(G)}M_J}$, we define $\beta^* = \beta^*(R_{ric})$ where $\beta^*(R_{ric})$ is defined by:

$$\beta^* = \frac{1}{4\|G\|_{\text{op}}(2\kappa(G)\|\Delta\|_{\text{op}}M_JR_{ric} + \|\Delta\|_G^2 + \kappa(G)M_J^2R_{ric}^2)} \quad (84)$$

for all $\alpha > 0$ and $\beta \leq \beta^*$ with $\frac{\beta}{\alpha} < \min\{\varepsilon_0^*, \varepsilon_1^*\}$ where $\varepsilon_0^*, \varepsilon_1^*$ are dependent on R_{ric} and defined in (74) and (75), the matrix $U_{\frac{\beta}{\alpha}} \in B_{R_{ric}}^{U_0}$ is a solution of the Riccati equation defined in (76) with $\varepsilon = \frac{\beta}{\alpha}$ and $U_0 = J_{xx}^{-1}J_{xy}$, and we have:

$$\left\|I - \beta(\Delta - J_{yx}(U_{\frac{\beta}{\alpha}} - U_0))\right\|_G \leq e^{-d\beta}$$

where $d = (8\|G\|_{\text{op}})^{-1}$.

Proof. Taking the Lyapunov norm

$$\begin{aligned}
\|I - \beta(\Delta - J_{yx}(U - U_0))\|_G^2 &= \sup_{\|u\|_G=1} \{u^T(I - \beta\Delta - \beta J_{yx}(U - U_0))^T G(I - \beta\Delta - \beta J_{yx}(U - U_0))u\} \\
&= \sup_{\|u\|_G=1} \{u^T G u - \beta u^T G \Delta u - \beta u^T G J_{yx}(U - U_0)u - \beta u^T \Delta^T G u \\
&\quad + \beta^2 u^T \Delta^T G \Delta u + \beta^2 u^T \Delta^T G J_{yx}(U - U_0)u \\
&\quad - \beta u^T (U - U_0)^T J_{yx}^T G u + \beta^2 u^T (U - U_0)^T J_{yx}^T G \Delta u \\
&\quad + \beta^2 u^T (U - U_0)^T J_{yx}^T G J_{yx}(U - U_0)u\} \\
&= 1 + \sup_{\|u\|_G=1} \{-\beta u^T u - 2\beta u^T G J_{yx}(U - U_0)u + 2\beta^2 u^T \Delta^T G J_{yx}(U - U_0)u \\
&\quad + \beta^2 u^T \Delta^T G \Delta u + \beta^2 u^T (U - U_0)^T J_{yx}^T G J_{yx}(U - U_0)u\} \\
&\leq 1 - \beta \|G\|_{\text{op}}^{-1} + 2\beta \sup_{\|u\|_G=1} \{-u^T G J_{yx}(U - U_0)u\} \\
&\quad + 2\beta^2 \sup_{\|u\|_G=1} \{u^T \Delta^T G J_{yx}(U - U_0)u\} + \beta^2 \|\Delta\|_G^2 + \beta^2 \|J_{yx}(U - U_0)\|_G^2 \\
&\leq 1 - \beta \|G\|_{\text{op}}^{-1} + 2\beta \left\| G^{1/2} J_{yx}(U - U_0) G^{-1/2} \right\|_{\text{op}} \\
&\quad + 2\beta^2 \left\| G^{-1/2} \Delta^T G J_{yx}(U - U_0) G^{-1/2} \right\|_{\text{op}} + \beta^2 \|\Delta\|_G^2 + \beta^2 \|J_{yx}(U - U_0)\|_G^2 \\
&\leq 1 - \beta \|G\|_{\text{op}}^{-1} + 2\beta \kappa(G)^{1/2} M_J \|U - U_0\|_{\text{op}} \\
&\quad + 2\beta^2 \left\| G^{-1/2} \Delta^T G J_{yx}(U - U_0) G^{-1/2} \right\|_{\text{op}} + \beta^2 \|\Delta\|_G^2 + \beta^2 \|J_{yx}(U - U_0)\|_G^2
\end{aligned}$$

Here we set the radius of convergence of the Riccati equation to calibrate for the negative term in the first line of the last inequality above. We want:

$$2\kappa(G)^{1/2} M_J \|U - U_0\|_{\text{op}} \leq (2\|G\|_{\text{op}})^{-1}$$

which is equivalent to:

$$\|U - U_0\|_{\text{op}} \leq \frac{1}{4\|G\|_{\text{op}} \kappa(G)^{1/2} M_J} = R_{ric}$$

So under this condition we have

$$\|I - \beta(\Delta - J_{yx}(U - U_0))\|_G^2 \leq 1 - \beta(2\|G\|_{\text{op}})^{-1} + \beta^2(2\kappa(G)\|\Delta\|_{\text{op}} M_J R_{ric} + \|\Delta\|_G^2 + \kappa(G)M_J^2 R_{ric}^2)$$

So the contraction is established for $\beta \leq \beta^*$ where β^* is defined in the body of the lemma and for $d' = (4\|G\|_{\text{op}})^{-1}$ we have

$$\|I - \beta(\Delta - J_{yx}(U - U_0))\|_G^2 \leq 1 - d\beta \leq e^{-d'\beta}$$

this implies that:

$$\|I - \beta(\Delta - J_{yx}(U - U_0))\|_G \leq e^{-\frac{d'}{2}\beta}$$

□

In the next lemma we handle the norms of the sum of deviations appearing in the proof of Theorem 3.

Lemma 7. Taking the sequence $\{x_k, y_k\}_{k \geq 0}$ defined by (1) and under assumptions A.1-2 and B.1. Assume also that the step sizes α and β satisfy $\alpha < \alpha^*$, $\beta < \beta^*$ and $\frac{\beta}{\alpha} < \omega^*$ such that $\alpha^*, \beta^*, \omega^*$ are defined in (8),(9),(10) then, the term (48) is bounded by:

$$(48) \leq \alpha A_1 + \beta(A_2 + B_1) + \alpha^2 A_3 + \beta^2 B_2 + \alpha\beta(A_4 + B_3) + \frac{\beta^2}{\alpha} B_4$$

in addition the term (52) is bounded by:

$$(52) \leq \alpha(C_2 + D_4) + \beta(C_1 + D_1) + \alpha\beta(C_4 + D_3) + \beta^2(C_3 + D_2)$$

where the constants are independent of α and β and are detailed in the proof.

Proof. We start with bounding the first term:

$$\alpha \left\| \sum_{j=0}^k (I - \alpha J_{xx}^{\alpha, \beta})^{k-j} \mathbb{E} [\psi_j] \right\| \quad (85)$$

In Lemma 2 we have seen that there exists a function v_f the solution of the Poisson equation (54), which allowed us to write:

$$\mathbb{E} [\psi_s] = \mathbb{E} [v_f(x_s, y_s, \xi_{s+1}) - \mathbb{E}_{\xi' \sim K_s(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')]]$$

we could write the term inside the expectation as:

$$\psi_s = v_f(x_s, y_s, \xi_{s+2}) - \mathbb{E}_{\xi' \sim K_{s+1}(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')] \quad (86)$$

$$+ v_f(x_{s+1}, y_{s+1}, \xi_{s+2}) - v_f(x_s, y_s, \xi_{s+2}) \quad (87)$$

$$+ v_f(x_s, y_s, \xi_{s+1}) - v_f(x_{s+1}, y_{s+1}, \xi_{s+2}) \quad (88)$$

$$+ \mathbb{E}_{\xi' \sim K_{s+1}(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')] - \mathbb{E}_{\xi' \sim K_s(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')] \quad (89)$$

where the first term (86) is a Martingale Difference Sequence and we denote $K_s \triangleq K_{(x_s, y_s)}$. By taking the expectation, we get:

$$\begin{aligned} \mathbb{E}[v_f(x_s, y_s, \xi_{s+2}) - \mathbb{E}_{\xi' \sim K_{s+1}(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')]] &= \mathbb{E}[\mathbb{E}_{\xi' \sim K_{s+1}(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')]] \\ &\quad - \mathbb{E}_{\xi' \sim K_{s+1}(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')] \\ &= 0 \end{aligned}$$

moving to the second term (87) and for legibility, we set $v_f(x_s, y_s, \xi_{s+2}) = v_s$ and $v_f(x_{s+1}, y_{s+1}, \xi_{s+2}) = v_{s+1}$ we have:

$$\begin{aligned} \alpha \left\| \sum_{s=0}^k (I - \alpha J_{xx}^{\alpha, \beta})^{k-s} (v_s - v_{s+1}) \right\| &\leq \alpha \sum_{s=0}^k \|I - \alpha J_{xx}^{\alpha, \beta}\|^{k-s} \|v_s - v_{s+1}\| \\ &\leq \alpha \sqrt{\kappa(\mathbb{L})} \sum_{s=0}^k e^{-c\alpha(k-s)} \|v_s - v_{s+1}\| \\ &\leq \alpha \sqrt{\kappa(\mathbb{L})} L_{v_f} \sum_{s=0}^k e^{-c\alpha(k-s)} (\|x_s - x_{s+1}\| + \|y_s - y_{s+1}\|) \\ &\leq \alpha \sqrt{\kappa(\mathbb{L})} L_{v_f} (\alpha M_f + \beta M_g) \sum_{s=0}^k e^{-c\alpha s} \\ &\leq \alpha \sqrt{\kappa(\mathbb{L})} L_{v_f} (\alpha M_f + \beta M_g) (1 + \int_0^\infty e^{-c\alpha t} dt) \\ &= \alpha c^{-1} \sqrt{\kappa(\mathbb{L})} L_{v_f} M_f + \beta c^{-1} \sqrt{\kappa(\mathbb{L})} L_{v_f} M_g \\ &\quad + \alpha^2 \sqrt{\kappa(\mathbb{L})} L_{v_f} M_f + \alpha \beta \sqrt{\kappa(\mathbb{L})} L_{v_f} M_g \end{aligned}$$

Next, we handle the pseudo-telescoping term (88), but before that we set $v_f(x_s, y_s, \xi_{s+1}) = v_s$, $v_f(x_{s+1}, y_{s+1}, \xi_{s+2}) = v_{s+1}$, $J_{xx}^{\alpha, \beta} = J_{xx} + \frac{\beta}{\alpha} U J_{yx}$ and $A_{\alpha, \beta} = I - \alpha J_{xx}^{\alpha, \beta}$. From the Neumann series convergence, $A_{\alpha, \beta}$ is invertible if we have a condition of the type

$$\alpha \|J_{xx}^{\alpha, \beta}\|_{\text{op}} < 1$$

since we are assuming

$$\alpha < \frac{1}{\|J_{xx}\|_{\text{op}} + (R_{ric} + \|U_0\|_{\text{op}}) \|J_{yx}\|_{\text{op}}} \quad (90)$$

then the condition is satisfied and $A_{\alpha, \beta}$ is invertible and the inverse can be given by the Neumann series $A_{\alpha, \beta}^{-1} = \sum_{k=0}^\infty (\alpha J_{xx}^{\alpha, \beta})^k$. However, we will not need the explicit form of the inverse, so we

use it directly:

$$\begin{aligned}
\sum_{s=0}^k A_{\alpha,\beta}^{k-s} (v_s - v_{s+1}) &= \sum_{s=0}^k A_{\alpha,\beta}^{k-s} v_s - A_{\alpha,\beta} A_{\alpha,\beta}^{k-(s+1)} v_{s+1} \\
&= \sum_{s=0}^k A_{\alpha,\beta}^{k-s} v_s - A_{\alpha,\beta}^{k-(s+1)} v_{s+1} + \alpha J_{xx}^{\alpha,\beta} A_{\alpha,\beta}^{-1} \sum_{s=0}^k A_{\alpha,\beta}^{k-s} v_{s+1} \\
&= A_{\alpha,\beta}^k v_0 - A_{\alpha,\beta}^{-1} v_{k+1} + \alpha J_{xx}^{\alpha,\beta} A_{\alpha,\beta}^{-1} \sum_{s=0}^k A_{\alpha,\beta}^{k-s} v_{s+1}
\end{aligned}$$

taking the norm:

$$\begin{aligned}
\alpha \left\| \sum_{s=0}^k A_{\alpha,\beta}^{k-s} (v_s - v_{s+1}) \right\| &\leq \alpha^2 \|J_{xx}^{\alpha,\beta}\|_{\text{op}} \|A_{\alpha,\beta}^{-1}\|_{\text{op}} \sum_{s=0}^k \|A_{\alpha,\beta}^{k-s}\|_{\text{op}} \|v_{s+1}\| + \alpha \|A_{\alpha,\beta}^k\|_{\text{op}} \|v_0\| \\
&\quad + \alpha \|A_{\alpha,\beta}^{-1}\|_{\text{op}} \|v_{k+1}\| \\
&\leq \alpha^2 \sqrt{\kappa(\mathbf{L})} \|J_{xx}^{\alpha,\beta}\|_{\text{op}} \|A_{\alpha,\beta}^{-1}\|_{\text{op}} M_\xi \sum_{s=0}^k e^{-c\alpha s} + \alpha e^{-c\alpha k} \sqrt{\kappa(\mathbf{L})} M_\xi \\
&\quad + \alpha \|A_{\alpha,\beta}^{-1}\|_{\text{op}} M_\xi \\
&\leq \sqrt{\kappa(\mathbf{L})} \|J_{xx}^{\alpha,\beta}\|_{\text{op}} \|A_{\alpha,\beta}^{-1}\|_{\text{op}} M_\xi \left(\alpha^2 + \frac{\alpha}{c} \right) + \alpha e^{-c\alpha k} \sqrt{\kappa(\mathbf{L})} M_\xi \\
&\quad + \alpha \|A_{\alpha,\beta}^{-1}\|_{\text{op}} M_\xi \\
&= \alpha^2 \sqrt{\kappa(\mathbf{L})} \|J_{xx}^{\alpha,\beta}\|_{\text{op}} \|A_{\alpha,\beta}^{-1}\|_{\text{op}} M_\xi + \alpha e^{-c\alpha k} \sqrt{\kappa(\mathbf{L})} M_\xi \\
&\quad + \alpha c^{-1} \sqrt{\kappa(\mathbf{L})} \|J_{xx}^{\alpha,\beta}\|_{\text{op}} \|A_{\alpha,\beta}^{-1}\|_{\text{op}} M_\xi + \alpha \|A_{\alpha,\beta}^{-1}\|_{\text{op}} M_\xi
\end{aligned}$$

where in the second inequality we used $\|D\|_{\text{op}} \leq \sqrt{\kappa(\mathbf{L})} \|D\|_{\mathbf{L}}$. The last term (89) is capturing the drift in the kernel and is bounded in the same way as (58) by

$$\left\| \mathbb{E}_{\xi' \sim K_{s+1}(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')] - \mathbb{E}_{\xi' \sim K_s(\xi_{s+1}, \cdot)} [v_f(x_s, y_s, \xi')] \right\| \leq M_\xi M_S L_K |\Xi| (\alpha M_f + \beta M_g)$$

which yields the bound

$$\alpha \left\| \sum_{s=0}^k (I - \alpha J_{xx}^{\alpha,\beta})^{k-s} (89) \right\| \leq \sqrt{\kappa(\mathbf{L})} M_S M_\xi L_K |\Xi| c^{-1} (\alpha M_f + \beta M_g)$$

We pack all the bounds together, under our assumptions on the step sizes there is C_A, C_J that satisfy

$\|A_{\alpha,\beta}^{-1}\|_{\text{op}} \leq C_A$ and $\|J_{xx}^{\alpha,\beta}\|_{\text{op}} \leq C_J$ so we have:

$$\begin{aligned}
(85) &\leq \underbrace{\alpha \sqrt{\kappa(\mathbf{L})} \left(c^{-1} M_f (L_{v_f} + M_S M_\xi L_K |\Xi|) + M_\xi \left(1 + C_A \left(c^{-1} C_J + \sqrt{\kappa(\mathbf{L})}^{-1} \right) \right) \right)}_{\triangleq A_1} \\
&\quad + \underbrace{\beta \left(c^{-1} \sqrt{\kappa(\mathbf{L})} M_g (L_{v_f} + M_S M_\xi L_K |\Xi|) \right)}_{\triangleq A_2} + \underbrace{\alpha^2 \sqrt{\kappa(\mathbf{L})} (L_{v_f} M_f + C_J C_A M_\xi)}_{\triangleq A_3} \\
&\quad + \underbrace{\alpha \beta \left(\sqrt{\kappa(\mathbf{L})} L_{v_f} M_g \right)}_{\triangleq A_4}
\end{aligned}$$

The second part of (48) concerns bounding the term:

$$\beta \left\| \sum_{j=0}^k (I - \alpha J_{xx}^{\alpha,\beta})^{k-j} U \mathbb{E} [\varphi_j] \right\| \tag{91}$$

that is similar to the first one, except that now we have a multiplication by β and the deviations concern the dynamics of the slow variable. There exists a function v_g solution of the Poisson equation:

$$v_g(x_j, y_j, \xi_{j+1}) - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_g(x_j, y_j, \xi')] = \varphi_j$$

so we can write:

$$\varphi_j = v_g(x_j, y_j, \xi_{j+2}) - \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)}[v_g(x_j, y_j, \xi')] \quad (92)$$

$$+ v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) - v_g(x_j, y_j, \xi_{j+2}) \quad (93)$$

$$+ v_g(x_j, y_j, \xi_{j+1}) - v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) \quad (94)$$

$$+ \mathbb{E}_{\xi' \sim K_{j+1}(\xi_{j+1}, \cdot)}[v_g(x_j, y_j, \xi')] - \mathbb{E}_{\xi' \sim K_j(\xi_{j+1}, \cdot)}[v_g(x_j, y_j, \xi')] \quad (95)$$

where term (92) is an MDS, under expectation it will equal zero. The second term (93), similarly to the previous part, we set $v_g(x_j, y_j, \xi_{j+2}) = v_j$ and $v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) = v_{j+1}$ we have

$$\begin{aligned} \beta \left\| \sum_{j=0}^k (I - \alpha J_{xx}^{\alpha, \beta})^{k-j} U(v_j - v_{j+1}) \right\| &\leq \beta^2 \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} L_{v_g} M_g + \alpha \beta \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} L_{v_g} M_f \\ &\quad + \frac{\beta^2}{c\alpha} \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} L_{v_g} M_g + \beta c^{-1} \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} L_{v_g} M_f \end{aligned}$$

for the third term (94) it will follow the same bound as the one used in (88), we just need to multiply it by β , we set $v_g(x_j, y_j, \xi_{j+1}) = v_j$, $v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) = v_{j+1}$:

$$\begin{aligned} \beta \left\| \sum_{j=0}^k A_{\alpha, \beta}^{k-j} U(v_j - v_{j+1}) \right\| &\leq \alpha \beta \sqrt{\kappa(\mathbf{L})} \|J_{xx}^{\alpha, \beta}\|_{\text{op}} \|U\|_{\text{op}} \|A_{\alpha, \beta}^{-1}\|_{\text{op}} M_\xi + \beta e^{-c\alpha k} \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} M_\xi \\ &\quad + \beta c^{-1} \sqrt{\kappa(\mathbf{L})} \|J_{xx}^{\alpha, \beta}\|_{\text{op}} \|U\|_{\text{op}} \|A_{\alpha, \beta}^{-1}\|_{\text{op}} M_\xi + \beta \|A_{\alpha, \beta}^{-1}\|_{\text{op}} \|U\|_{\text{op}} M_\xi \end{aligned}$$

Finally, we bound the term

$$\beta \left\| \sum_{s=0}^k (I - \alpha J_{xx}^{\alpha, \beta})^{k-s} U(95) \right\| \leq \frac{\beta}{\alpha} \|U\|_{\text{op}} \sqrt{\kappa(\mathbf{L})} M_S M_\xi L_K |\Xi| c^{-1} (\alpha M_f + \beta M_g)$$

Putting all the bounds together, we get:

$$\begin{aligned} (91) &\leq \underbrace{\beta \|U\|_{\text{op}} \sqrt{\kappa(\mathbf{L})} \left(c^{-1} (L_{v_g} + M_S M_\xi L_K |\Xi|) M_f + M_\xi \left(C_A \left(c^{-1} C_J + \sqrt{\kappa(\mathbf{L})}^{-1} \right) + 1 \right) \right)}_{\triangleq B_1} \\ &\quad + \underbrace{\beta^2 \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} L_{v_g} M_g}_{\triangleq B_2} + \underbrace{\alpha \beta \sqrt{\kappa(\mathbf{L})} \left(C_J \|U\|_{\text{op}} C_A M_\xi + \|U\|_{\text{op}} L_{v_g} M_f \right)}_{\triangleq B_3} \\ &\quad + \underbrace{\frac{\beta^2}{\alpha} \sqrt{\kappa(\mathbf{L})} \|U\|_{\text{op}} M_g c^{-1} (L_{v_g} + M_S M_\xi L_K |\Xi|)}_{\triangleq B_4} \end{aligned}$$

We merge the major two bounds together to get:

$$(85) + (91) \leq \alpha A_1 + \beta(A_2 + B_1) + \alpha^2 A_3 + \beta^2 B_2 + \alpha \beta(A_4 + B_3) + \frac{\beta^2}{\alpha} B_4$$

ending the first part of the proof. Now we start with the second part to bound (52). We start by the first term:

$$\alpha \left\| \sum_{j=0}^k (I - \beta J_{yy}^{\alpha, \beta})^{k-j} X \mathbb{E}[\psi_j] \right\| \quad (96)$$

as we have discussed above about rewriting the inside terms with the solutions of the Poisson equation, we set $v_f(x_s, y_s, \xi_{s+2}) = v_s$ and $v_f(x_{s+1}, y_{s+1}, \xi_{s+2}) = v_{s+1}$ we have:

$$\begin{aligned}
\alpha \left\| \sum_{s=0}^k (I - \beta J_{yy}^{\alpha, \beta})^{k-s} X(v_s - v_{s+1}) \right\| &\leq \alpha \sum_{s=0}^k \|I - \beta J_{yy}^{\alpha, \beta}\|^{k-s} \|X\| \|v_s - v_{s+1}\| \\
&\leq \alpha \sqrt{\kappa(G)} \|X\| \sum_{s=0}^k e^{-d\beta(k-s)} \|v_s - v_{s+1}\| \\
&\leq \alpha \sqrt{\kappa(G)} L_{v_f} \|X\| (\alpha M_f + \beta M_g) (1 + \frac{1}{d\beta}) \\
&\leq \beta R_{syl} \sqrt{\kappa(G)} L_{v_f} (\alpha M_f + \beta M_g) (1 + \frac{1}{d\beta}) \\
&= \alpha \beta R_{syl} \sqrt{\kappa(G)} L_{v_f} M_f + \beta^2 R_{syl} \sqrt{\kappa(G)} L_{v_f} M_g \\
&\quad + \alpha d^{-1} R_{syl} \sqrt{\kappa(G)} L_{v_f} M_f + \beta d^{-1} R_{syl} \sqrt{\kappa(G)} L_{v_f} M_g
\end{aligned}$$

where in the fourth inequality we have used the bound (41), now we target the almost telescoping term. We set $v_f(x_s, y_s, \xi_{s+1}) = v_s$, $v_f(x_{s+1}, y_{s+1}, \xi_{s+2}) = v_{s+1}$, $J_{yy}^{\alpha, \beta} = \Delta - J_{yx}(U - U_0)$ and $B_{\alpha, \beta} = I - \beta J_{yy}^{\alpha, \beta}$. Since we are assuming

$$\beta < \frac{1}{\|\Delta\|_{\text{op}} + R_{ric} \|J_{yx}\|_{\text{op}}} \quad (97)$$

then $B_{\alpha, \beta}$ is invertible using the Neumann series, we use the inverse to have:

$$\begin{aligned}
\sum_{s=0}^k B_{\alpha, \beta}^{k-s} X(v_s - v_{s+1}) &= \sum_{s=0}^k B_{\alpha, \beta}^{k-s} X v_s - B_{\alpha, \beta} B_{\alpha, \beta}^{k-(s+1)} X v_{s+1} \\
&= \sum_{s=0}^k B_{\alpha, \beta}^{k-s} X v_s - B_{\alpha, \beta}^{k-(s+1)} X v_{s+1} + \beta J_{yy}^{\alpha, \beta} B_{\alpha, \beta}^{-1} \sum_{s=0}^k B_{\alpha, \beta}^{k-s} X v_{s+1} \\
&= B_{\alpha, \beta}^k X v_0 - B_{\alpha, \beta}^{-1} X v_{k+1} + \beta J_{yy}^{\alpha, \beta} B_{\alpha, \beta}^{-1} \sum_{s=0}^k B_{\alpha, \beta}^{k-s} X v_{s+1}
\end{aligned}$$

applying the norm:

$$\begin{aligned}
\alpha \left\| \sum_{s=0}^k B_{\alpha, \beta}^{k-s} X(v_s - v_{s+1}) \right\| &\leq \alpha \|B_{\alpha, \beta}^k\|_{\text{op}} \|X\|_{\text{op}} \|v_0\| + \alpha \|B_{\alpha, \beta}^{-1}\|_{\text{op}} \|X\|_{\text{op}} \|v_{k+1}\| \\
&\quad + \alpha \beta \|J_{yy}^{\alpha, \beta}\|_{\text{op}} \|B_{\alpha, \beta}^{-1}\|_{\text{op}} \|X\|_{\text{op}} \sum_{s=0}^k \|B_{\alpha, \beta}\|_{\text{op}}^{k-s} \|v_{s+1}\| \\
&\leq \beta \sqrt{\kappa(G)} R_{syl} e^{-d\beta k} M_\xi + \beta \sqrt{\kappa(G)} R_{syl} \|B_{\alpha, \beta}^{-1}\| M_\xi \\
&\quad + \beta^2 \sqrt{\kappa(G)} R_{syl} \|J_{yy}^{\alpha, \beta}\| \|B_{\alpha, \beta}^{-1}\| M_\xi \sum_{s=0}^k e^{-d\beta(k-s)} \\
&\leq \beta R_{syl} \sqrt{\kappa(G)} M_\xi \left(e^{-d\beta k} + \|B_{\alpha, \beta}^{-1}\| + \|J_{yy}^{\alpha, \beta}\| \|B_{\alpha, \beta}^{-1}\| d^{-1} \right) \\
&\quad \beta^2 R_{syl} \sqrt{\kappa(G)} M_\xi \|J_{yy}^{\alpha, \beta}\| \|B_{\alpha, \beta}^{-1}\|
\end{aligned}$$

and we get our bound for the first part, under our assumptions on the step sizes there is C_A, C_J that satisfy $\|B_{\alpha,\beta}^{-1}\| \leq C_B$ and $\|J_{yy}^{\alpha,\beta}\|_{\text{op}} \leq C_J$ so we have:

$$\begin{aligned}
(96) \leq & \underbrace{\beta R_{syl} \sqrt{\kappa(\mathbf{G})} (d^{-1}(L_{v_f} + M_S M_\xi L_K |\Xi|) M_g + M_\xi (1 + C_B + \|J_{yy}^{\alpha,\beta}\| C_B d^{-1}))}_{\triangleq C_1} \\
& + \underbrace{\alpha R_{syl} \sqrt{\kappa(\mathbf{G})} (L_{v_f} + M_S M_\xi L_K |\Xi|) M_f d^{-1}}_{\triangleq C_2} + \underbrace{\beta^2 R_{syl} \sqrt{\kappa(\mathbf{G})} (M_\xi C_J C_B + L_{v_f} M_g)}_{\triangleq C_3} \\
& + \underbrace{\alpha \beta R_{syl} \sqrt{\kappa(\mathbf{G})} L_{v_f} M_f}_{\triangleq C_4}
\end{aligned}$$

our last bound is for the term :

$$\beta \left\| \sum_{j=0}^k (I - \beta J_{yy}^{\alpha,\beta})^{k-j} I' \mathbb{E}[\varphi_j] \right\| \quad (98)$$

from the Poisson decomposition the first term will equal zero under expectation. For the second term (93), similarly to the previous part, we set $v_g(x_j, y_j, \xi_{j+2}) = v_j$ and $v_g(x_{j+1}, y_{j+1}, \xi_{j+2}) = v_{j+1}$ we have:

$$\begin{aligned}
\beta \left\| \sum_{j=0}^k (I - \beta J_{yy}^{\alpha,\beta})^{k-j} I'(v_j - v_{j+1}) \right\| \leq & \alpha \beta \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} L_{v_g} M_f + \beta d^{-1} \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} L_{v_g} M_g \\
& + \alpha d^{-1} \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} L_{v_g} M_f + \beta^2 \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} L_{v_g} M_g
\end{aligned}$$

and the last term, we set $v_j = v_g(x_j, y_j, \xi_{j+1})$, $v_{j+1} = v_g(x_{j+1}, y_{j+1}, \xi_{j+2})$:

$$\begin{aligned}
\beta \left\| \sum_{s=0}^k B_{\alpha,\beta}^{k-s} I'(v_s - v_{s+1}) \right\| \leq & \beta \|I'\|_{\text{op}} M_\xi \left(\sqrt{\kappa(\mathbf{G})} e^{-d\beta k} + \|B_{\alpha,\beta}^{-1}\|_{\text{op}} + \sqrt{\kappa(\mathbf{G})} \|J_{yy}^{\alpha,\beta}\|_{\text{op}} \|B_{\alpha,\beta}^{-1}\|_{\text{op}} d^{-1} \right) \\
& + \beta^2 \|I'\|_{\text{op}} M_\xi \sqrt{\kappa(\mathbf{G})} \|J_{yy}^{\alpha,\beta}\|_{\text{op}} \|B_{\alpha,\beta}^{-1}\|_{\text{op}}
\end{aligned}$$

which implies the bound on the term:

$$\begin{aligned}
(98) \leq & \underbrace{\beta \|I'\|_{\text{op}} M_\xi \left(C_B + \sqrt{\kappa(\mathbf{G})} \left(1 + d^{-1} \left(C_J C_B + \|I'\|_{\text{op}} (L_{v_g} + M_S M_\xi L_K |\Xi|) M_g \right) \right) \right)}_{\triangleq D_1} \\
& + \underbrace{\beta^2 \left(\|I'\|_{\text{op}} M_\xi \sqrt{\kappa(\mathbf{G})} C_J C_B + \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} L_{v_g} M_g \right)}_{\triangleq D_2} \\
& + \underbrace{\alpha \beta \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} L_{v_g} M_f}_{\triangleq D_3} + \underbrace{\alpha d^{-1} \|I'\|_{\text{op}} \sqrt{\kappa(\mathbf{G})} (L_{v_g} + M_S M_\xi L_K |\Xi|) M_f}_{\triangleq D_4}
\end{aligned}$$

which implies the following bound:

$$(96) + (98) \leq \alpha(C_2 + D_4) + \beta(C_1 + D_1) + \alpha\beta(C_4 + D_3) + \beta^2(C_3 + D_2)$$

which concludes the proof. $\square$

Lastly, this lemma is needed to bound the weighted sum of the MSEs of the fast and slow iterates. We need this result in the proof of Theorem 3.

Lemma 8. *Taking the sequence $\{x_k, y_k\}_{k \geq 0}$ defined by (1) and under assumptions A.1-4, let q, γ be positive real numbers, we have:*

$$\sum_{s=0}^k e^{-q\gamma(k-s)} \mathbb{E} \left[\|\hat{y}_s\|^2 + \|\hat{x}_s\|^2 \right] \leq \left(1 + \frac{1}{q\gamma} \right) (\text{MSE}_\infty^x(\alpha, \beta) + \text{MSE}_\infty^y(\alpha, \beta)) + u_k^{q,\gamma} + w_k^{q,\gamma}$$

where $\text{MSE}_\infty^x(\alpha, \beta)$ and $\text{MSE}_\infty^y(\alpha, \beta)$ are defined in Theorem 1 and

$$u_k^{q,\gamma} = \|\hat{x}_0\|^2 e^{-kq\gamma} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f))^s$$

$$w_k^{q,\gamma} = \sum_{s=0}^k e^{-q\gamma(k-s)} (1 - \beta\mu_g)^s \|\hat{y}_0\|^2 + \sum_{s=0}^k e^{-q\gamma(k-s)} D_1 \beta s (1 - \alpha\mu_f \wedge \beta\mu_g)^s \|\hat{x}_0\|^2$$

are vanishing sequences as $k \rightarrow \infty$.

Proof. First, we start by analyzing the fast iterate, we invoke the result of Theorem 1 to bound its MSE, we start by the sum of MSEs of the fast iterate:

$$\begin{aligned} \sum_{s=0}^k e^{-q\gamma(k-s)} \mathbb{E} [\|\hat{x}_s\|^2] &\leq \sum_{s=0}^k e^{-q\gamma(k-s)} \left((1 - \alpha\mu_f)^s \|\hat{x}_0\|^2 + \text{MSE}_\infty^x(\alpha, \beta) \right) \\ &\leq \sum_{s=0}^k e^{-q\gamma(k-s)} \left((1 - \alpha\mu_f)^s \|\hat{x}_0\|^2 \right) + \text{MSE}_\infty^x(\alpha, \beta) \left(1 + \int_0^\infty e^{-q\gamma t} dt \right) \\ &= \sum_{s=0}^k e^{-q\gamma(k-s)} \left((1 - \alpha\mu_f)^s \|\hat{x}_0\|^2 \right) + \text{MSE}_\infty^x(\alpha, \beta) \left(1 + \frac{1}{q\gamma} \right) \\ &\leq \|\hat{x}_0\|^2 e^{-kq\gamma} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f))^s \\ &\quad + \left(1 + \frac{1}{q\gamma} \right) \text{MSE}_\infty^x(\alpha, \beta) \end{aligned} \tag{99}$$

Now from our assumptions we have $1 - \alpha\mu_f \in (0, 1]$. If $e^{q\gamma}(1 - \alpha\mu_f) = 1$ then the sum equals $e^{-kq\gamma} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f))^s = e^{-kq\gamma}(k+1)$ which will go to 0 as $k \rightarrow \infty$. For the other case $e^{q\gamma}(1 - \alpha\mu_f) \neq 1$ so the sum :

$$e^{-kq\gamma} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f))^s = \frac{e^{-kq\gamma} - e^{q\gamma}(1 - \alpha\mu_f)^{k+1}}{1 - e^{q\gamma}(1 - \alpha\mu_f)}$$

which will also disappear as k goes to infinity. So we set the sequence $u_k^{q,\gamma}$ to be defined by:

$$u_k^{q,\gamma} = \|\hat{x}_0\|^2 e^{-kq\gamma} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f))^s$$

such that

$$\lim_{k \rightarrow \infty} u_k^{q,\gamma} = 0$$

Same reasoning goes with the MSE of the slow iterate, from Theorem 1 following the same steps in the fast MSE above we get:

$$\begin{aligned} \sum_{s=0}^k e^{-q\gamma(k-s)} \mathbb{E} [\|\hat{y}_s\|^2] &\leq \sum_{s=0}^k e^{-q\gamma(k-s)} (1 - \beta\mu_g)^s \|\hat{y}_0\|^2 \\ &\quad + \sum_{s=0}^k e^{-q\gamma(k-s)} D_1 \beta s (1 - \alpha\mu_f \wedge \beta\mu_g)^s \|\hat{x}_0\|^2 \end{aligned} \tag{100}$$

$$+ \sum_{s=0}^k e^{-q\gamma(k-s)} \text{MSE}_\infty^y(\alpha, \beta) \tag{101}$$

The first term follows the same reasoning as the term (99), we focus on the second term (100):

$$(100) \leq \|\hat{x}_0\|^2 D_1 \beta k e^{-q\gamma k} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f \wedge \beta\mu_g))^s$$

if $e^{q\gamma}(1 - \alpha\mu_f \wedge \beta\mu_g) = 1$:

$$ke^{-q\gamma k} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f \wedge \beta\mu_g))^s = e^{-q\gamma k} k(k+1)$$

which vanishes as $k \rightarrow \infty$.

otherwise, if $e^{q\gamma}(1 - \alpha\mu_f \wedge \beta\mu_g) \neq 1$:

$$ke^{-q\gamma k} \sum_{s=0}^k (e^{q\gamma}(1 - \alpha\mu_f \wedge \beta\mu_g))^s = \frac{ke^{-q\gamma k} - ke^{q\gamma}(1 - \beta\mu_g \wedge \alpha\mu_f)^{k+1}}{1 - e^{q\gamma}(1 - \beta\mu_g \wedge \alpha\mu_f)}$$

which also vanishes as $k \rightarrow \infty$. We define the new sequence $w_k^{q,\gamma}$ as:

$$w_k^{q,\gamma} = \sum_{s=0}^k e^{-q\gamma(k-s)} (1 - \beta\mu_g)^s \|\hat{y}_0\|^2 + \sum_{s=0}^k e^{-q\gamma(k-s)} D_1 \beta s (1 - \alpha\mu_f \wedge \beta\mu_g)^s \|\hat{x}_0\|^2$$

such that

$$\lim_{k \rightarrow \infty} w_k^{q,\gamma} = 0$$

and the last term (101) is bounded by

$$(101) \leq \left(1 + \frac{1}{q\gamma}\right) \text{MSE}_\infty^y(\alpha, \beta)$$

now we can write the final bound:

$$\sum_{s=0}^k e^{-q\gamma(k-s)} \mathbb{E} \left[\|\hat{y}_s\|^2 + \|\hat{x}_s\|^2 \right] \leq \left(1 + \frac{1}{q\gamma}\right) (\text{MSE}_\infty^x(\alpha, \beta) + \text{MSE}_\infty^y(\alpha, \beta)) + u_k^{q,\gamma} + w_k^{q,\gamma}$$

□

F Illustrations

F.1 First example

We investigate the empirical performance of a TTSA where both variables are 1-dimensional: $x_k \in \mathbb{R}$ and $y_k \in \mathbb{R}$ and satisfy the following recursion

$$\begin{aligned} x_{k+1} &= x_k - \alpha(x_k - y_k + \xi_{k+1}^x) \\ y_{k+1} &= y_k - \beta(2y_k - e^{-y_k} - x_k + \xi_{k+1}^y), \end{aligned} \quad (102)$$

where the noise terms $\xi_k^x, \xi_k^y \in \{-1, 1\}$ form the Markovian noise with transition probability kernel $\mathbb{P}[\xi_{k+1} = s' \mid \xi_k = s] = 0.3$ for $s \neq s'$. Initially, we set $x_0 = y_0 = 0$. For this example, $h(y) = y$ and y^* is the solution of $y = \exp(-y)$ which is equal to $W_0(1) \approx 0.567$, where W_0 is the standard Lambert function.

In Figure 2, we display a trajectory of the TTSA for three values of $\alpha \in \{0.1, 0.01, 0.002\}$ and a fixed $\beta = 0.001$. We observe two phenomena that are shown in Theorem 1:

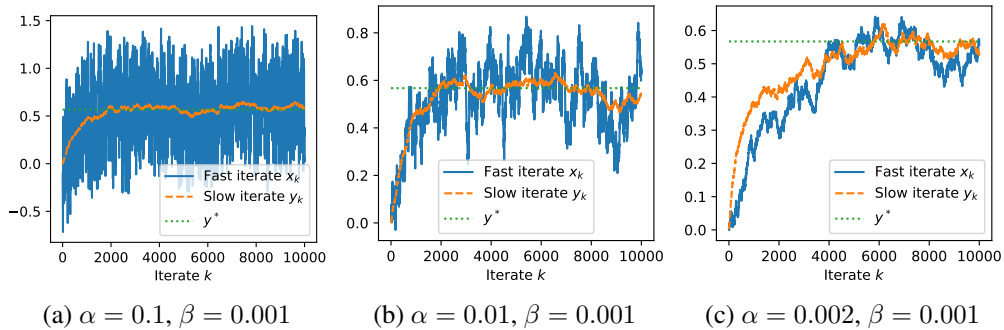
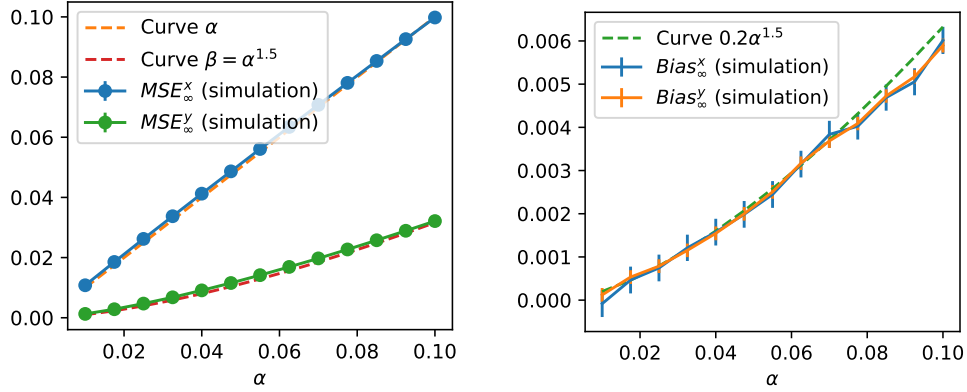


Figure 2: Sample trajectory of the TTSA defined in (102).

1. First, the vanishing term vanishes at a slower rate as α gets smaller. For instance, when $\alpha = 0.002$ (Figure 2(c)), the fast variable is lagging behind $h(y_k) = y_k$ where when $\alpha = 0.01$ (Figure 2(b)), the variable x_k has no lag y_k .
2. Second, the larger is α , the larger are the oscillations of x_k and y_k : in Figure 2(a), the variable x_k is not lagging behind $h(y_k) = y_k$ but it is oscillating a lot.

In Figure 3, we focus on the asymptotic behavior of the MSE and of the bias as a function of the step size α and for the case where there is a time scale separation $\beta = \alpha^{1.5}$. For each value, we simulate a trajectory of length $k = 1.2 \cdot 10^5$ steps and compute the average values of x_k for the last half of the trajectory. We perform 5000 independent trajectories to obtain the confidence intervals (they are shown on both panels but are too small to be visible on the MSE plot in Figure 3(a)). This figure shows that for this example, MSE^x seems to be of the order of α whereas MSE^y and the bias terms are of the order of $\beta = \alpha^{1.5}$.



(a) MSE as a function of α (for $\beta = \alpha^{3/2}$) (b) Bias as a function of α (for $\beta = \alpha^{3/2}$)

Figure 3: MSE and Bias of the TTSA defined in (102) for $\beta = \alpha^{3/2}$

F.2 Example showing a lower bound in $\Omega(\beta^2/\alpha^2)$

The example used in the paper (Figure 1(c)) to illustrate the $\Omega(\beta^2/\alpha^2)$ term in the MSE is a TTSA whose recurrence equation is given by

$$\begin{aligned} x_{k+1} &= x_k - \alpha(x_k - y_k) \\ y_{k+1} &= y_k - \beta \frac{(x_k - y_k)}{|x_k - y_k|^\gamma}, \end{aligned} \quad (103)$$

with $x_0 = 0$ and $y_0 = 1$.

In this example, x_k chases the variable y_k ($h(y_k) = y_k$), while y_k tries to escape x_k . This example can be easily analyzed: Rewriting $\delta_k \triangleq y_k - x_k$, one obtains

$$\delta_{k+1} = (1 - \alpha)\delta_k + \beta \frac{(\delta_k)}{|\delta_k|^\gamma}$$

As k goes to infinity, this shows that $\lim_{k \rightarrow \infty} \delta_k = (\beta/\alpha)^{1/\gamma}$. This shows that:

$$\lim_{k \rightarrow \infty} \text{MSE}^x = \left(\frac{\beta}{\alpha} \right)^{2/\gamma}.$$

When γ is close to 1, this is essentially equal to $(\beta/\alpha)^2$. This is what is illustrated in Figure 4: for this example, the quantity MSE^x does not depend on α and is equal to $(\beta/\alpha)^{2/\gamma}$ for all α , β and γ .

Discussion on the assumptions The TTSA (103) satisfies the assumptions about the kernel and the fast iterate, since here we care about the tracking error of the fast iterate that is of order $\frac{\beta^2}{\alpha^2}$ and

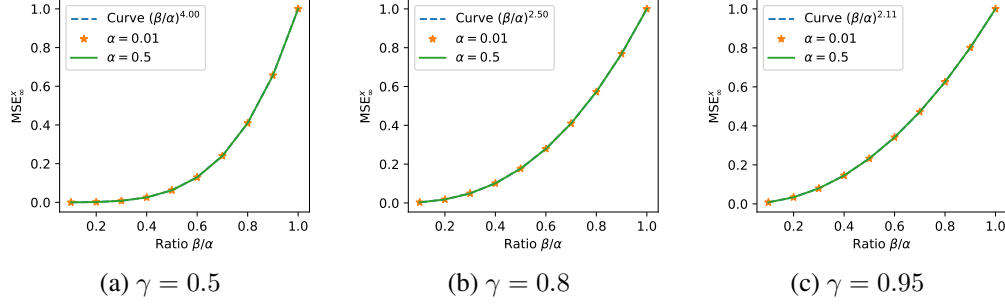


Figure 4: MSE^x of the TTSA given by (103) as a function of the ratio β/α . We observe that MSE^x does not depend on α and is equal to $(\beta/\alpha)^{2/\gamma}$ for all α, β and γ .

we got it from the MSE analysis of x without using information about Lipschitzness nor the strong monotonicity of g so A.4 was not needed in that analysis. In addition to that we relax the assumption on the boundedness of the iterates as we could not find an example that guarantees boundedness.

The fact that x_k and y_k do not live in a compact set is more problematic but can be solved by slightly modifying the above example. Note that this new example cannot be easily solved in closed form which is why we decided to keep the TTSA (103) as the main example. To construct this new example, we consider 2-dimensional variables $x_k, y_k \in \mathbb{R}^2$ with $x_0 = [1, 0]$, $y_0 = [1, 1]$ and

$$\begin{aligned} x_{k+1} &= x_k - \alpha(x_k - y_k + 0.1\xi_{k+1}) \\ y_{k+1} &= y_k - \beta \left(\frac{(x_k - y_k)}{\|x_k - y_k\|^\gamma} + y_k \tanh(\|y_k\|^2 - 1) \right), \end{aligned} \quad (104)$$

where ξ_{k+1} is a sequence of Rademacher random variables.

This example is essentially the same as the TTSA in (103) with two differences. First, we add a small white noise in the dynamics of x_k (this does not change anything, but shows that the counter-example does not depend on the dynamics being deterministic). Second, and more importantly, the evolution of y_k contains the term $y_k \tanh(\|y_k\|^2 - 1)$. The role of this term is to push y_k towards the unit circle. As an example, we display sample trajectories of (x_k, y_k) in Figure 5 for various values of α and $\beta = 0.1\alpha$. We observe that both iterates x_k and y_k oscillate on the circle (with more noise when α is larger).

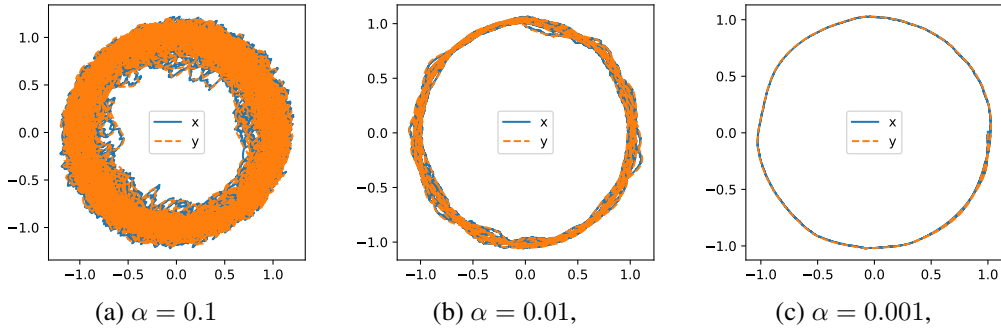


Figure 5: Sample trajectories of x_k and y_k for the TTSA (104) for various values of α and $\beta = \alpha/10$

For this dynamics, we again have $h(y_k) = y_k$. In Figure 6(a), we plot a sample trajectory of the tracking error $x_k - h(y_k) = x_k - y_k$ for various values of α while keeping the ratio $\beta/\alpha = 0.1$ fixed. We observe that, as α goes to 0, the error does not vanish but seems to converge to a circular dynamics. Similarly to what was shown in Figure 4 for the TTSA (103), we observe in Figure 6(b) that the MSE of x grows like $(\beta/\alpha)^{2/\gamma}$, independently of α .

Note that this model does not satisfy the strong monotonicity assumption for $\bar{g}$ (A.4). It is an open question if one can construct an example that satisfies A.4, and whose asymptotic MSE for x has a dependence on (β/α) .

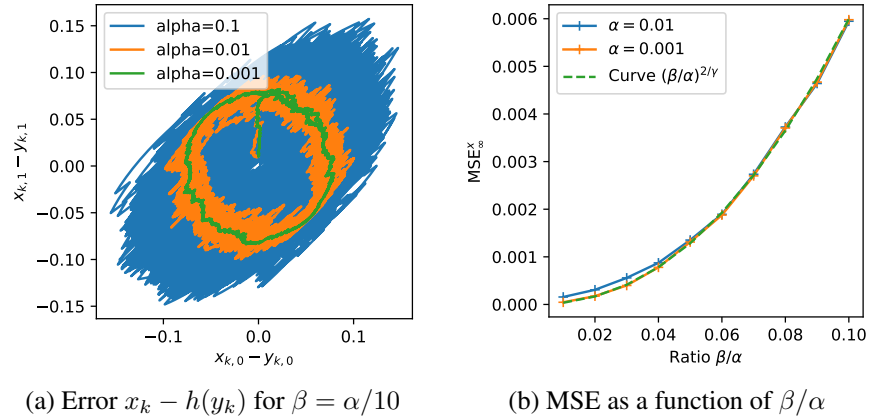


Figure 6: Error and MSE of the TTSA (104).